

ANNEX A

CleanHub — Functional Specifications Document

Technology Development Contract — Initialled Copy

This document is the CleanHub Functional Specifications Document (89 pages). It is attached to the Technology Development Contract as Annex A. Both parties have initialled every page and signed below, confirming that this document constitutes the sole contractual reference for the scope of work to be delivered by Party B.

■ **IMPORTANT:** No feature may be added, removed or modified from this document without a written amendment signed by both parties. The developer (Party B) undertakes to deliver ALL features described in this document within the total budget of RMB 160,000.

Document details:

Document title	CleanHub — Functional Specifications Document
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Provided by	Cheikh Diaw — iWash Group — Dakar, Senegal
Provided to	Party B (Developer — Ying Jiahui)
Date of provision	1 May 2026
Status	CONTRACTUAL REFERENCE — initialled and signed by both parties

Initialling instructions:

Each party must write their initials at the bottom right corner of every page of this document (pages 1 to 89). The initials confirm that both parties have read and agreed to the content of that page. A page without both parties' initials is not considered part of the contractual reference.

Signatures:

Name: Cheikh Diaw: _____ _____
Title / Role: _____
Date of signing: _____ _____
Signature: _____ _____
Party A initials: ____ ____ ____ (initial each page)

Name Ying Jiahui: _____ _____
Title / Role: _____
Date of signing: _____ _____
Signature: _____ _____
Party B initials: ____ ____ ____ (initial each page)

By initialling each page and signing above, both parties confirm that they have read, understood and agreed to the full content of this Functional Specifications Document, and that it shall serve as the sole contractual reference for the scope of work covered by the Technology Development Contract.

CLEANHUB — REQUIREMENTS CONFIRMATION

CleanHub POS System — Senegal | Date: 10/04/2026

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1. HARDWARE INTEGRATION

1.1 Printers

Three distinct printers are required for different use cases. All will be sourced directly from Chinese manufacturers. No specific brand (Zebra/Brother) is imposed.

Receipt printer — Bluetooth or WiFi connection:

The receipt printer uses Bluetooth or WiFi instead of USB so that it works easily with Android tablets and smartphones as well as Windows PCs. One printer, any device — no cable required.

Connection type	How it works	Best for
Bluetooth	Printer pairs directly with the device (tablet, smartphone or PC)	Single counter — 10 tablets or smartphones or 1 PC
WiFi	Printer connects to the local WiFi network — any device on the same network can print	Multiple counters or multiple devices (tablets or PCs)
USB (fallback)	Printer connects via cable to the PC only — no tablet or smartphone	Fixed Windows PC counter only — not recommended

■ *Recommended approach: select a thermal 80mm printer that supports BOTH Bluetooth AND WiFi (many Chinese models support both). This gives maximum flexibility: use Bluetooth for a single tablet counter, switch to WiFi if a second device is added later. The developer must test Bluetooth and WiFi printing from both the Android app and the Windows Electron app before delivery.*

■ **IMPORTANT — Cash drawer compatibility: the cash drawer connects to the PRINTER via RJ11/RJ12 cable regardless of whether the printer uses USB, Bluetooth or WiFi. The drawer opens via an ESC/POS command sent through the printer — this works identically on all connection types. The developer must verify that the selected Bluetooth/WiFi printer model supports the ESC/POS cash drawer open command.**

Printer	Usage	Recommended Type	Status
Receipt Printer	Customer receipts at POS	Thermal 80mm counter — Bluetooth or WiFi — single tablet or Windows PC, Android tablet	To Be Defined
A4 Laser	B2B invoices	A4 USB Laser — mono or colour	To Be Defined
Matrix / Label Printer	Garment tickets (attached to clothes)	Thermal — narrow roll — durable ink	To Be Defined
Connection (all)	USB — all printers	USB HID — no driver complexity	Confirmed
Portable Bluetooth Printer	Collection slips — printed at the counter	Portable Bluetooth thermal 58mm — paired to a staff smartphone (Android & iOS)	To Be Defined

■ *Garment ticket printer must use durable ink or water-resistant thermal transfer technology — labels must survive washing machine cycles.*

5th Printer — Adhesive Labels (Finished Articles)

Printer	Usage	Recommended Type	Status
Adhesive Label Printer	Print stickers placed on each finished article	Thermal adhesive roll labels (50mm or 80mm) — USB — sourced from China	To Be Defined

■ *Adhesive label must contain: order number + barcode, client name, article type, service performed, collection date, storage location number.*

1.1c Economy Mode — Matrix Printer for Both Receipts AND Garment Labels

For pressing businesses with a limited budget, CleanHub supports a SINGLE-PRINTER MODE where the matrix printer prints BOTH the customer receipt AND the garment marking tickets. The client receives a receipt printed on the same matrix printer used for garment labels — one printer, zero additional cost.

Mode	Printers required	Best for	Limitation
Standard mode (recommended)	2 printers: thermal 80mm for receipt and large expensing label	Medium and large pressing, high daily volume	Higher professional presentation
Economy mode	1 printer: matrix printer ONLY	Small pressing with tight budget, garment labels	Receipts on low-quality paper (narrower, do

How economy mode works in CleanHub:

Feature	Description
Single printer configuration	Manager activates Economy Mode from Settings → Printers. CleanHub switches to single-printer mode.
Customer receipt on matrix paper	The receipt is formatted to fit the matrix printer's narrow paper roll. CleanHub automatically adjusts the layout.
Garment labels unchanged	Garment marking labels are printed identically whether in standard or economy mode — same watermarks.
Print order at intake	When an order is created, CleanHub prints in sequence on the same printer: (1) the customer receipt, then (2) the garment labels.
Cash drawer still works	The matrix printer can still trigger the cash drawer via RJ11/RJ12 cable — the ESC/POS drawer opener command is supported.
Upgrade path	When the pressing grows and wants a dedicated thermal receipt printer, the manager simply deactivates Economy Mode.

- Economy mode is ideal for a pressing that is just starting out or operating in a market where the cost of a second printer is a barrier — the matrix printer alone is sufficient to run a complete CleanHub pressing operation
- The manager can switch between Standard and Economy mode at any time from Settings — no developer intervention, no reinstallation, no data loss
- If the pressing later adds a thermal receipt printer, all historical receipts printed in economy mode are still stored digitally in CleanHub — the transition is seamless

■ *Hardware cost comparison: Standard mode requires 2 printers (matrix + thermal receipt) = approximately 80,000–120,000 XOF total. Economy mode requires 1 printer (matrix only) = approximately 40,000–60,000 XOF. A pressing can start with economy mode and upgrade to standard mode when revenue allows.*

■ **DEVELOPER REQUIREMENT: the developer must implement a printer configuration screen in CleanHub Settings that allows the manager to choose between Standard Mode (separate printers for receipts and labels) and Economy Mode (single matrix printer for both). The receipt layout must be adapted to fit the matrix printer's paper width in economy mode. Both modes must be tested and validated before delivery.**

1.1b EMV Android All-In-One Handheld Terminal — Multi-Client Deployment

CleanHub supports deployment on EMV Android all-in-one handheld terminals. These devices combine a touchscreen, built-in receipt printer, NFC contactless payment reader, Bluetooth and 4G/WiFi in a single portable device. One terminal replaces three separate pieces of hardware: tablet + receipt printer + TPE payment terminal.

Device description — EMV PCI 6-inch Android 11 All-In-One:

Specification	Detail
Screen	6-inch touchscreen — runs CleanHub Android app directly — full POS interface
Built-in printer	Integrated thermal receipt printer — prints CleanHub receipts and collection slips without any external printer

Specification	Detail
NFC payment	Contactless NFC reader — accepts Wave tap, Orange Money tap and bank card contactless payment d
Connectivity	Bluetooth + WiFi + 4G SIM — works anywhere: at the counter, in the field for delivery/collection, in the c
Camera	Built-in camera — used to scan barcodes and QR codes directly from the device — no external Bluetooth
Operating system	Android 11 — CleanHub installed as APK — same application as on tablets and smartphones
EMV / PCI certified	Certified for card payments — compliant with international payment security standards
Battery	Rechargeable battery — full day of use — suitable for mobile agents
Replaces	Android tablet + Bluetooth receipt printer + TPE payment terminal — 3 devices in 1

Multi-client deployment — several terminals in the same centre:

A pressing or car wash can deploy multiple EMV handheld terminals simultaneously — one per agent, one per counter, one for field delivery staff. All terminals connect to the same CleanHub account and synchronise in real time.

Deployment scenario	How it works
Main counter	1 terminal at the reception counter — cashier uses it as the primary POS — prints receipts and acce
Second counter	2nd terminal for a second cashier — both work simultaneously — orders appear on both in real time
Field delivery agent	Delivery staff uses a terminal in the field — collects garments at client location, prints collection slip c
Car wash intake agent	Agent at the car wash entrance registers the vehicle, takes photos, records fuel level and belongings
Roaming manager	Manager moves around the centre with their terminal — can process payments, check orders and pr
Laundry agent	Laundry staff updates wash statuses (In washing / Drying / Ironing) directly on their terminal as they

Advantage	Description
One device — three functions	Tablet + receipt printer + TPE in a single portable device — lower cost per agent station, less hardwa
No cables	Fully wireless — Bluetooth, WiFi or 4G — agents move freely — no counter required for field operati
Same app on all devices	The same CleanHub Android APK runs on tablets, smartphones and EMV terminals — no separate
NFC payment anywhere	Clients can pay by Wave/Orange Money tap or contactless bank card directly on the agent's termina
Camera as scanner	The terminal camera scans garment barcodes and QR codes — the Bluetooth scanner is not needed
Offline capable	The terminal works offline if the network drops — orders queued locally and synced when connection
Multi-language	Full CleanHub interface in French and English — same as all other devices

■ *Recommended sourcing: EMV PCI 6-inch Android 11 all-in-one terminals sourced directly from China — the model shown (similar to Telpo TPS900, Newland N910, or equivalent) is widely available on Alibaba/1688 at competitive prices. The developer must test CleanHub APK installation, NFC payment API integration and Bluetooth printer pairing on this terminal type before delivery.*

■ **DEVELOPER REQUIREMENT: CleanHub Android app must be fully compatible with EMV Android terminals. The receipt printer integration must use the terminal's built-in printer driver (ESC/POS via internal API — not Bluetooth) when running on an all-in-one EMV terminal, and switch automatically to Bluetooth printer when running on a regular Android tablet or smartphone. The NFC payment reader on the terminal must be integrated with the same Wave/Orange Money payment flow as all other devices.**

CleanHub uses a BLUETOOTH barcode scanner — not USB. Bluetooth pairing works on Windows PC, Android tablet and Android smartphone without any driver or cable. One scanner can be used across multiple devices in the same centre.

Specification	Detail	Status
Connection	Bluetooth — pairs directly with Windows PC, Android tablet or Android smartphone	Confirmed
Compatibility	HID Bluetooth profile — recognised as a keyboard by all devices — no driver or app required	Confirmed
Range	Up to 10 metres from the paired device — agent can scan garments or vehicles anywhere in the centre with no line of sight	Confirmed
Multi-device	The scanner can be paired with different devices during the day — tablet or smartphone in the morning, PC in the afternoon	Confirmed
Battery	Rechargeable via USB-C — minimum 8 hours of continuous use per charge	Confirmed
Fallback	If the Bluetooth scanner is unavailable, the agent can manually enter the code number or scan via the device camera	Confirmed

■ *Recommended scanner type: Bluetooth HID barcode scanner, rechargeable, compatible with Android and Windows. To be sourced from China (Taobao / Alibaba). Examples: Inateck BCST-70, Eyoyo or equivalent. The developer must test Bluetooth HID pairing on both the Android app and the Windows Electron app before delivery.*

1.3 Conveyor Belt (Future Integration)

To be confirmed when the hardware is selected. The protocol (HTTP API vs RS232/RS485) will be defined at that time.

2. OPERATING ENVIRONMENT

The system will be designed to support the following device types simultaneously:

Device	Specification	Status
POS Principal	PC Windows (egalement ecran 24") + tablette Android 10"	Confirmed
Appareils Mobiles	Telephones Android & iPhones	Confirmed
Android Version	Recent versions only — no legacy support required	Confirmed
Version iOS	Versions iOS actuelles supportees	Confirmed
Acces Distant Proprietaire	Rapports mobiles via application responsive	Confirmed
Production Line Scan	Android tablet or smartphone camera	Confirmed

3. OFFLINE OPERATIONS & POWER CUT PROTECTION

Power cuts are a daily reality in Senegal and Côte d'Ivoire. CleanHub must be designed from the ground up to handle power cuts, internet outages and device restarts without ANY data loss, transaction corruption or operational interruption. This is not optional — it is a core architectural requirement.

Requirement	Answer	Status
Offline Duration	Full day or more — autonomous 24h+ operation required	Confirmed
Offline Search	Maximum client history accessible offline — full client database	Confirmed
Offline Payment	Wave & Orange Money APIs require network — cash payments processed fully offline — Wave/OM APIs are not used	Confirmed
Power cut survival	All data written to local persistent storage BEFORE connection loss	Confirmed
Device restart recovery	After any unexpected restart (power cut, crash), the app resumes exactly where it left off	Confirmed
Server power cut	Server UPS provides minimum 4 hours of autonomy — data backup runs every 15 minutes — zero data loss	Confirmed

3.1 Power Cut Protection — Technical Requirements:

Protection layer	What it protects	How it works
Local persistent storage on POS device	Payments recorded, Every POS has printed item receipt, SQL table in device DB, BEFORE sending to server	
Transaction atomicity	Payment + order status update always together	Database transactions use one with either both the payment record and order status update
Auto-resume on restart	POS app restarts after a power cut	On start, app checks for any unreported transactions in the local DB
Duplicate transaction prevention	Power cut during a Wave/OM payment	Each payment is assigned a unique idempotency key before being sent to server
Offline order queue	Orders created without internet are queued	Queue and queue size automatically and the queue order. Sync is done when internet is restored
Label printing after power cut	Garment labels already printed are not reprinted	Every system has a timestamp of last power printed. On restart, only unprinted labels are reprinted
Cash drawer on restart	Cash drawer does not open automatically	Cash drawer open command is only sent in response to a confirmed payment
Server backup every 15 minutes	All server data is backed up to the cloud	Have if the African backup server is destroyed (fire, theft, flood), no more than 15 minutes of data loss

3.2 Power Cut Scenarios — How CleanHub Behaves:

Scenario	What happens	What the cashier sees
Power cuts while entering a new order (not yet saved)	Order not yet saved	On restart: 'You have unsaved orders, please save' — no duplicate order created
Power cuts after 'Confirm Order' is pressed	Order saved locally	On restart: job is processed automatically — no duplicate order created
Power cuts during a cash payment (drawer closed)	Payment recorded locally	On restart: drawer is correctly marked as paid, drawer state is verified
Power cuts during a Wave/OM payment (QR code displayed)	Payment (QR code displayed) — payment status unknown	On restart: system queries Wave/OM API to verify payment status — no duplicate payment
Internet outage (no power cut)	POS continues working fully — all cash payments processed	On restart: 'Operations overflows Wave/OM queue' — syncs all pending payments
Server power cut (UPS engaged)	UPS powers server for 4+ hours	On restart: POS remains available on local network even if server is down
Server power cut (UPS exhausted)	POS device switches to pure local mode	Cashier sees 'Server offline' — working in local mode' — all orders are processed locally

■ **DEVELOPER REQUIREMENT — NON-NEGOTIABLE: Power cut resilience is not a nice-to-have feature — it is a core architectural requirement for the African market. The developer must implement and test all 8 protection layers above before delivery. The acceptance test for power cut resilience is: the tester will pull the power cable from the device at any random moment during 50 different operations, restart the device, and verify that no data was lost, no transaction was duplicated, and the app resumed correctly. ALL 50 tests must pass. Any failure is a blocking bug that must be fixed before final sign-off.**

Wave & Orange Money offline payment queue:

■ Wave and Orange Money APIs require an active internet connection to process payments. When internet is unavailable: (1) Cash payments work fully offline — no internet required. (2) Wave/OM payments are queued locally with a pending status — the cashier sees 'Payment pending sync'. (3) When internet is restored, the queue is processed automatically — each payment is sent with its unique idempotency key to prevent duplicates. (4) The client receives their WhatsApp confirmation when the payment is confirmed by the API. This approach requires formal client validation that a queued Wave/OM payment carries a small reconciliation risk — the payment may fail on sync if the client's account has insufficient funds. The cashier is notified immediately in this case.

4. LOCALISATION & COMPLIANCE

Topic	Answer	Status
Fiscal Authority (Black Box)	Not currently required. Integration capability will be planned for future compliance.	Not a
SMS / WhatsApp Provider	International provider preferred — Twilio or equivalent	Confirmed

5. BUSINESS MODEL & MULTI-TENANT ARCHITECTURE

Topic	Answer	Status
Data Isolation	Row-level isolation retained — faster development, easier long-term maintenance	Confirmed
Server Hosting	Own on-premise server, hosted in Senegal at the client's office	Confirmed
Data Sovereignty	All data remains in Senegal on the client's infrastructure	Confirmed

6. INFRASTRUCTURE & SERVER COST ESTIMATE

Option A — HP ProLiant Server (Purchased in China)

High-performance HP ProLiant server purchased directly from China (via Alibaba.com or authorised HP distributors). Installed at the client's office in Senegal. One-time investment — no monthly fees beyond electricity.

Component	Recommended Specification (1,000+ clients)
HP ProLiant DL380 Gen10/11 (Primary)	256GB RAM, 2x Intel Xeon 16-core, 8TB NVMe SSD RAID
HP ProLiant DL360 Gen10 (Backup/Failover)	128GB RAM, 2x Intel Xeon 8-core, 4TB SSD
HP Network Switch	48-port Gigabit managed switch
UPS (Power Backup)	Online UPS 6000VA — protects against Senegal power cuts
Structured Cabling	Cat6 cabling, patch panels, rack cabinet
Shipping & Import (China to Senegal)	Freight + customs duties — to be quoted
On-site Setup & Configuration	Server OS, network, security hardening
Electricity / month	Both servers running 24/7

■ *Why HP ProLiant? Professional reliability (5-year MTBF), iLO remote management, redundant power supplies, spare parts available worldwide. Break-even vs cloud: approximately 24-30 months.*

Option B — Cloud Hosting (Chinese Providers, Africa Datacentre)

Alibaba Cloud, Huawei Cloud and Tencent Cloud operate datacentres in Africa (Johannesburg / Nigeria), offering low latency for Senegal. No hardware to purchase — monthly payment, instant scale-up.

Clients	Specification	Alibaba Cloud	Huawei Cloud	Tencent Cloud
1-5 clients	4 Go RAM, 2 vCPU	To be quoted	To be quoted	To be quoted
6-20 clients	8 Go RAM, 4 vCPU	To be quoted	To be quoted	To be quoted
20-50 clients	16 Go RAM, 8 vCPU	To be quoted	To be quoted	To be quoted
1 000 clients	256 Go RAM, 32 vCPU + BDD + Storage	To be quoted	To be quoted	To be quoted

■ *Huawei Cloud recommended for the African market — strong local presence, local partnerships, competitive pricing. Payments accepted in USD.*

Option C — Recommended Hybrid Strategy

Combine the best of both worlds: HP ProLiant as the primary production server in Senegal for data sovereignty, backed by a cloud instance for disaster recovery and owner remote access.

Element	Detail
HP ProLiant DL380 (Primary)	Main production server — Senegal office

Element	Detail
Huawei Cloud Africa (DR)	Disaster recovery + daily automated backups
Electricity	Primary server 24/7 power cost

OPEN POINTS — CLARIFICATION REQUIRED

■ The following points require final confirmation before development begins:

- Exact printer models to source from China — technical specs required (DPI, paper width, connection type)
- Conveyor belt control protocol — to be defined when hardware is selected
- Offline payment reconciliation — formal client validation required on the queue approach

CleanHub

Technical Specification for Development

Multi-Tenant SaaS Platform — African Market | Version 1.0 | April 2026

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1. Project Overview

This document is the complete technical specification for building CleanHub — a native application for pressing and car wash management, designed for the African market. It covers all functional requirements, system architecture, database design, security rules, UI screens, integrations, and the delivery roadmap.

1.1 Business Objectives

- Digitise and automate day-to-day pressing operations
- Enable real-time garment tracking from drop-off to pickup
- Integrate local mobile payment methods: Wave and Orange Money
- Support businesses with multiple branches from a single platform
- Offer the system as a subscription-based SaaS (monthly/annual plans)

1.2 Target Users

Role Utilisateur	Responsabilites
Owner / Administrator	Full system access, SaaS subscription management, reporting
Branch Manager	Branch operations, staff management, local reports
Cashier / Agent	Order creation, payment processing, customer check-in
Delivery Staff	Pickup and delivery scheduling, garment handoff confirmation

2. System Architecture

The platform must be built as a NATIVE APPLICATION installed on client terminals and PCs — NOT a website. The user never opens a browser. They double-click the CleanHub icon (on PC) or tap it (on tablet) and the application opens directly — exactly like GestiClean or WhatsApp.

2.0 Native Application Model — Core Principle

Aspect	REJECTED Model (website)	RETAINED Model (native app)
Access	www.cleanhub.sn in a browser	Double-click the CleanHub icon
Identification	Domain name visible	Unique pressing code (e.g. SN-0042) + password
Browser	Chrome / Firefox required	Never a browser
Offline	Dependent on browser cache	Fully autonomous without internet
Installation	Nothing to install	App installed once, used forever
Update	User reloads the page	Automatic in background like WhatsApp

Aspect	REJECTED Model (website)	RETAINED Model (native app)
Lock	Cannot block the browser	Kiosk mode: terminal only runs CleanHub
Looks like	A Chrome tab	GestiClean, Word, WhatsApp

How the client accesses CleanHub:

Step	On Windows PC	On Android Terminal
1	Double-click CleanHub icon on desktop	Tap CleanHub icon on home screen
2	CleanHub login screen opens	CleanHub login screen opens
3	Enter: Pressing Code (e.g. SN-0042) + password	Enter: Pressing Code + password OR scan loyalty card
4	CleanHub opens on the dashboard	CleanHub opens on the dashboard
5	Work normally — offline if needed	Work normally — offline if needed

■ *The Pressing Code is assigned by CleanHub at registration. It is printed on all receipts and invoices. It also serves as a reference for technical support: 'My pressing code is SN-0042'.*

■ **NO web page, NO browser, NO domain name is visible to the end user. The CleanHub server is invisible — it is the plumbing in the background. The user sees only the CleanHub application.**

2.1 Technology Stack

Layer	Technology
Backend	Node.js with NestJS (recommended) or Django/Python — centralised REST API hosted on the client
Windows Application	Electron.js — desktop application installed on Windows PC. Double-click CleanHub icon -> app open
Android Application	React Native or Flutter — app installed directly on Android terminals sourced from China. CleanHub
Database	PostgreSQL (primary relational database)
Cache	Redis (session management, real-time updates)
Cloud Infrastructure	AWS, GCP or Azure (developer's choice)
File Storage	AWS S3 (garment photos, invoices, documents)

■ *Architecture : Native Windows app (Electron) + Native Android app (React Native/Flutter) + Central backend server (NestJS). Each terminal and PC installs the CleanHub application locally. Data is synchronised with the central server. No web browser, no domain name exposed to end users.*

■ **Exigences critiques : 1. App must work WITHOUT a browser — user never opens Chrome or Safari. 2. App must launch in kiosk mode on client terminals — cannot exit without admin code. 3. Automatic background updates — like WhatsApp — user does nothing. 4. APK (Android) and Windows installer distributed directly by CleanHub — no mandatory Google Play or Microsoft Store.**

2.2 Multi-Tenant Management — Without Domain Name

- Each tenant (pressing) identified by a unique PRESSING CODE assigned by CleanHub (e.g. SN-0042) — user enters this code + password on the application login screen
- Complete database-level data isolation between tenants — each pressing sees only its own data
- Tenant-specific configuration: logo, colours, pricing, language — configurable from the application
- Shared infrastructure with logical separation (tenant_id at row level) — the server is invisible to the end user
- A single Android APK and single Windows installer distributed to all clients — the pressing code determines which account opens

2.2.1 Business Hours — Automatically Configured at Subscription

Each pressing subscribed to CleanHub has its own business hours — one pressing may open at 7am and close at 9pm, another from 8am to 6pm, another 24/7. These hours are defined ONCE at account creation and apply automatically to all software features.

Feature	Description
Configured at subscription	When the account is created, CleanHub support configures the pressing's business hours: opening t
Hours unique to each client	Each pressing has its own hours — completely independent from others. A Dakar pressing may open
Automatic appointment calculation	When the system automatically calculates a pickup date, it takes into account the pressing's opening
Block outside business hours	If a cashier attempts to create an order or process a payment outside configured business hours, the
WhatsApp / SMS respects hours	Automatic reminders sent to clients (order ready, appointment tomorrow...) take into account busines
Z Report linked to hours	The end-of-day Z report is automatically generated at the configured closing time — not at midnight,
Editing hours	If the pressing changes its hours (e.g. earlier closing during Ramadan), ONLY THE OWNER can mo
Hours per branch	For multi-branch pressings, each branch can have its own business hours — configurable independe

Exemple d'heures d'ouverture configurees a l'abonnement :

Pressing	Opening Time	Closing Time	Working Days	Rest Days
Pressing SN-0042 (Dakar)	07:00	21:00	Monday — Saturday	Sunday
Pressing SN-0075 (Abidjan)	08:00	19:00	Monday — Sunday	None — open 7 days
Laundry SN-0089 (Dakar)	09:00	18:00	Tuesday — Saturday	Sunday + Monday
Pressing SN-0103 (Saint-Louis)	06:30	22:00	Every day	None — open 7 days

■ *Business hours are configured by CleanHub support at account creation. The owner can modify them from the Settings panel at any time — but every change is logged with server timestamp. Business hours automatically affect: pickup appointments, WhatsApp/SMS notification sending, daily Z reports and overdue order alerts.*

■ **Business hours are linked to the CleanHub server clock — not the local computer clock. It is impossible for a staff member to bypass business hours by changing their PC time.**

2.3 CleanHub Internal Clock — Independent Date & Time

The date and time used by CleanHub must be completely independent of the clock of the computer or tablet on which the software is installed. CleanHub synchronises its own internal clock with the CleanHub server via internet. It is IMPOSSIBLE for a local user to modify the date or time in the software.

Rule	Description
Server time only	CleanHub uses ONLY the time from the central CleanHub server — never the local computer time. E
Automatic sync	At every app start and every 5 minutes, CleanHub syncs its clock with the server via internet (NTP p
Date modification forbidden	No user — not even the Owner — can modify the date or time in CleanHub from the application.
Only CleanHub support can correct	a date/time correction is needed for an exceptional reason (e.g. wrong timezone configured at insta
Anti-fraud protection	This clock independence protects against fraud: a staff member cannot roll back the date to create fa
Offline	In offline mode, CleanHub uses its last synced time and increments automatically. Upon reconnection

■ **The developer must implement a complete clock lock: the application ignores the device system time entirely and uses only the NTP time from the CleanHub server. This requirement is NON-NEGOTIABLE — it**

is a fundamental security measure.

3. Core Modules — Functional Requirements

3.1 Customer Management

- Create, edit and search customer profiles (name, phone, address, email)
- Customer categorisation: Regular, VIP, Corporate
- Loyalty points system: points earned per order, redeemable as discounts
- Order history and spending summary per customer
- Automatic SMS/WhatsApp notifications for order status updates

3.1.0 Loyalty Card — Automatic Benefits Even Without the Card

In Africa, clients often forget their loyalty card AND their receipt. CleanHub must NEVER turn away a client who has neither their card nor their receipt. The system identifies the client and retrieves all their orders by phone number — the phone number is the universal identifier.

Finding a client and their orders without a receipt:

Identification method	What the cashier does	What CleanHub shows
Phone number (primary)	Cashier types the client's phone number — Client profile '7 ALL' shows all matching clients, past orders, o	
Client name	Cashier types 2-3 letters of the name — filtered list appears instantly	Search results instantly
Old receipt barcode scan	Cashier scans the barcode of any old receipt — Order linked to barcode found instantly — client profile opene	
Digital card / QR code	Client shows personal QR code on their phone — Same — full profile opened	
Vehicle plate (car wash)	Agent types the licence plate	Vehicle file with all wash history — even without client name

What the cashier sees when a client is found by phone number:

Information visible	Detail
Client name + category	DIALLO Mamadou — VIP — 77 123 45 67
Orders ready to collect	Order N2026-147 — 3 shirts — Ready since 2 days — 9,000 XOF to pay
Orders in progress	Order N2026-156 — 2 suits — In ironing — Pickup: tomorrow
Outstanding balance	Total owed: 14,500 XOF across 2 orders
Loyalty points	1,250 points — equivalent to 6,250 XOF of vouchers
Last visit	Last visit: 8 days ago — 3 shirts
Birthday promo	■ Birthday promo unused: -15% valid until [date]

- The client does not need a receipt, a card, or an appointment number — their phone number is enough to retrieve everything
- If the client does not know their own number, the cashier can search by name — even with a partial name (3 letters minimum)
- A client who registered under a different name can be found by their second number, spouse's number, or old receipt barcode
- The order number and barcode on ANY old receipt (even a year old) links back to the complete client profile — nothing is ever lost

- Duplicate phone detection: if two different clients have the same phone number, CleanHub shows both profiles side by side — cashier selects the correct one

■ **FUNDAMENTAL RULE: loyalty benefits belong to the CLIENT, not the card. A client who forgets their card or receipt must never lose their points or discounts. CleanHub identifies the client — not the document.**

3.1.1 Customer Categories & Category-Based Pricing

The system must support fully customisable customer categories, each with its own dedicated pricing grid. The owner can create, rename and delete categories at any time. Each category defines a price for each individual article — not a price per family. When an order is created for a customer of that category, CleanHub automatically applies the correct per-article price for that category.

Category	Target Client	Pricing Logic
Standard	Regular walk-in customers	Base price — default price list applies
VIP	Loyal high-spending customers	Custom reduced price per service — configured by owner
Corporate / B2B	Companies, offices	Negotiated flat rates — monthly invoicing
Hotel	Hotels and guesthouses	Volume rates — price per kg or per item for linen service
Embassy / NGO	Embassies, international organisations	Special contract rates — invoiced periodically
Staff	Pressing employees	Internal discount rates
[Custom]	Any category created by owner	Fully configurable price grid

- Owner can create unlimited custom categories — no limit on number of categories
- Each category defines a price for every individual article — prices are set article by article, not per family. A family is only an organisational group.
- When a new order is created, the system automatically applies the correct price grid based on the customer's category — no manual selection needed
- Price override: cashier can manually adjust a price on a specific order if needed — with manager authorisation
- Promotional discounts stack on top of category prices — VIP customer gets category price PLUS any active promotion
- Reports filterable by customer category — see revenue per category (Standard vs VIP vs Corporate)
- Bulk pricing: categories like Hotel can have prices per kg or per m2 instead of per item

3.1.1a Individual Client Pricing — Custom Price Per Client

Beyond category-based pricing, CleanHub allows the owner to set a PERSONALISED price grid for a specific individual client — independently of their category. This is the most granular level of pricing: one client, one custom price list. Useful for a key corporate account with negotiated rates, a long-standing VIP client with a specific arrangement, or a hotel with a unique price per item.

Feature	Description
Individual price grid per client	The owner can open any client profile and define custom prices for specific articles or services — the
Partial override	The individual price does not need to cover all articles — only the articles with a custom price are ov
Automatic application	When a cashier creates an order for a client with individual pricing, CleanHub automatically applies t
Price priority hierarchy	CleanHub applies prices in this order: (1) Individual client price — if set, always wins. (2) Category pr
Only Owner can set individual prices	Individual client pricing is set ONLY by the Owner from the client profile — Manager and Cashier can

Feature	Description
Visible on quote/invoice	The invoice shows the prices actually applied — no indication of 'discount' or 'special rate' unless the
Price review report	Owner can run a report showing all clients with individual pricing, the articles concerned and the cust
Expiry date (optional)	The owner can set an expiry date on an individual price arrangement — e.g. 'Custom rates valid until

■ *Example: Hotel Teranga has a corporate account (Corporate category = -10% on standard prices). In addition, the owner has negotiated a special rate of 800 XOF per shirt specifically for Hotel Teranga (vs 1,200 XOF standard and 1,080 XOF corporate). CleanHub applies 800 XOF automatically every time Hotel Teranga is selected — no manual override needed.*

■ **Individual client pricing is the most powerful pricing tool in CleanHub. It must be used carefully — ONLY THE OWNER can set, modify or delete individual client prices. Every change is logged in the audit trail with the server timestamp.**

3.1.2 Admin Settings — Prices, Colours, Labels & Reserves

Administrators and branch managers must be able to update prices, article labels, colours and reserve rules directly from the settings panel — instantly, without technical intervention. All changes take effect immediately on all devices.

1. Article Families — Catalogue Structure:

All articles in CleanHub are organised into FAMILIES. A family groups similar articles — e.g. the family 'Tops' contains Shirt, T-shirt, Boubou, Jacket, Blazer. Families serve as an ORGANISATIONAL STRUCTURE for the catalogue, the production reports and the label printing rules. IMPORTANT: a family has no price of its own — prices are always defined article by article within each family.

Feature	Description
Unlimited families	Owner creates as many families as needed: Tops, Trousers, Dresses, Suits, Linen, Accessories, Ca
Articles within families	Each family contains its own articles. The cashier browses by family on the POS screen to find article
Article-level pricing	Prices are defined ARTICLE BY ARTICLE within each family. A family is only an organisational grou
Label rule per family	Each family has its own label printing rule — see section below
Family colour	Each family has a display colour on the POS grid — instant visual recognition
Family order on POS	Owner configures the display order of families — most used families at the top

2. Label Printing Rule Per Family — No Label for Certain Families:

Each article family is configured with a label rule. Some families need a matrix label per article (pressing garments). Others need no label at all (laundry, retail products, car wash). The manager sets this once per family — CleanHub applies it automatically at every intake. The NO LABEL option is essential for families like laundry (blanchisserie), retail sales and car wash services where individual garment labels make no sense.

Real-world family examples and their label rules:

Family	Label rule	Why
Pressing (garments)	One label per article	Each shirt, boubou, jacket needs its own waterproof matrix label for indi
Blanchisserie (laundry)	NO LABEL — or one label per batch	Laundry items (bedsheets, towels, uniforms) are washed as a batch. Or
Tapis (carpets)	One label per roll	Large label attached to the carpet roll — carpet tracked as one unit
Divers (miscellaneous)	Adhesive sticker or NO LABEL	Miscellaneous items — configurable by the manager per order

Family	Label rule	Why
Ventes (retail sales)	NO LABEL	Retail products (detergent, hangers, softener) already have their own E
Automobiles (car wash)	NO LABEL	Vehicle tracked by plate number — no garment label
F1, F2, F3, F4, F5, F6... (custom families)	Owner defined freely by owner	The owner creates as many custom families as needed — each gets its

Label rule	Description	Typical families
One label per article	Waterproof matrix label on each individual article	Pressing garments, coats, suits, trousers, dresses, jackets
One label per bag / bundle	Single label on the whole bag — no label per article (inside)	Laundry bags, hotel linen, kilo washing, towels
One label per roll	Large label attached to the whole roll	Tapis (carpets), rugs, curtains
Adhesive sticker only	Fast adhesive sticker instead of matrix label	Express service, express alterations, divers
NO LABEL — no marking required	Label printed at all — article tracked by order number (retail products), no families (car wash), any fam	

- Label rule configured once per family by the manager — the cashier never decides at intake — CleanHub applies the rule automatically
- If an order mixes families with different label rules (e.g. pressing garments + laundry bag + retail product), CleanHub prints the correct label type for each family automatically in one print job
- Families marked NO LABEL still appear in the order, on the receipt and in reports — no physical label is simply not printed
- The owner can change the label rule of any family at any time from Settings — the change applies immediately to all new orders

3. Price Management — Article-by-Article Pricing:

- Admin can update any service price at any time — one click to edit, one click to save
- Price changes apply instantly on all POS screens and mobile devices — no restart required
- Price history kept: system records who changed which price and when — full audit trail
- Bulk price update: admin can apply a percentage increase or decrease to all article prices at once — for all families, one selected family, or a selection of individual articles. The update always applies at the article level — never at the family level.
- Scheduled price change: admin can set a new price to activate automatically on a future date
- Different prices per branch: each branch can have its own price overrides

Copy prices from one family to another — price export:

The owner can copy all article prices from one family and apply them to another family in one click. Since prices are defined article by article, CleanHub copies the price of each article individually. This saves significant time when setting up a new family or when two families have similar pricing.

Feature	Description
Copy all prices	Owner selects source family (e.g. 'Trousers') and target family (e.g. 'Skirts') — clicks 'Copy prices' —
Copy with adjustment	Owner copies prices AND applies an automatic adjustment: e.g. 'Copy Trousers prices to Skirts with
Copy between categories	Copy the price grid of one CUSTOMER CATEGORY to another: e.g. copy the VIP price grid to a new
Copy between branches	Copy the price grid of one branch to another — useful when opening a new branch with the same pri
Preview before applying	Owner sees a preview of the copied prices before confirming — can review and adjust individual arti
Copy history	Every price copy operation is logged: who copied what, from which family/category to which, date and
Excel import / export	Export any price grid to Excel — modify prices in the spreadsheet — re-import to CleanHub in one cl

■ *Example: the owner has configured all prices for the 'Shirts' family across 4 customer categories (Standard, VIP, Corporate, Hotel). They create a new 'Kaftans' family. Instead of entering 4×N prices manually, they copy the 'Shirts' price grid to 'Kaftans' — then adjust only the prices that differ. Time saved: 90%.*

4. Article Colours:

Feature	Description
Colour per article	Each article can be assigned a display colour — shown on POS grid and production board
Colour picker	Admin selects from a visual colour picker — no colour codes required
Colour on labels	Article colour indicator printed on garment label for easy identification at production stage
Colour on Kanban board	Production board cards colour-coded by article type for instant visual recognition

5. Article Label / Name Customisation:

- Admin can rename any article or service at any time — e.g. rename 'Shirt' to 'Kaftan' for local market
- Add, edit or delete article families and sub-families — fully flexible catalogue structure
- Short label for receipts (max 20 characters) + full label for screens — both configurable separately
- Archive articles that are no longer offered — hidden from POS but history preserved

Article Creation by Duplication — Creating New Articles That Don't Yet Exist in the Database:

When a client brings an article that does not exist in the CleanHub catalogue (a new garment type, a specific item, a custom service), the cashier or manager can CREATE it instantly by DUPLICATING an existing similar article and modifying it. No need to start from scratch — no need to call the developer.

Feature	Description
Duplicate any article	Select an existing article, press 'Duplicate' — a copy is created instantly with all the same properties: family,
Edit the duplicate	Modify the name, price, label rule and any property of the duplicate — the original article is unchanged
Available immediately	The new article is available at the POS immediately after creation — no restart, no sync delay
Who can create	Manager and Owner can create new articles by duplication — Cashier cannot create new articles
Examples of use	'Boubou 3-piece' duplicated from 'Boubou' with a different price / 'Express Shirt' from 'Shirt' with Express sur
One-time article	For a truly unique one-off item (e.g. a wedding dress with a special price agreed with the client), the cashier
Catalogue cleanup	Owner reviews all articles created by duplication periodically and can archive those no longer needed — ke
Import from Excel	For bulk article creation (e.g. creating 50 new articles at once), manager exports the price template Excel, a

■ *Duplication is the fastest way to expand the catalogue without errors. The new article inherits all settings from the original — the manager only needs to change what is different. A new article can be created and ready to use in under 60 seconds.*

■ **Every article created by duplication is logged in the audit trail: who created it, when, which original article it was duplicated from. The developer must ensure that duplicated articles are fully independent of their source — modifying the duplicate never affects the original.**

6. Reserve Management:

Setting	Description
Reserve delay	Admin configures after how many days an uncollected order is flagged as 'In Reserve' (e.g. 30 days)
Reserve alert	Automatic notification sent to customer when their order enters reserve status

Setting	Description
Storage fee	Option to configure a daily or weekly storage fee applied automatically after reserve threshold
Write-off delay	Admin sets after how many days a reserve item can be written off — requires manager authorisation
Reserve report	One-click report listing all items in reserve with customer contact, days in reserve and estimated value
Reserve location	Each reserve item can be assigned a physical shelf/slot number for easy retrieval
Feature	Description
Colour per article	Each article in the catalogue can be assigned a display colour — shown on the POS grid and production board
Colour picker	Admin selects colour from a visual colour picker — no colour codes required
Colour on labels	Article colour indicator printed on garment label/ticket for easy identification at production stage
Colour on Kanban board	Production board cards colour-coded by article type for instant visual recognition
Colour on customer display	Article colours shown on customer-facing screen during order entry

3. Article Label / Name Customisation:

- Admin can rename any article or service at any time — e.g. rename 'Shirt' to 'Kaftan' depending on local market
- Add, edit or delete article families and sub-families — fully flexible catalogue structure
- Short label for receipts (max 20 characters) + full label for screens — both configurable separately
- Duplicate an article to create a variant quickly — e.g. 'Express Shirt' from 'Shirt' with a different price
- Archive articles that are no longer offered — hidden from POS but history preserved

4. Reserve Management:

Setting	Description
Reserve delay	Admin configures after how many days an uncollected order is flagged as 'In Reserve' (e.g. 30 days)
Reserve alert	Automatic notification sent to customer when their order enters reserve status
Storage fee	Option to configure a daily or weekly storage fee applied automatically after reserve threshold
Write-off delay	Admin sets after how many days a reserve item can be written off — requires manager authorisation
Reserve report	One-click report listing all items in reserve with customer contact, days in reserve and estimated value
Reserve location	Each reserve item can be assigned a physical shelf/slot number for easy retrieval

■ All settings changes (prices, colours, labels, reserves) must take effect instantly on all connected devices with no restart required. Non-technical staff must be able to make all these changes themselves — no developer intervention needed.

3.1.3 Administrative Management — Receivables, Pricing & Lists

Administrators must have full control over three critical management screens: customer receivables (dues), price list editing, and client/product list editing. All editable in real time without technical knowledge.

1. Customer Receivables — Verification of Dues:

The receivables module gives the owner and manager a complete, real-time view of all outstanding amounts owed by clients — per client, per order, per category and across the entire business. This is the primary tool for monitoring unpaid balances and following up on credit.

Feature	Description
Full receivables dashboard	One screen showing ALL customers with an outstanding balance — sorted by amount owed, oldest to newest
View dues per client	Click any client to see: total amount owed, list of every unpaid or partially paid order (date, articles, total)
View dues per order	Each order shows its own payment status: Fully paid / Partially paid (amount paid + balance remaining)
Multiple unpaid orders per client	If a client has 3 unpaid orders, the dashboard shows the total across all 3 — click to expand and see details
Search a specific client	Cashier types the client name or phone number at the counter — CleanHub instantly shows if this client has a balance
Balance alert at order creation	When creating a new order for a client with an existing balance, CleanHub displays a prominent alert
Record a payment	Admin or cashier records a full or partial payment against a client balance directly from the receivables dashboard
Collect multiple orders in one payment	Client pays for several orders at once — cashier selects which orders are being paid — single receipt generated
Overdue flag	Debts older than a configurable number of days flagged in red — e.g. 'OVERDUE — 45 days'
Instalment tracking	Clients on payment plans: CleanHub shows each instalment due date, amount, and whether it was paid
Write-off	Owner can write off an irrecoverable debt with mandatory reason — logged permanently in audit trail
Export receivables	Export full receivables list to PDF or Excel — for accountant, owner review or bank presentation

Automatic reminders and owner reports:

Report / Action	Trigger	Channel	Recipient
Weekly receivables summary	Every Monday at 8am	WhatsApp + Email	Owner — full list sorted by highest amount
Overdue alert	When a debt exceeds the configurable threshold (e.g. 30 days)	WhatsApp	Owner — client name, amount, days overdue
Payment reminder to client	Configurable — not sent automatically	Manually WhatsApp by Cashier	Client — only when cashier chooses to send
Balance confirmation to client	After each payment recorded	WhatsApp receipt	Client — new balance remaining after payment
Monthly receivables report	1st of each month	Email PDF	Owner — total outstanding, new debts, debts collected

Receivables dashboard — example view:

Client	Orders with balance	Oldest debt	Total owed	Days overdue	Status
DIALLO Mamadou	3 orders	15 Jan 2026	32,500 XOF	62 days	■ OVERDUE
NDIAYE Fatou	1 order	03 Mar 2026	8,000 XOF	18 days	■ Due soon
Hotel Teranga	12 orders	01 Feb 2026	185,000 XOF	48 days	■ OVERDUE
FALL Amadou	2 orders	10 Mar 2026	15,500 XOF	11 days	■ Recent
TOTAL	18 orders	—	241,000 XOF	—	—

- Receivables dashboard accessible in one tap from the Owner mobile app — the owner sees the total outstanding amount from their phone at any time
- Cashier can check a client's balance before any new order — prevents accumulation of large unpaid debts
- The system does NOT automatically send payment reminders to clients — cashier sends manually when appropriate
- Receivables by category: Owner can filter by Corporate / Hotel / VIP / Standard — see which client category generates the most credit risk
- Year-to-date receivables trend: how the total outstanding amount has evolved month by month — is it growing or shrinking?

■ The receivables module is **READ-ONLY** for Cashiers — they can **SEE** balances but cannot write off debts or modify payment records. Only the Owner can write off a debt. Manager can record a payment. Cashier can record a payment at the counter during a client visit.

2. Price List Editing:

Feature	Description
Full price list screen	All articles and services displayed in an editable grid — click any price to edit inline
Edit by category	Switch between price grids: Standard / VIP / Corporate / Hotel — edit each independently
Bulk update	Select multiple articles and apply a percentage increase or decrease to all at once
Scheduled update	Set new prices to activate on a future date automatically
Price history	See the full history of price changes per article — who changed it and when
Export price list	Export full price list to PDF — printable for display at the counter
Import prices from Excel	Manager can enter all prices in an Excel file provided by CleanHub and import it directly into the software
Excel price template	CleanHub provides a pre-formatted Excel template with columns: Family, Article, Service, Standard Price

3. Client List Editing:

Feature	Description
Search & filter	Search by name, phone, category, branch, loyalty points, last visit date
Inline editing	Edit customer name, phone, email, address, category directly from the list
Bulk category change	Select multiple customers and change their category at once
Merge duplicates	Detect and merge duplicate customer profiles (same phone number)
Archive / deactivate	Archive inactive clients — hidden from POS search but history preserved
Export client list	Export full client list to Excel — with all fields including loyalty points and balance
Export phone contacts	Export ONLY client names and phone numbers in vCard format (.vcf) — importable directly into any phone
Selective contact export	Export only certain clients: by category (VIP only), by branch, or by last visit date
Import clients	Import a client list from Excel — for migration from another system

■ All management screens must be accessible from any device. Changes apply instantly across all POS terminals with no restart required. All edits are logged in the audit trail with the user's name and timestamp.

Feature	Description
Full price list screen	All articles and services displayed in an editable grid — click any price to edit inline
Edit by category	Switch between price grids: Standard / VIP / Corporate / Hotel — edit each independently
Bulk update	Select multiple articles and apply a percentage increase or decrease to all at once
Scheduled update	Set new prices to activate on a future date automatically
Price history	See the full history of price changes per article — who changed it and when
Export price list	Export full price list to PDF — printable for display at the counter
Import prices from Excel	Manager can enter all prices in an Excel file provided by CleanHub and import it directly into the software

Feature	Description
Excel price template	CleanHub provides a pre-formatted Excel template with columns: Family, Article, Service, Standard

3. Client List Editing:

Feature	Description
Search & filter	Search by name, phone, category, branch, loyalty points, last visit date
Inline editing	Edit customer name, phone, email, address, category directly from the list
Bulk category change	Select multiple customers and change their category at once
Merge duplicates	Detect and merge duplicate customer profiles (same phone number)
Archive / deactivate	Archive inactive clients — hidden from POS search but history preserved
Export client list	Export full client list to Excel — with all fields including loyalty points and balance
Export phone contacts	Export ONLY client names and phone numbers in vCard format (.vcf) — importable directly into any
Selective contact export	Export only certain clients: by category (VIP only), by branch, or by last visit date — to create targeted
Import clients	Import a client list from Excel — for migration from another system

■ All four management screens must be accessible from any device. Changes apply instantly across all POS terminals with no restart required. All edits are logged in the audit trail with the user's name and timestamp.

3.1.4 Pre-Orders — Advance Booking for Future Deposits

Customers must be able to pre-book a deposit appointment for a future date — by phone, WhatsApp or directly at the counter. The pre-order is recorded in the system and appears on the planning calendar.

Feature	Description
Create pre-order	Staff creates a pre-order: customer name, phone, planned deposit date, estimated number of garments
Pre-order on calendar	Pre-order appears on the planning calendar on the scheduled date — staff see it when they arrive in
Automatic reminder	System sends automatic WhatsApp/SMS reminder to the customer the day before their planned dep
Convert to order	When the customer arrives, cashier clicks 'Convert Pre-order' — all pre-filled information transferred
Pre-order status	Status: Pending -> Confirmed -> Converted to Order -> Cancelled
Recurring pre-order	Option to create a recurring pre-order (e.g. every Monday — for hotel or corporate clients with weekl
No-show tracking	If customer does not show up, staff marks as 'No-show' — tracked per customer for follow-up
Pre-order history	All pre-orders (past and upcoming) visible on the customer profile — with conversion rate tracking

- Pre-orders visible on the main dashboard as a 'Today's pre-orders' widget — staff see at a glance who is expected
- Pre-orders can be created up to 30 days in advance
- Corporate and hotel clients can schedule recurring weekly pre-orders automatically

3.1.5 Order Correction — Reopening, Modification & Deletion

Staff must be able to correct order mistakes and reopen delivered orders when needed. These sensitive operations require strict authorisation and are fully logged in the audit trail.

1. Modify a Registered Order:

Modification type	Who can do it	Condition	Logged
Add an article to an existing order	Owner only	Order not yet delivered	Yes — mandatory reason
Remove an article from an order	Owner only	Order not yet delivered	Yes — mandatory reason
Change the service type on an article	Owner only	Article not yet processed	Yes — mandatory reason
Change the pickup date	Owner, Manager, Cashier	Anytime before delivery	Yes — server timestamp + who changed
Change the price on a specific order	Owner only	Any time before payment	Yes — mandatory reason
Add or modify order comment	Manager or Owner	Any time	Yes
Change payment method after payment	Owner only	Within same day only	Yes — mandatory reason

2. Reopen a Delivered Order:

Feature	Description
Reopen delivered order	Owner can reopen any order previously marked as Delivered — status returns to In Progress or Ready for Delivery
Reason required	Mandatory reason before reopening: Garment returned by customer / Quality issue / Wrong item delivered
Audit trail	Reopen event logged permanently: who, when, why
Customer notification	Optional: automatic WhatsApp sent to customer
Payment adjustment	If refund or additional charge needed, cashier can adjust payment after reopening
New garment labels	System reprints labels automatically when order reopened — new labels for reprocessing

3. Delete an Order:

Feature	Description
Delete unpaid order	Owner can delete an order not yet paid — reason mandatory
Delete paid order	Owner can delete a paid order — triggers automatic refund workflow — reason mandatory
Soft delete	Deleted orders are NEVER permanently erased — archived as Cancelled — visible to admin in reports
Stock reversal	If products were deducted from stock for this order, automatically restocked on deletion
Audit trail	Deletion logged permanently — traceable even years later

4. Absolute Security Rule — Full Permission Table:

Action	Owner	Manager	Cashier	Delivery
Delete unpaid order	Yes — reason	No	No	No
Delete paid order	Yes — reason	No	No	No
Modify article in order	Yes — reason	No	No	No
Modify price in order	Yes — reason	No	No	No
Reopen delivered order	Yes — reason	No	No	No
Reprint garment marking labels	Yes — logged	No	No	No
Add comment to order	Yes	Yes	Yes	Yes
Change pickup date	Yes — logged	Yes — logged	Yes — logged	No

■ **ONLY THE OWNER can delete, modify articles/prices or reopen a registered order — and ONLY THE OWNER can reprint garment marking labels. However, the PICKUP DATE can be changed by Owner, Manager and Cashier (all roles) — because a client may call at any moment to reschedule, and the cashier must be able to update it immediately without waiting for the owner. Every pickup date change is permanently logged: who changed it, old date, new date, server timestamp.**

3.1.5a Sales & Collection Statistics Per Cashier / User

The owner and manager can see the complete sales and payment performance of every staff member who uses CleanHub. This is the primary tool for monitoring individual productivity, detecting anomalies and calculating bonuses.

Statistic	Description	Available for
Orders created per user	Total number of orders created by each cashier/agent per period — daily, weekly, monthly	All cashier/agent users
Revenue collected per user	Total amount collected (all payment methods) by each cashier — by day, week or month	All cashier/agent users
Revenue by payment method per user	Breakdown per cashier: how much collected by cash, Wave, Orange Money / card — detects discrepancies	All cashier/agent users
Average order value per user	Average amount per order per cashier	All cashier/agent users
Number of discounts given per user	How many discounts or price overrides applied by agent/cashier — alerts if excessive	All cashier/agent users
Number of cancellations per user	Orders cancelled or deleted by each cashier/agent — cancellation rate triggers alert to owner	All cashier/agent users
Cash drawer openings per user	How many times the cash drawer was opened per cashier per shift — unusual openings flagged	All cashier/agent users
Retail products sold per user	Number and value of retail products sold per cashier/agent — identifies top sellers	All cashier/agent users
Peak hours per user	Which hours each cashier processes the most orders — useful for shift planning	All cashier/agent users
Comparison between users	Side-by-side comparison of all cashiers — overall averages, average ticket, discounts — ranking	Owner/Manager view

Per-cashier report — example view:

Cashier	Orders	Revenue collected	Avg ticket	Discounts	Cancellations	Cash	Wave
DIALLO Moussa	47	285,000 XOF	6,064	2	1	120,000	165,000
NDIAYE Fatou	38	210,500 XOF	5,539	0	0	85,000	125,500
FALL Ibou	52	318,000 XOF	6,115	5	3	130,000	188,000
TOTAL	137	813,500 XOF	5,938	7	4	335,000	478,500

- Report filterable by: date range, branch, specific user, payment method, service type
- Owner receives automatic weekly summary by WhatsApp or email: top cashier of the week, total revenue, any anomaly detected
- Anomaly detection: if a cashier's cancellation rate or discount rate exceeds a configurable threshold, the owner receives an instant alert
- All per-cashier statistics are linked to the server clock — a cashier cannot manipulate their statistics by changing the device time
- Per-cashier report exportable to Excel or PDF — useful for salary calculations and performance reviews

■ **The per-cashier statistics module is one of the most important management tools for a pressing owner who is not always present on site. It enables remote monitoring of every staff member's performance from a smartphone — from anywhere.**

3.1.6 Garment Condition Reservations & Webcam Photo Capture

At intake, staff must be able to document any pre-existing defects on a garment (stains, tears, discolouration or other damage) and take a photo. This protects the pressing from false claims and gives the customer full transparency.

1. Garment Condition Reservations — Predefined Defect Categories:

Defects are organised into three main categories — each with a predefined list. The cashier selects from this list in one tap — no typing needed. The manager can add, rename or delete any defect at any time.

Category	Defects included
■ STAINS	Grease stain / Oil stain / Ink stain / Rust stain / Food stain / Coffee or tea stain / Blood stain / Make-up stain / Sweat stain
■ TEARS & DAMAGE	Tear / Hole / Fraying / Worn-through fabric / Broken zip / Missing button / Loose button / Broken clasp / Broken seam
■ DISCOLOURATION & FADING	Fading / Discolouration / Bleach damage / Colour bleeding (another garment bled onto this one) / Sun damage

Feature	Description
Add reservation	Staff selects one or more defects from the predefined list — one tap per defect — no typing required
Defect location on garment	Staff indicates WHERE on the garment using a visual body diagram: Front / Back / Collar / Sleeve / Pocket
Severity level	Each defect rated: Minor (barely visible) / Moderate (visible) / Severe (prominent) — shown on production board
Free text note	Staff can add a free-text note to any defect: 'Old stain — pre-existing' / 'Client aware' / 'Attempted removal' / 'Pre-existing tear' / 'Pre-existing hole' / 'Pre-existing damage' / 'Pre-existing discoloration' / 'Pre-existing fading' / 'Pre-existing wear' / 'Pre-existing tear' / 'Pre-existing hole' / 'Pre-existing damage' / 'Pre-existing discoloration' / 'Pre-existing fading' / 'Pre-existing wear'
Printed on receipt — clear wording	All reservations printed on the customer receipt in clear language: 'Pre-existing stain on collar (Minor)' / 'Pre-existing tear on sleeve (Moderate)' / 'Pre-existing hole on back (Severe)' / 'Pre-existing discoloration on front (Minor)' / 'Pre-existing fading on sleeve (Moderate)' / 'Pre-existing wear on collar (Severe)'
Printed on garment label	Defect symbol printed on garment label — alerts production staff at each processing stage
Visible on production board	Garments with reservations flagged in yellow/orange on the Kanban board — production staff see them
Customer digital signature	Customer taps to confirm on the customer-facing screen — digital signature of reservation — same as receipt
Auto reminder to production	When a garment with a 'Severe stain' or 'Colour issue' reservation reaches the Ironing stage, an auto reminder is sent to production

2. Webcam & Camera Photo Capture at Intake:

Feature	Description
Webcam integration (PC)	On Windows PC, staff clicks 'Take Photo' — webcam activates directly in CleanHub — photo taken and stored
Camera integration (tablet/phone)	On Android/iOS, staff taps 'Take Photo' — device camera opens inside the app — photo taken and stored
Multiple photos per garment	Staff can take multiple photos per garment: front, back, defect close-up — all stored and linked to the garment
Photo at intake	Photo taken when garment is received — documents condition BEFORE cleaning
Photo at delivery	Optional photo taken when garment is returned — documents condition AFTER cleaning
Photo sent to customer	Optional: intake photo sent to customer via WhatsApp — full transparency on garment condition received
Photo storage	All photos stored securely in cloud storage (S3) — linked to order and garment ID — never lost

■ This combination of defect categories + photos is the pressing's best legal protection. If a customer claims the pressing caused a stain or tear, the intake photo and signed reservation prove the damage was pre-existing.

3.2 Order Management

- Create orders linked to a customer; each order contains one or more garments
- Assign a unique order number and QR code / barcode label per order

- Track order status: Received -> In Progress -> Ready -> Delivered
- Specify garment type, service type (wash, iron, dry-clean), colour and special notes
- Support for partial order delivery (some garments ready before others)

3.2.0a Linen Washing by the Kilo — Weight-Based Pricing

In addition to the standard pressing service (per item), CleanHub must manage WASHING BY THE KILO — the client deposits a bag or pile of laundry, staff weigh it, the price is automatically calculated by weight. Very common in neighbourhood laundries in Africa.

Feature	Description
Weighing at intake	Cashier enters the weight of the deposited laundry in kg (e.g. 3.5kg) — system automatically calculates
Configurable per-kg rate	Manager sets the per-kg price from the Settings panel: e.g. 500 XOF/kg — modifiable at any time. C
Service types per kilo	Wash only / Wash + Dry / Wash + Dry + Iron / Express — each with its own per-kg rate
Weight printed on receipt	Client receipt shows: weight (3.5kg), per-kg rate (500 XOF/kg), calculated total (1,750 XOF), estimat
No label per garment	For kilo washing, the system prints ONE SINGLE matrix label on the bag or bundle — no individual l
Automatic appointment	As with standard pressing, a pickup date is automatically calculated and assigned — confirmation W
Storage by bag	When laundry is ready, the bag is stored at a numbered location (e.g. A-3-14) — same system as oth
Min / Max weight	Manager can configure a minimum weight (e.g. 1kg) and maximum (e.g. 15kg) per order — system a
Kilo revenue report	Dedicated report: total kg processed per day/week/month, revenue generated, N vs N-1 comparison
Partial payment possible	Client can pay a deposit at intake and the balance at pickup — same workflow as standard pressing

Kilo order example:

Element	Detail
Client	DIALLO Mamadou — 77 123 45 67
Weight	3.5 kg
Service	Wash + Dry
Rate	500 XOF/kg
Total	1,750 XOF
Pickup date	Tomorrow from 2:00pm
Printed label	N2026-0147 DIALLO 3.5 kg Pickup: tomorrow 2pm [Barcode]

■ *Kilo washing and standard pressing (per item) coexist in CleanHub — a pressing can offer both services simultaneously. A client can have a kilo order AND a standard order at the same time in the system.*

3.2.0b Laundromat — Machine and Service Package Management

In Africa, an 'automatic' laundry is NOT full self-service like in Europe. C'est un etablissement avec des machines a laver professionnelles ou le personnel joue un role actif : il aide le client a choisir son programme, charge parfois les machines, et propose des formules avec differents niveaux de service (lavage seul / lavage + sechage / formule complete avec repassage et pliage). CleanHub est conçu pour cette realite africaine.

■ **The African 'automatic' laundry is a HYBRID: the machines are automatic but the service is assisted by staff. The packages are the real value-add — the client chooses how much they want**

to delegate.

Laundry service levels:

Level	What the client does	What the staff does	Pricing
Assisted self-service	Client loads the machine themselves	Staff advise, help if needed	Cheapest — token only
Partial service	Client drops off and collects	Staff sort, start, then machines	Medium — wash by kg
Full service	Client drops off and collects	Staff washing, iron and fold all the laundry	Most expensive — full service package

How it works — General flow:

Step	Client Action	Staff / System Action
1. Arrival	Client arrives at the counter with their laundry	Cashier welcomes the client and presents available packages
2. Package selection	Client chooses their service level: assisted self-service / package wash / full service with laundry and folding	Staff select the package wash / full service with laundry and folding — price calculated
3. Payment	Client pays in cash, Wave or Orange Cashier	Cashier collects payment — receipt printed. If assisted self-service: tokens handed over.
4. Process	Self-service: client inserts tokens with machine. Staff assist with drop-off package / staff handles everything	Client handles everything / staff handles everything status updated by staff at each step
5. Additional options	Client can add drying, ironing or folding	Cashier adds options to the order — price recalculated instantly
6. End and collection	Client comes back to collect their clean laundry	WhatsApp / SMS sent to client when ready. Staff hand over laundry in bag or folded.

Machine and Token Management:

Feature	Description
Machine creation	Manager creates machines in CleanHub: type (washing machine / dryer), capacity (5kg, 7kg, 10kg...)
Programmes et tarifs	Le gerant configure les programmes et leurs prix depuis le panneau Parametres. Trois types de programmes: 1. LAVAGE AU KG : tarif configure au kilo (ex. 500 XOF/kg) — le client est facturé selon le poids de la lessive 2. REPASSAGE : tarif au kilo ou à la pièce — le linge lavé est repassé par le personnel avant remise au client 3. PLIAGE : tarif au kilo ou forfait — le linge lavé (et éventuellement repassé) est soigneusement plié
Token types	CleanHub can manage multiple token types by value — e.g. Wash token (500 XOF), Drying token (400 XOF)
Token sales at counter	Cashier selects programme, CleanHub calculates number of tokens to hand over and total amount — receipt printed
Token receipt	Receipt printed with: available machine number(s), chosen programme, number of tokens handed over
Machine dashboard	Real-time view of all machines: FREE (green) / OCCUPIED (red) / BROKEN DOWN (grey) — manager can filter by type, capacity, status
Machine assignment	When client buys tokens, system automatically assigns a free machine matching their capacity — display on machine
Token stock	Manager tracks physical token stock in circulation — how many distributed, how many returned — alert when low
Lost or stolen tokens	If a client loses their tokens, only the OWNER can authorise replacement — action logged in audit trail
Daily report	Total tokens sold, revenue by programme, most used machines, peak hours of the laundry
Weekly / monthly report	Laundry revenue evolution, N vs N-1 comparison, machine occupancy rate (% of time each machine is occupied)
Machine maintenance	Manager records breakdowns and maintenance — a BROKEN DOWN machine is automatically hidden from client view

Real-time Laundromat Dashboard:

Machine	Type	Capacity	Status	Client	Since	Est. End
M-01	Wash	7 kg	OCCUPIED	FALL Amadou	14:30	15:00

Machine	Type	Capacity	Status	Client	Since	Est. End
M-02	Wash	5 kg	FREE	—	—	—
M-03	Wash	10 kg	OCCUPIED	NDIAYE Fatou	14:45	15:20
M-04	Drying	7 kg	FREE	—	—	—
M-05	Wash	7 kg	BROKEN DOWN	Since yesterday, technician called		
M-06	Drying	5 kg	OCCUPIED	DIALLO M.	15:00	15:30

Plans d'abonnement — Laverie :

Laundry managers can subscribe to CleanHub according to the size of their establishment:

Plan	Machines gerees	Utilisateurs	Prix	Ideal pour
Starter Laundry	Up to 5 machines	2 cashiers	To be defined	Small neighbourhood laundry
Pro Laundry	Up to 15 machines	5 cashiers	To be defined	Medium laundry
Business Laundry	Up to 30 machines	15 employees	To be defined	Large multi-service laundry
Enterprise Laundry	Unlimited	Unlimited	To be defined	Multi-site laundry chain

3.2.0c Full Laundry Service — Wash by Kg + Ironing + Folding

The Full Laundry Service is the premium offering: the client drops off their laundry at the counter and leaves empty-handed. Staff handle everything — wash by kg + ironing + folding. It is the most profitable service for the manager and the most convenient for the client.

The 3 services of the Full Laundry Package:

Service	Description	Pricing	Label
1. Wash by kg	Laundry is weighed at intake and machine washed on the 50°C SE/kr. Drying is done in hot air (at least 60°C) for 1h.	500 XOF/kg	Wash by kg
2. Ironing	After washing and drying, staff iron each garment.	300 XOF/10 items	Ironing
3. Folding	After ironing (or directly after drying if no ironing) staff fold each item.	200 XOF/5kg	Folding

Available packages — possible combinations:

Package	What is included	Example total price
Wash only	Wash by kg only — client gets clean dry laundry (not ironed)	500 XOF/kg
Wash + Iron	Wash by kg + ironing of all garments — clean and ironed laundry	800 XOF/kg
Wash + Fold	Wash by kg + careful folding — clean, dry and folded laundry	700 XOF/kg
Full Service	Wash + Iron + Fold — clean, ironed and folded laundry. Client has nothing to do.	1,000 XOF/kg
Express	Full service in under 4h — Express surcharge configurable	1,500 XOF/kg

Full Laundry Service — Order Flow:

Etape	Action	Systeme
1. Intake	Client drops off laundry bag — cashier weighs, selects package, collects payment	Weight, selected package, collected payment calculated automatically — matrix label printed
2. Washing	Staff start the machine wash with the client's bag	Status: 'Washing' — visible on the dashboard
3. Drying	Machine or air drying	Status: 'Drying'

Etape	Action	Systeme
4. Ironing (if package includes)	Staffage ironing	Status: 'Ironing' — manager sees number of items remaining to iron
5. Folding (if package fully folds)	Staffage fully folds each item	Status: 'Folding in progress'
6. Ready	Bag stored at its location (rack/locker)	Status: 'Ready' — automatic WhatsApp/SMS sent to client: 'Your laundry is ready. C
7. Pickup	Client comes to collect their laundry	Cashier finds order by name or barcode — payment if balance remaining — bag ha

- One bag = one order = one matrix label — no individual garment label for kilo laundry
- Weight recorded at intake — if weight changes after washing (dry fabric = lighter), the intake weight is the reference for billing
- Manager configures estimated time per package from Settings: e.g. Wash only (3h), Wash + Iron (6h), Full Service (8h) — pickup date calculated automatically
- Cashier can add special notes to the order: 'No heat on silk', 'Shirts only to iron', 'Sort by colour' — visible to staff throughout processing
- If client wants to collect laundry before processing is complete, cashier can mark partial delivery: e.g. deliver washed but not yet ironed laundry — balance adjusted accordingly
- Daily laundry report: number of orders, kg processed, best-selling packages, revenue per package, average processing time per package

■ **Ironing and folding are performed by pressing STAFF — CleanHub manages status tracking and billing, but these are human services, not automated. The quality of ironing and folding depends on staff training.**

Full Laundry Service — Real-time Dashboard:

Order	Client	Weight	Package	Status	Planned Pickup
LC-001	DIALLO Mamadou	4.5 kg	Wash only	Ready — stored A-3	14:00
LC-002	NDIAYE Fatou	3.0 kg	Wash + Iron	Ironing	17:00
LC-003	FALL Amadou	6.2 kg	Full Service	Folding	Tomorrow 9:00
LC-004	BA Mariama	2.5 kg	Express	Washing	In 2h
LC-005	SOW Ibrahima	5.0 kg	Wash + Fold	Drying	16:30

■ *Summary of laundry services in CleanHub: 1. Assisted self-service (tokens + staff guidance) + 2. Drop-off kilo wash (staff handle the machines) + 3. Full Service (wash + iron + fold). The same establishment can offer all 3 simultaneously — one CleanHub subscription covers everything. This is the African reality: automatic machines with human service.*

■ *Un pressing peut proposer les TROIS services simultanément dans CleanHub : pressing classique (à la pièce) + lavage au kilo + laverie automatique (jetons). Chaque service a son propre module, ses propres tarifs et ses propres rapports — mais tout est géré depuis une seule application avec un seul abonnement.*

■ **The developer must provision the following database tables: wash_machines (machines), token_types (token types), token_sales (token sales), machine_programs (programmes and pricing), kilo_orders (kilo orders) — separate from standard order tables but linked to the same tenant_id.**

When each order is created, the software automatically assigns the client a PICKUP DATE and an APPOINTMENT NUMBER. This information is printed on the customer receipt and on the garment labels. The client knows exactly when to come back to collect their garments.

Element	Description	Printed on
Order number	Unique number generated automatically (e.g. SN-2026-00142)	Receipt and garment labels through collection system
Pickup date	Date calculated automatically by service type: simple wash (2h), iron + steam (4h), ironing (24h), express (fe	Receipt and garment labels

Element	Description	Printed on
Appointment number	Daily collection order number (e.g. APT-47) — allows staff to prepare orders in appointment order	Receipt
Pickup time slot	Optional: configurable pickup time slots (e.g. 9am-12pm or 2pm-6pm) — avoids peak congestion	Receipt
Delay per service	Manager configures default delays by service type from the Settings panel — adjustable per order if needed	Settings panel

- At order creation, the cashier can manually adjust the pickup date if the client has a special need (e.g. express service, urgent order)
- The client automatically receives a WhatsApp/SMS confirmation with their pickup date and order number
- When the order is ready, the client receives: 'Hello [Name], your order N[XXXX] is ready. Your appointment is on [Date]. Please come and collect it!'
- 'Today's appointments' dashboard: staff see at a glance all orders expected today, sorted by time — they can prepare the garments in advance
- Delay alert: if an order is not ready by the promised pickup date, the system alerts the manager

3.2.0b Garment Labels — Printed on Matrix Printer

The labels sewn or attached to garments must be printed on a MATRIX PRINTER (dot matrix / needle) — not a thermal printer. The reason is simple: matrix labels resist water, soap and the machine washing cycle. A thermal label fades in the wash.

Criterion	Thermal Printer	Matrix Printer (RETAINED)
Water resistance	No — ink fades in the wash	Yes — ink penetrates the paper, resists washing
Heat resistance	No — blackens with heat	Yes — resists industrial machines
Label lifespan	A few hours in water	Several days even in a washing machine
Recommended use	Customer receipts, ready-article sticker labels	Labels attached directly to garments
Why GestiClean uses it	N/A	Industry standard for pressing for 20+ years

Content of the matrix label sewn onto the garment:

Information	Description	Mandatory
Order number	E.g. SN-2026-00147 — identifies the order	Yes
Barcode / QR code	Scannable by production staff to update status	Yes
Client name	Surname in large letters (e.g. DIALLO)	Yes
Garment type	E.g. Shirt, Suit, Trousers	Yes
Service type	E.g. Dry-clean, Wash, Iron	Yes
Pickup date	Date promised to the client	Yes
Piece number	E.g. 2/5 (garment 2 of 5 in the order)	Yes
Reservation / warning	Symbol if the garment has a pre-existing defect reservation	If applicable

- The Windows application must integrate a direct print driver for the matrix printer — no configuration dialog at each print
- Label format configurable from the Settings panel: roll width, character size, content to display
- Automatic printing of all garment labels in one order with one click at intake

- Reprint a label if lost: ONLY THE OWNER can reprint garment marking labels — cashier cannot. Each reprint is logged in the audit trail with server date/time and owner name.
- Number of labels printed corresponds exactly to the number of garments in the order — one label per garment

■ **The matrix printer sourced from China must use narrow paper rolls compatible with pressing labels. The developer must test compatibility with at least 2 Chinese matrix printer models before MVP delivery.**

3.2.0c Articles Without Individual Labels — Tracking Without Matrix Label

Not all articles handled by a pressing or laundry require an individual matrix label. Many items — retail products for sale, bulk laundry items, accessories, carpet rolls — are too numerous, too small or simply not suitable for individual labelling. CleanHub manages these articles differently depending on their type.

Article type	Why no individual label	How CleanHub tracks them
Retail products for sale	Products sold at the counter (detergent, Tangled, wasp perfume) are sold by cashier or stock clerk	Handled by stock module — barcode scanned or by sales clerk
Bulk laundry items (kilo wash)	When a client deposits 3.5 kg of mixed linen	One single impossible printed label is attached to the bag or bundle
Bed linen / household laundry	Hotels and households deposit large quantities of sheets, pillowcases, towels	One label per batch is attached to the bundle
Carpet and rug rolls	Large carpets or rugs cannot receive a standard matrix label	One large matrix label attached to the roll with a tag or tied with string
Accessories (belts, ties, scarves)	Very small items — label attachment is difficult	Grouped in a labelled bag or envelope attached to the main order
Alterations / repair items	Buttons, zips, patches — too small for individual labels	Noted in the order comments — tracked as a service line on the main order
Express items (same-day service)	Service to print and attach individual labels	Sticker adhesive label printer used on the outside of a transparent bag

Label strategy configuration — manager sets the rule per service type:

Label strategy	When to use	CleanHub setting
One label per garment	Standard pressing (shirts, suits, trousers, dresses)	Default — matrix label printed per article at intake
One label per bag / bundle	Kilo laundry, household linen, bulk items	Manager configures: kilo orders → 1 label per bag
One label per roll	Carpets and rugs	Manager configures: carpet/rug orders → 1 label per roll
Grouped in labelled envelope	Small accessories (belts, ties, scarves)	Cashier selects 'Group accessories' at intake — 1 envelope per group
No label — tracked by order only	Retail product sales, alterations	Stock module or order comment — no matrix label printed
Adhesive sticker label	Express same-day service	Adhesive label printer used instead of matrix printer — fast

- The manager configures the default label strategy per service type from the Settings panel — cashier does not need to decide at each intake
- For orders mixing labelled and unlabelled articles (e.g. 3 shirts with individual labels + 1 bag of socks with 1 group label), CleanHub prints the correct number and type of labels automatically
- Retail products are never assigned matrix labels — they are tracked through the stock module using their own product barcodes (EAN/UPC already printed on packaging)
- For hotels and corporate clients with large volumes of unlabelled linen, CleanHub supports batch intake: cashier enters total weight or total piece count — one order created with one label per batch

■ **The label strategy does not affect billing — CleanHub always calculates the correct price regardless of whether the article has an individual label or not. The label is a tracking tool, not a billing requirement.**

3.2.1 Order Comments — Visible on Invoice & Labels

Staff must be able to add free-text comments at any level of an order — on the full order, on individual garments, or as internal staff notes. Comments print on the customer invoice, on garment labels and are

visible on the production board.

Comment Type	Where it appears	Who can add
Order-level comment	Printed on customer receipt & PDF invoice — visible to all staff	Cashier, Manager, Owner
Garment-level comment	Printed on garment label — visible to production staff	Cashier, Manager, Owner
Internal staff note	Visible only on production board — NOT printed on receipt or invoice	Manager, Owner only
Delivery comment	Added at delivery — printed on delivery receipt sent to client	Delivery staff, Manager

- Free-text box on every order — 200 characters max for printed comments, unlimited for internal notes
- Comment templates: frequently used comments saved as presets — staff select from dropdown (e.g. Express service, Fragile fabric, No bleach, Handle with care, Silk — no heat, Return if stain not removable)
- Comments editable until order is In Production — after that, only Manager can edit
- Garment comment printed on the garment label next to the barcode
- Internal notes visible on production board Kanban card — highlighted in yellow

Custom Messages on Receipts — Owner Configures Printed Messages:

The owner can configure custom text that appears automatically on every customer receipt — footer messages, promotional messages, thank-you notes, business conditions or seasonal greetings. These messages print on all thermal receipts and PDF invoices.

Message type	Description	Example
Receipt footer	Fixed text always printed at the bottom of every receipt	Thank you for your purchase! Follow us on WhatsApp: +91 98765 43210
Promotional message	Seasonal or campaign message printed on all receipts	🎉 Special 15% Off on all items! Valid until [date]. Show this message at checkout.
Thank-you message	Personalised thank-you printed on the receipt	Thank you, [Name]! Your loyalty means everything to us.
Service conditions	Legal or service terms printed on receipts — responsible for dissection, payment 60 days	Responsible for dissection, payment 60 days will be placed in reserve.
Loyalty points reminder	Message showing the client's current points balance	Your points balance is 250 points. 250 more to earn a 5,000 discount!
Seasonal greeting	Holiday or event message active for a limited period	Happy Ramadan! Wishing you and your family a blessed month.

- Owner configures all receipt messages from Settings → Receipts & Messages — no developer required
- Multiple messages can be active simultaneously: footer + promotional + loyalty reminder all print on the same receipt
- Messages can be set with an active period (start date / end date) — automatically activate and deactivate without manual intervention
- Different messages per branch — each branch can have its own receipt footer (address, phone) while sharing a global promotional message
- Messages available in French and English — the receipt language matches the language configured for that client or branch

3.2.2 Partial Delivery & Partial Payment (Deposit)

Clients must be able to collect only part of their order at any time, and pay a deposit at intake or at any point. ALL staff (cashier, manager, owner and delivery staff) can perform a partial delivery or collect a partial payment — this is NOT restricted to the administrator.

Multi-Order Management — Client with Several Pending Orders:

When a client has several orders pending simultaneously, the seller must be able to process one order, several selected orders, or all orders at the same time in a single payment.

Scenario	Description	Result
Deliver one order	Seller selects a specific order from several pending	Delivery receipt for selected order only — balances of other orders remain
Deliver several orders	Seller ticks several orders — delivers them together	Single grouped receipt for all selected orders — combined total
Deliver all at once	Seller clicks 'Deliver All' — all ready orders for the client are delivered	Single grouped receipt — all orders closed simultaneously
Collect payment for one order	Select a specific order and collect its amount only	Payment recorded on selected order — other orders not affected
Collect payment for several orders	Select several orders and collect the combined total	Single payment automatically split across selected orders
Collect all at once	'Collect All' — system calculates total of all unpaid orders	Single global payment — balance of all orders reset to zero

- On the customer file, a list view shows ALL pending orders with their status, amount and balance — seller sees everything at a glance
- Checkboxes next to each order — seller selects freely: one, several or all
- 'Select All' and 'Deliver All / Collect All' buttons always visible when client has several active orders
- Combined total calculated in real time as the seller ticks orders
- A single grouped receipt printed — lists all included orders with each garment's detail and the global total
- If client pays partially on the grouped total, the system allocates the payment in chronological order (oldest first) — or seller can choose manually which order to prioritise

■ *Example: client DIALLO has 3 pending orders (8,000 XOF, 5,000 XOF, 12,000 XOF). He comes to collect and pay only the first 2. Seller ticks orders 1 and 2, clicks 'Collect & Deliver' — total 13,000 XOF — single Wave payment — single grouped receipt printed. The 3rd order remains pending, unchanged.*

Feature	Description	Who can do it
Select garments to deliver	Seller selects which garments the client wants to take	All sellers
Partial delivery receipt	Receipt printed showing items taken and items remaining	All sellers
Deposit at intake	Client pays a deposit when dropping off garments — amount recorded, balance tracked automatically	All sellers
Partial payment at any time	Any seller can record a partial payment at any time on a pending order	All sellers
Block delivery if unpaid	Configurable option: system blocks delivery if balance outstanding	Overfully: Manager and Owner

3.3 Garment Tracking

- Each garment receives a unique QR code or barcode label at intake
- Staff scan garments at each processing stage to update status
- Real-time garment location visible to manager and client (via SMS/WhatsApp)
- Photo capture at intake and at delivery to document garment condition

3.3.1 Physical Storage of Orders — Location in the Pressing

When garments are ready, staff must store them in the pressing and record their exact physical location in the software. When the client arrives to collect, staff immediately know where the garments are — no manual searching.

Storage layout structure:

Element	Description	Example
Zone	Large physical zones of the pressing	Zone A (simple garments), Zone B (suits), Zone C (express), Zone D (special)
Rack / Row	Subdivision of the zone — identified by a letter or number	Rack 1, Rack 2, Rack 3...
Position	Exact position on the rack — hanger or locker number	Position 14, Position 15...
Full location	Combination Zone + Rack + Position	A-3-14 = Zone A, Rack 3, Position 14

How storage works in the software:

Step	Staff Action	What the software does
1. Garment ready	Staff scans the finished garment's label	Status changes to 'Ready — awaiting storage'
2. Physical storage	Staff places garment on the hanger/locker and enters the location (e.g. A-3-14) in the app	Location recorded (e.g. A-3-14) in the app
3. Adhesive label	App automatically prints an adhesive label with full location order number on the physical protective cover	Full location order number printed on the physical protective cover
4. Client arrives	Cashier scans client loyalty card or enters their name	App immediately displays: 'Order N[XXX] — Location: A-3-14 — Zone A, Rack 3, Position 14'
5. Staff retrieval	Staff goes directly to the indicated location	No manual searching — pressing always organised
6. Delivery	Cashier scans label at the moment of handover	Status changes to 'Delivered' — location automatically freed

- Storage layout configurable: manager creates zones, racks and positions from the Settings panel according to the actual layout of their pressing
- Location assigned in one scan: staff scans the garment and simply enters the location (e.g. A-3-14) — software records and updates the status
- Location printed on adhesive label: each stored garment receives a label with its physical location
- Location visible on client file: when a client arrives, cashier immediately sees all their garments and their locations
- Overview dashboard: manager sees all occupied locations — colour-coded (red=occupied, green=free)
- Automatic release: when a garment is delivered, its location is automatically marked as free and available

■ **Developer must integrate the 'physical location' field (storage_location) in the garments table from the start — even if the full location management interface is delivered later.**

3.3.2 Daily Pickup List — Orders to Deliver Each Day

Each morning, the software automatically generates the complete list of all orders whose pickup appointment is scheduled for that day. Staff prepare the garments in advance before clients arrive — less waiting, better service.

Feature	Description
Morning auto-list	Each day at opening, the software automatically displays all orders with an appointment scheduled for that day
Detail per order	For each order: client name, phone, number of garments, physical location (e.g. A-3-14), amount to be paid
Real-time status	Each order shows its status: Ready and stored / In progress / Late — colour-coded green/orange/red
Advance preparation	Staff can retrieve garments from their locations in the morning and group them in a 'Ready to handover' box
Print the list	Print the daily list in one click — route sheet for staff preparing the day's orders
Filter by type	Filter the list by type: counter collections / home deliveries / express — each category managed separately
Client no-show	If a client has not arrived by closing time, the software alerts the manager and offers to send a WhatsApp message

Feature	Description
Daily collection rate	End of day: automatic report of orders collected vs not collected — uncollected orders automatically

'Today' dashboard view:

Information	Example
Total appointments today	E.g. 23 orders scheduled today
Already collected	E.g. 8 orders collected this morning
Awaiting collection	E.g. 12 orders ready — clients not yet arrived
Not yet ready	E.g. 3 orders still in production — must be finished before 2pm
Home deliveries	E.g. 5 deliveries scheduled this afternoon — delivery staff assigned
No-shows	E.g. 2 clients did not honour yesterday's appointment — WhatsApp follow-up proposed

- The software can also generate the appointment list for the next 3 days — useful for anticipating production workload and organising staff in advance.

3.3.3 Locker Storage — Numbered Locker Management

In addition to the rack system, the pressing can use numbered closed lockers to store ready garments. Each locker has a unique number. The client receives their locker number on their receipt — they can collect their garments even if staff are busy.

Feature	Description
Create lockers	Manager creates lockers in the software: number, size (small/medium/large), zone — from the Settings
Automatic assignment	When a garment is ready, the software automatically suggests a free locker of the right size — staff can accept or reject
Manual assignment	Staff can also manually choose the locker — useful if the client has several orders (same locker)
Number on receipt	The locker number is printed on the client receipt and sent by WhatsApp: 'Your order is ready — Locker N2026-0147'
Locker tracking	Real-time dashboard: occupied lockers (red) / free lockers (green) — overview of capacity
Multi-order locker	A client with several orders can have all their garments in one large locker — software manages the different orders
Locker release	At delivery, cashier scans the order — locker automatically marked as free and available
Full locker alert	If all lockers are occupied, the software alerts the manager — signal to accelerate pending deliveries
Locker history	Each locker keeps the history of all orders stored in it — complete traceability

Example real-time locker screen:

Casier	Taille	Statut	Client	Depuis	Commande
C-01	Grand	Occupe	DIALLO M.	2 days	N2026-0147
C-02	Moyen	Libre	—	—	—
C-03	Petit	Occupe	NDIAYE F.	Ce matin	N2026-0198
C-04	Grand	Occupe	FALL A.	3 jours — ALERTE	N2026-0132
C-05	Moyen	Libre	—	—	—

3.3.4 Automatic Conveyor Integration (Future)

The conveyor is a motorised hanger system that automatically retrieves garments and brings them to the counter when the client arrives. CleanHub must be architected to support this integration from the start — even if the hardware integration is implemented later.

Feature	Description	Phase
Architecture prepared	The conveyor_slot field is integrated in the database from the start	Free start — no migration needed later
Conveyor position	When a garment is stored on the conveyor, staff scans and enters the position number (e.g. Conv-24)	Minimal integration
Automatic retrieval	When integrated, the cashier scans the client's loyalty card	Full integration
Brand compatibility	CleanHub will be compatible with conveyors: ACS, Zetolink, Metal Integration, Artimecc — HTTP API or	Full integration
Delivery confirmation	Staff confirms handover to client in the software — conveyor records that position as freed	Full integration
Multi-garment management	If the client has 5 garments on the conveyor, the system brings them all back in the correct order	Full integration
Unknown position alert	If a garment is marked 'Ready' but has no conveyor position assigned, the software alerts the manager	Full integration

Comparison of the 3 storage systems in CleanHub:

System	Investment	Retrieval speed	Ideal for	Available
Numbered racks / hangers	Low — just labels	Fast — 1 min to find	Small and medium pressing	Included
Numbered lockers	Medium — locker purchase	Very fast — client goes directly	Pressings with self-service	Included
Automatic conveyor	High — industrial equipment	Instant — automatic	Large high-volume pressing	Future integration

■ We recommend beginning with the numbered rack system — simple, effective and free. Lockers and conveyors can be added progressively as the pressing grows.

3.4 Payment Module

- Accept multiple payment methods: cash, Wave, Orange Money, credit card, credit (tab)
- Support partial payments and instalment tracking
- Automatically generate PDF invoices and receipts
- Integrate Wave and Orange Money APIs for mobile payment confirmation
- Daily, weekly and monthly payment reconciliation reports
- After every confirmed payment: receipt automatically sent to client via WhatsApp AND email simultaneously
- Receipt is a PDF document including: pressing logo, order number, garment list, services, unit prices, discounts, payment method, cashier name, branch, date/time and estimated pickup date
- If client has no WhatsApp: SMS fallback automatically sent with a link to download the PDF receipt
- Owner receives a copy of every receipt by email for full traceability

3.4.0 Cash Drawer — Automatic Opening on Cash Payment

A cash drawer must be connected to the receipt printer via a RJ11/RJ12 cable (standard on all 80mm thermal receipt printers). The drawer opens automatically every time the cashier clicks 'Cash Payment' — no manual action required. Note: the RJ11/RJ12 cable connects the drawer to the printer, not to the PC or tablet — this is compatible with both USB, Bluetooth and WiFi receipt printers.

Rule	Description
Automatic opening — cash only	The cash drawer opens AUTOMATICALLY only when the cashier selects CASH as the payment method
Connected to receipt printer	The cash drawer is connected to the thermal receipt printer via a standard RJ11/RJ12 cable — the p

Rule	Description
Compatible hardware	Any standard cash drawer with RJ11/RJ12 port sourced from China — compatible with all 80mm the
Manual opening by owner	The owner can manually open the drawer at any time from the application (e.g. to add change at the
Manager override	The manager can also open the drawer manually with their PIN — for end-of-day cash counting. Acti
Cashier cannot open manually	The cashier CANNOT open the drawer manually without processing a cash payment — this prevents
Drawer open alert	If the cash drawer remains open for more than 30 seconds after a transaction, the software displays
Z Report includes drawer events	The end-of-day Z report lists every drawer opening: time, cashier name, reason (cash payment, man

Cash drawer opening rules summary:

Action	Drawer opens?	Who triggers it?
Cash payment confirmed	YES — automatically	System (triggered by receipt printer)
Wave payment confirmed	NO	N/A
Orange Money payment confirmed	NO	N/A
Card payment confirmed	NO	N/A
Mixed payment (cash + Wave)	YES — automatically (cash payment)	System
Manual opening — start of shift	YES — with PIN	Owner or Manager only
Manual opening — cash counting	YES — with PIN	Owner or Manager only
Cashier manual opening (no transaction)	NO — blocked	Not permitted

■ The developer must implement the cash drawer integration via the ESC/POS command sent through the receipt printer driver. The command must be sent simultaneously with the receipt print job — drawer opens at the exact moment the receipt is printed. This is a NON-NEGOTIABLE requirement for any business handling cash.

3.4.1 TPE Integration — Electronic Payment Terminals

CleanHub must integrate with physical payment terminals (TPE) to automatically capture Wave, Orange Money and bank card payments directly into the system — no manual entry required by staff.

Recommended TPE Hardware (sourced from China — Android-based):

Terminal	Specs	Best For
PAX A920 Pro	Android TPE 5.5 inch touchscreen, built-in printer, 4G/WiFi, battery power	Full-service POS — high volume
PAX A50	Compact Android TPE 4 inch screen, NFC, contactless, 4G/WiFi	Secondary counter or small branch
SUNMI P3 Air	Smart Android POS 6.75 inch screen, 8GB RAM, scanner, battery power	Delivery staff, mobile payments on-site
SUNMI V3 Mix	Countertop Android large touchscreen, built-in printer, WiFi	Fixed counter POS with receipt printing

African Payment Aggregators — API Integration:

Provider	Payments Supported	Coverage	Integration
PayTech (.sn)	Wave + Orange Money + Free Money + card	Senegal — full mobile money	API + TPE
SenePay	Wave + Orange Money + Free Money	Senegal — 1.8% flat fee	API
CinetPay	Wave + Orange Money + card (Visa/MC)	West Africa multi-country	API + webhook

Provider	Payments Supported	Coverage	Integration
PayDunya	Wave + Orange Money + card	Senegal + West Africa	API + TPE
KKiaPay	Mobile money + card	West Africa francophone	API + physical TPE

TPE Integration Flow with CleanHub:

- Cashier selects payment method on CleanHub POS screen -> TPE triggered automatically with the exact order amount
- Client pays on TPE (card tap, QR scan, or phone number for Wave/Orange Money)
- TPE sends webhook confirmation to CleanHub API in real time
- Order automatically marked as PAID in CleanHub — no manual entry
- Receipt printed automatically + PDF sent to client via WhatsApp
- Transaction logged in cash register journal + owner email triggered instantly
- All TPE transactions reconciled automatically in daily Z report

■ **Developer must implement TPE integration via the chosen aggregator API (PayTech or CinetPay recommended). PAX A920 Pro and SUNMI Android terminals are the preferred hardware — both manufactured in China.**

3.4.2 Custom Messages — Invoices & Customer Display

Message Type	Where it appears	Examples
Invoice footer message	Printed on every customer receipt and PDF invoice	'Thank you for your trust! Offer: -10% on your next order'
B2B invoice message	On corporate/hotel invoices only	'Payment due within 30 days. Bank details: ...'
Customer display message	Shown on customer-facing screen between transactions	'Special offer this week: express service in 24h!'
Promotional banner	Rotating banner on customer-facing screen	'Loyalty card: earn points on every order!'
Seasonal message	Scheduled for specific dates	'Happy Eid! -15% on all orders this week'
WhatsApp footer	Added to the bottom of every WhatsApp notification	'Follow us on WhatsApp for exclusive offers!'

- Owner and branch manager can write, edit and delete messages directly from the settings panel — no technical knowledge required
- Messages can be scheduled with a start date and end date — activate and deactivate automatically
- Multiple messages can be configured simultaneously — system rotates them on the customer display
- Preview button: owner can see exactly how the message will appear on the receipt before saving

3.4.3 Cash Management — Expense Statements, Cash Notes & Payment Method Correction

1. Expense Statement:

Feature	Description
Record expense	Staff records any business expense: category, amount, description, date, payment method, receipt photo
Expense categories	Configurable list: Rent, Electricity, Water, Staff salary, Supplies, Equipment repair, Delivery, Miscellaneous
Expense statement	Full statement of all expenses for any period — daily, weekly, monthly, custom range
By branch	Expenses tracked per branch — consolidated view across all branches for owner
Receipt photo	Staff takes photo of expense receipt with webcam or smartphone — stored with the expense record

Feature	Description
Revenue vs expenses	Dashboard shows revenue vs expenses side by side — net profit calculated automatically
Export	Export expense statement to PDF or Excel — for accountant or owner
Weekly auto email	Weekly expense summary automatically emailed to owner every Monday

2. Cash Notes:

Feature	Description
Create cash note	Cashier creates a note for any cash movement not linked to a customer order: petty cash advance, cash on hand, etc.
Cash note types	Cash In (entry) or Cash Out (exit) — each tracked separately in the cash register
Reason required	Reason mandatory: who gave/received cash, purpose, manager authorisation
Appears in Z Report	All cash notes appear in the end-of-day Z report — cash expected vs actual calculation includes them
Manager authorisation	Cash Out notes above a configurable threshold require manager PIN confirmation
Printable note	Cash note can be printed as a small receipt — given to the person who received/gave cash

3. Payment Method Correction:

Feature	Description
Correct payment method	Manager can change the payment method recorded on any payment — e.g. change Cash to Wave, or vice versa
Reason required	Reason mandatory before correction: who made the error, what happened
Amount unchanged	Only the payment METHOD changes — the amount stays the same — no financial impact
Audit trail	Correction logged permanently: original method, corrected method, who corrected it, when, why
Authorisation	Only Manager and Owner can correct payment methods — Cashier cannot
Time limit	Correction only allowed on same-day payments — previous day corrections require Owner authorisation

■ All cash management operations (expenses, cash notes, payment corrections) are permanently logged in the audit trail and appear in the daily Z report sent automatically to the owner. No cash movement can be hidden or deleted.

3.5 Inventory & Stock Management

- Track consumables: dry-cleaning solvent, detergent, starch, plastic garment bags, hangers, label rolls, packaging
- Set minimum stock thresholds and generate low-stock alerts
- Record stock purchases from suppliers with cost tracking
- Link stock consumption to orders for cost-of-goods reporting

3.5.0 Garment Inventory — One Click to See Everything in the Pressing

When the manager clicks the INVENTORY button, the software immediately displays the complete list of ALL garments currently in the pressing — that is, all garments RECEIVED but NOT YET DELIVERED to the client.

■ Simple definition: a garment is 'in stock' from the moment it is dropped off by the client until the moment it is returned to them. Everything between these two moments appears in the inventory.

The list displayed with one click:

#	Client Name	Phone	Garments	Status	Location	Total Amount	Amount Still to Pay
1	DIALLO Mamadou	77 123 45 67	3	Ready — stored	A-3-14	12,000 XOF	12,000 XOF
2	NDIAYE Fatou	76 234 56 78	1	In progress	Workshop	5,000 XOF	0 XOF (paid)
3	FALL Amadou	70 345 67 89	5	Ready — stored	B-2-07	28,500 XOF	18,500 XOF
4	BA Mariama	78 456 78 90	2	In reserve	Zone D-01	8,000 XOF	8,000 XOF
...	...	...	...	...	...	...	...

- Click on the client name -> opens their full file with all their orders
- Click on the amount still to pay -> opens the payment screen directly for that client
- Sort options: by name, by amount, by status, by deposit date, by days in the pressing
- Quick filter: All / Ready only / In progress / In reserve / Unpaid only
- Total displayed at the bottom: total number of garments + total amount to collect from all clients
- One-click export: print the list or export to Excel

■ The list must display in under 2 seconds after the click — even if the pressing has hundreds of orders in progress. This is a simple database query filtered on status != Delivered.

3.5.1 Stock Entry & Retail Sales Module

In addition to tracking consumables, CleanHub must support a full retail sales module allowing pressing owners to sell physical products directly to clients at the counter (e.g. detergent, laundry bags, hangers, fabric softener, stain remover).

Stock Entry — Recording Purchases for Resale:

When the owner or manager buys products to resell at the counter, they record the stock entry in CleanHub. This updates the stock level, calculates the cost price and enables automatic margin calculation on each sale.

Feature	Description
Stock entry screen	Admin records new stock arrivals: product name, quantity received, unit purchase cost, supplier, invoice
Batch entry	Enter multiple products in one stock entry session — submit all at once — useful when recording an
Product catalogue	Each product is created once in the product catalogue: name, barcode (EAN/UPC), unit of sale, selling
Barcode on products	Each product has a barcode — scan with the Bluetooth scanner to add to stock entry or to sell at PO
Purchase price per entry	The purchase price can vary per stock entry (price changes from one supplier delivery to the next) —
Stock valuation	Current stock value calculated automatically: at purchase cost price (actual cost of goods in stock) a
Margin per product	CleanHub calculates the gross margin per product: selling price minus purchase cost — visible on th
Low stock alert	Automatic dashboard alert and WhatsApp to owner when any product falls below the minimum thresh
Stock history	Full history of all stock entries, sales and manual adjustments per product — who entered what, whe
Stock adjustment	Admin can manually adjust stock quantity for loss, breakage, stock count correction — reason requir
Multi-location stock	For multi-branch businesses: each branch has its own stock level — stock entry recorded per branch
Stock count	Admin can launch a physical stock count: CleanHub displays expected quantities — admin scans or
Print available stock	Admin can print the current stock list at any moment: one click generates a printable A4 or thermal re

Retail Sales at POS — (works for both pressing businesses and car wash centres):

- Cashier can add retail products to any order OR create a standalone product sale — mixed orders (service + product) supported
- Scan product barcode at counter — product added to order instantly with correct price
- Every product sold automatically deducted from stock — no manual update required
- Products sold appear on the customer receipt alongside pressing services

Product	Unit	Stock tracked	Business
Detergent / Laundry powder	Per kg or per bag	Yes	Pressing
Fabric softener	Per bottle	Yes	Pressing
Stain remover	Per bottle	Yes	Pressing
Laundry bags / covers	Per unit	Yes	Pressing
Hangers	Per unit or per dozen	Yes	Pressing
Motor oil / Engine oil	Per litre or per bottle	Yes	Car wash
Car air freshener / perfume	Per unit	Yes	Car wash
Car wax / polish	Per unit or per bottle	Yes	Car wash
Tyre shine product	Per bottle	Yes	Car wash
Windscreen cleaning fluid	Per bottle	Yes	Car wash
Microfibre cloths	Per unit or per pack	Yes	Car wash
Car shampoo	Per litre or per bottle	Yes	Car wash
Dashboard protector	Per bottle	Yes	Car wash
Any other retail product	Configurable unit	Yes	Both

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Motor oil / Engine oil	Per litre or per bottle	Yes	Car wash
Car air freshener / perfume	Per unit	Yes	Car wash
Car wax / polish	Per unit or per bottle	Yes	Car wash
Tyre shine product	Per bottle	Yes	Car wash

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Windscreen cleaning fluid	Per bottle	Yes	Car wash
Microfibre cloths	Per unit or per pack	Yes	Car wash
Car shampoo	Per litre or per bottle	Yes	Car wash
Dashboard protector	Per bottle	Yes	Car wash
Any other retail product	Configurable unit	Yes	Both

3.5.2 Employee HR Management

CleanHub must include a complete employee management module allowing the owner and branch manager to manage all staff directly from the software — contracts, attendance, salaries, advances, performance and disciplinary records.

Feature	Description
Employee profile	Full name, phone, address, emergency contact, date of birth, photo, role, branch, start date, contract
Shift scheduling	Admin sets weekly schedule per employee: working days, start time, end time
Clock in / clock out	Employee clocks in/out via PIN on the CleanHub POS — time recorded automatically
Monthly attendance report	Complete monthly attendance sheet per employee: days worked, late arrivals, absences, total hours
Salary calculation	Base salary automatically adjusted for absences and overtime based on attendance records
Salary advances	Record salary advance given to employee — deducted automatically from next salary
Payslip generation	Auto-generate PDF payslip per employee per month — printable and sendable via WhatsApp
Account deactivation	When an employee leaves, their CleanHub access is automatically deactivated instantly — no manu

■ The HR module is available as an optional paid add-on. When an employee account is terminated, their CleanHub login is automatically deactivated instantly. All their historical transactions and performance data are preserved for audit purposes.

3.5.3 Artificial Intelligence (AI) Features

CleanHub must integrate AI-powered features to automate garment defect detection from photos and predict future order volumes. These features reduce staff workload, improve accuracy and give the owner a strategic advantage.

1. AI Stain & Defect Detection (from webcam/camera photos):

Feature	Description
Auto stain detection	When staff takes a photo of a garment at intake, AI automatically analyses the image and detects vis
Defect highlighting	AI draws a coloured box around detected defects on the photo
Defect classification	AI classifies each defect: Stain / Tear / Discolouration / Burn / Hole / Missing button — with confiden
Auto-fill reservation	Detected defects automatically pre-fill the reservation form — staff just confirms or edits — saves tim
Treatment suggestion	AI suggests the appropriate treatment for each detected stain: pre-soak, dry-clean, hand wash, spec
Staff override	Staff can always add, edit or remove AI-detected defects — AI assists, never replaces human judger
Fabric recognition	AI identifies fabric type from photo (cotton, silk, wool, synthetic) — suggests appropriate service auto
Colour detection	AI detects garment colour and flags risk of colour bleeding — alerts production staff

2. AI Order & Demand Forecasting:

Feature	Description
Customer return prediction	AI predicts when each customer is likely to return based on their visit history — shown on dashboard
Demand forecast by day	AI predicts how many orders to expect each day of the week — helps staff planning and stock prep
Peak period prediction	AI identifies upcoming busy periods (holidays, events, end of month) and alerts owner in advance
Stock demand prediction	AI predicts which consumables will run low based on forecasted order volume — triggers early res
Revenue forecast	AI projects expected monthly revenue based on historical trends — shown as a target line on repo
Dormant customer prediction	AI identifies customers at risk of churning before they stop coming — triggers automatic WhatsApp
Intelligent appointment scheduling	AI analyses each client's visit habits (e.g. always comes Saturday morning, every 2 weeks) and au
Capacity-based appointments	AI analyses the workshop workload in real time and distributes pickup appointments intelligently: if
Slot optimisation	AI identifies slow hours of the week and encourages clients to come during these slots via a What
Overload alerts	If too many orders have the same pickup date, AI alerts the manager and proposes to redistribute

■ **No existing pressing SaaS on the African market currently offers AI-powered stain detection, demand forecasting or intelligent appointment scheduling. These features will make CleanHub the most advanced pressing management platform in Africa.**

3.6 Planning, Agenda & Delivery Management

- Calendar view for scheduling pickups and deliveries per day and branch
- Assign delivery tasks to specific staff members
- Map/zone-based delivery routing (optional enhancement)
- Staff shift scheduling and attendance tracking

3.6.0a Field Order Creation — Delivery Staff Registers Orders at Client's Home

A delivery agent can create a COMPLETE order directly at the client's home using their smartphone or EMV terminal. The order is instantly transmitted to the pressing server — the production team sees it immediately on their dashboard. The client receives a receipt on the spot. No need to return to the pressing to register the order.

Feature	Description
Full order creation in the field	The delivery agent opens the CleanHub app on their smartphone or EMV terminal, creates an order
Instant server sync	The order is transmitted to the pressing server in real time as soon as it is created — the production
Receipt printed on the spot	The delivery agent prints a receipt immediately at the client's home using the portable Bluetooth ther
Garment collection	The agent collects the garments from the client — notes any existing defects or reservations directly
Matrix label at the pressing	Since the field agent does not have a matrix printer, garment labels are printed automatically at the p
Pickup appointment assigned	The pickup appointment date is automatically calculated and assigned by the server as soon as the c
Order attributed to delivery agent	The order is linked to the delivery agent who created it — visible in their performance statistics (orde
Client order history updated	The order appears immediately in the client's profile — next time they visit the counter or are called,

3.6.0b Payment Collection by Delivery Staff — Full Cashless Payment in the Field

The delivery agent can collect payment directly at the client's home — at garment collection OR at delivery of finished garments. The agent does not need to be at the counter. The payment is recorded on the pressing server instantly.

Payment method	How the agent collects	What CleanHub records
Wave (mobile money)	Agent presents a QR code on their phone screen confirmed via Wave API mobile link by WhatsApp	Payment confirmed via Wave API mobile link by WhatsApp
Orange Money	Same as Wave — agent shares OM payment link on QR code	Payment confirmed via Orange Money API — order updated
NFC contactless card	If agent has an EMV terminal — client taps card on terminal — processed instantly	Payment confirmed — processed instantly — receipt printed on EMV terminal
Cash	Agent collects cash — enters amount received in the CleanHub app — order updated	Cash payment entered in CleanHub app — order updated
Deposit at collection	Client pays a deposit when the agent collects the garment — balance is payable on delivery	Deposit recorded — balance payable on delivery — agent sees remaining balance

- Payment collected by the delivery agent is immediately visible to the manager at the pressing on their dashboard — real-time cash flow visibility
- The delivery agent CANNOT modify a price or apply a discount — only the Owner can. The agent collects the exact amount shown by the system
- If a client refuses to pay or cannot pay, the agent marks the order as 'Payment pending' — the manager is notified immediately
- All payments collected in the field are included in the daily Z report and cashier statistics — attributed to the delivery agent's user account

■ **The delivery agent's smartphone must be registered and authorised in CleanHub by the Owner before they can create orders or collect payments. An unregistered device cannot access CleanHub. The owner can deactivate a delivery agent's access in one click from the Settings panel.**

3.6.0c Security — Protection Against Hacking and Unauthorised Access

CleanHub is a business-critical application handling financial transactions and client data. The following security measures are NON-NEGOTIABLE and must be implemented at every level of the application.

Security layer	Measure	Detail
Authentication	JWT tokens with short expiry	Access tokens expire after 15 minutes — automatically refreshed if user is active
Authentication	Device registration	Each device (PC, tablet, smartphone, EMV terminal) must be registered and authorised
Authentication	4-digit PIN for quick lock	Staff lock the screen with a 4-digit PIN when stepping away — screen locks
Authentication	Biometric (optional)	On supported Android devices: fingerprint or face unlock as alternative to PIN
Data in transit	HTTPS / TLS 1.3	All communications between the app and the server are encrypted — no plaintext
Data at rest	Database encryption	Sensitive fields encrypted in the database: phone numbers, emails, payment details
Access control	Strict RBAC	4 roles with different permissions — no role can exceed its authorised actions
Anti-tampering	Server-side validation	ALL business rules enforced on the server — the client app never makes financial decisions
Anti-tampering	Clock on server only	All timestamps come from the server (NTP) — a hacked device cannot manipulate time
Audit trail	Immutable logs	Every action by every user is logged with server timestamp, user ID and device ID
Brute force	Rate limiting	Failed login attempts trigger a lockout: 5 failed attempts = account locked for 15 minutes
Session hijacking	IP binding (optional)	Session token can be bound to the device's network signature — suspicious logins blocked
Data backup	Encrypted backups	Daily automated backups encrypted before storage — stored on the CleanHub cloud
Penetration testing	Pre-launch security audit	The developer must perform a security audit (or hire a security specialist) before launch

Security layer	Measure	Detail
Code protection	Obfuscation	Android APK obfuscated before distribution — source code cannot be extracted
API protection	API key + signature	All API calls require a valid API key + request signature — replay attacks blocked

Specific protections for field operations (delivery agents):

Risk	Protection
Delivery agent creates fake orders	All orders created in the field are visible to the pressing manager in real time — any suspicious orders are flagged
Delivery agent collects payment but does not pay for the order	All payments must be recorded in CleanHub before the receipt is generated — the client receives a receipt
Delivery agent uses CleanHub on an unauthorised device	Device must be pre-registered and authorised by the Owner — new device requires Owner approval
Delivery agent shares their login credentials	Each agent has a unique account — sharing credentials is detectable (simultaneous sessions) — flagged
Someone steals the delivery agent's phone	Remote device lock: Owner can deactivate the device from the Settings panel — CleanHub becomes unusable
Hacker intercepts network traffic	TLS 1.3 encryption on all API calls — certificate pinning on the Android app — intercepted traffic is unusable
SQL injection or API attack	All inputs validated and sanitised server-side — parameterised queries — API rate limiting — automatic blocking of suspicious IP addresses

■ **Security is not an optional feature — it is a fundamental requirement of CleanHub. The developer must treat every security item in this section as a hard technical deliverable. Security failures in a financial application directly harm the owner, the clients and the reputation of CleanHub. A security audit report must be delivered alongside the MVP.**

3.6.0d Order Ready Notification — Automatic Client Alert

As soon as a pressing staff member marks an order as READY in CleanHub, the client automatically receives a notification — without any manual action by the cashier. This is one of the most valued features by pressing clients in Africa: knowing the exact moment their garments are ready.

Feature	Description
Automatic trigger	The notification is sent the instant the production staff member taps 'Mark as Ready' on the Kanban board
WhatsApp (primary channel)	Client receives a WhatsApp message: 'Hello [First name], your order N[XXX] is ready at [Pressing name]'
SMS (fallback)	If the client does not have WhatsApp or the WhatsApp API fails, an SMS is automatically sent as fallback
Client with outstanding balance	If the client has an outstanding balance, the notification adds: 'Please note: you have an outstanding balance of [Amount]'
Multiple orders	If the client has multiple orders ready simultaneously: 'Your 3 orders are ready — total to pay: [X] XOF'
Delivery order ready	If the garments are to be delivered (not collected by the client), the notification says: 'Your order N[XXX] is ready for delivery at [Address]'
Notification timing	Notifications respect the pressing's configured business hours — if an order is marked ready at 11pm, the notification is sent at 12pm
Notification history	Every notification sent is logged: date, time, channel (WhatsApp/SMS), content, delivery status (sent/read)
Resend notification	If the client says they didn't receive the notification, cashier can resend with one tap from the order screen
Custom notification message	Owner can customise the notification template from the Settings panel — add the pressing name, address, phone number
Client confirmation (optional)	Client can reply to the WhatsApp notification to confirm they will collect — cashier sees confirmation in CleanHub

- The order ready notification is the single most important automated message in CleanHub — it eliminates the need for the cashier to manually call every client when their order is done
- Notification sent even when the cashier is busy or the pressing is crowded — the system never forgets a client
- If the notification API (Twilio / WhatsApp Business) fails, CleanHub retries automatically 3 times before marking the notification as failed and alerting the manager

■ *Statistics: notification sent report — how many notifications sent per day, delivery rate (% received by client), channel breakdown (WhatsApp vs SMS), orders collected within 24h of notification vs those not collected.*

3.6.1a Delivery with Real-Time GPS Tracking

When a delivery person leaves with garments or a vehicle, the client and manager can track their position in real time on a map. The client knows exactly where their delivery person is and when they will arrive.

Feature	Description
Real-time position	Delivery staff GPS position updated every 30 seconds on the map — visible to manager from dashboard
Client tracking link	At delivery confirmation, client receives a WhatsApp with a tracking link: 'Track your delivery in real time'
Automatic ETA	System calculates the estimated arrival time (ETA) in real time based on delivery staff position and route
Arrival alert	Client receives automatic WhatsApp when delivery staff is 5 minutes away: 'Your delivery is arriving in 5 minutes'
Manager dashboard	Manager sees all their delivery staff on one map: who is where, which delivery they are doing, how many deliveries they have completed
Route history	Each delivery route recorded: departure time, arrival time, distance covered — useful for calculating fuel costs
Delivery zone	Manager configures the maximum delivery zone (radius in km around the pressing) — software blocks deliveries outside zone
Route optimisation	If several deliveries are scheduled the same day, the system suggests the optimal order to minimise travel time

■ **GPS tracking uses the delivery staff's smartphone location (CleanHub mobile app). Delivery staff must accept location sharing at installation. Location is active only during configured working hours.**

3.6.2 Owner Mobile App — Full Business Control from Your Smartphone

The owner has a dedicated mobile application that gives them COMPLETE control over their pressing business from their smartphone — wherever they are, at any time. This is not just a dashboard to read numbers — it is a fully operational interface to monitor, act, approve and manage in real time without being physically present at the pressing.

What the owner can do from their phone	Available on iOS & Android
See the live dashboard for each branch	Revenue today, orders in progress, orders ready, outstanding
See all orders in real time	Full list of all orders from all branches — filterable by status, branch, date
Approve sensitive operations	A cashier requests a refund or a price modification — the owner approves or rejects
Receive fraud alerts	If a cashier cancels too many orders, grants unusual discounts
View the Z Report from any branch	At the end of each day, the owner taps to see the complete Z Report
Track delivery staff on the map	Real-time GPS position of each delivery agent — the owner can see who is where
Send a message to any branch	Owner sends a text instruction to any branch manager or all staff
View and respond to complaints	All open complaints from all branches visible on the phone — owner can respond
Check receivables (dues)	Total outstanding balance across all clients and all branches — owner can chase
Monitor stock alerts	Low stock notifications from any branch — owner can trigger reorder
View cashier statistics	Performance of each cashier at each branch — revenue collected, orders completed
Receive daily summary every morning	Automatic WhatsApp sent to the owner every morning at 8am
Multi-branch view	If the owner has several branches, they can switch between branches

Mobile app — available on Android and iPhone:

Platform	Installation	Notes
Android	APK installed directly — same APK as the POS application	Application launch on owner's phone sees only the owner dashboard interface
iPhone (iOS)	Installed from the App Store — owner version	Requires Apple Store submission by the developer before availability
Access from any smartphone	The owner does not need a specific phone model — CleanHub mobile works on any recent Android or iPhone	
No separate app to install	The owner uses the same CleanHub application as the cashiers — but their role (Owner) gives them access to	

Notification settings — owner controls what they are alerted about:

Notification	Default	Owner can turn off
Daily morning summary (WhatsApp)	ON	Yes
Order ready notifications	OFF (client only)	Can turn ON for monitoring
Payment received (above threshold)	ON — e.g. any payment	Yes — above 50,000 VND threshold
Fraud alert (unusual cancellation/discount)	ON — always	No — mandatory security alert
New complaint opened	ON	Yes
Low stock at any branch	ON	Yes
Delivery agent leaves zone	ON	Yes
Z Report generated	ON	Yes
Prepaid account top-up	ON	Yes
Overdue order (not collected in X days)	ON	Yes

■ *The owner mobile app gives the pressing owner total peace of mind when they are away from the business — travelling, in a meeting, at home. Everything that happens at the pressing is visible, auditable and actionable from their phone in real time.*

■ **DEVELOPER REQUIREMENT: the owner mobile dashboard must be the SAME CleanHub Android/iOS app with role-based display. When the user logs in with an Owner account, the app shows the full management dashboard. When logged in with a Cashier account, the app shows only the POS. No separate app to build — one app, role-based interface. This is a core deliverable.**

3.6.3 Customer-Facing Screen — Order Display & Advertising

A second screen is placed facing the client during order entry. The client sees in real time what the cashier is recording — prices, items, total — and can verify that everything is correct. Between transactions, the screen displays advertisements and promotions from the pressing.

Mode	When	What is displayed
Order entry mode	During order recording	List of items entered, unit prices, running total, discounts applied, client name
Payment mode	When cashier requests payment	Total to pay, available payment methods (Wave, Orange Money, cash), Wave
Satisfaction survey mode	After each payment	Satisfaction survey (stars, smileys) — client responds directly on the screen
Standby / advertising mode	Between transactions	Active promotions, special offers, welcome message, monthly subscription an
Birthday mode	If it is the client's birthday	Personalised happy birthday message displayed on customer-facing screen o

- Advertising content is configured by the manager from the Settings panel — add images, text, promotions — activates automatically
- The customer-facing screen can be a second monitor connected to the PC (via HDMI) or an Android tablet on a rotating stand

- All advertisements and messages are in French — manager can also add messages in Wolof or the local language

3.6.4 Real-Time Production Tracking — Workshop Kanban Board

The Kanban board allows the workshop manager to track each garment at every processing stage in real time. It is the central tool for the production team — visible on a large tablet or wall screen in the workshop.

Kanban Column	Garments in this column	Status
Received	Garments deposited and labelled — awaiting processing	Not yet processed
In washing	Garments currently in the machines	In progress
Drying	Garments currently drying	In progress
Ironing	Garments currently being ironed	In progress
Folding	Garments currently being folded (laundry)	In progress
Dry-cleaning	Garments in the dry-cleaning machine	In progress
Ready — to be stored	Garments processed, not yet stored to their location	Not yet stored
Stored — awaiting pickup	Garments stored, waiting for the client	Ready
In reserve	Garments not collected for more than 10 15 days	Archived

- Each Kanban card shows: client name, garment type, colour, planned pickup date, any reservations or special notes
- Urgency colour coding: green (on time), orange (appointment tomorrow), red (overdue)
- Staff update the status with one scan: scan the garment label and select the new stage
- Manager can view the Kanban from their phone — they don't need to be in the workshop
- Automatic alerts: if a garment stays stuck in the same column for too long, the manager is notified
- Workshop statistics: average time per stage, number of garments processed per hour, bottlenecks automatically identified

3.6.5 Customer Complaints & Incident Management on Garments

A client reports a problem: damaged garment, lost item, colour that bled, shrinkage, tear caused during processing, button torn off, stain that was not removed. CleanHub opens a COMPLAINT FILE, tracks every step of the resolution and protects the pressing with all evidence collected at intake — photos, signed reservations, processing notes.

1. Complete complaint workflow:

Step	Who	Description
1. Complaint	Cashier or Manager	Client reports a problem at the counter or by phone/WhatsApp. Cashier opens a complaint from the order
2. Order evidence	System	Automatically immediately displays: intake photo of the garment, all reservations signed by the client at intake
3. Complaint description	Cashier/Manager	Garment damaged during processing / Garment lost or not found / Colour bled or ran / Shrinkage during processing
4. Severity assessment	Manager	Minor (cosmetic, barely visible) / Moderate (visible, client notices) / Severe (garment unusable or significantly damaged)
5. Evidence review	Manager	Manager compares the intake photo with the current state of the garment — sees all signed reservations and processing notes
6. Decision	Manager or Owner	Accepted (pressing at fault) / Rejected (pre-existing damage proven) / Partially accepted (shared responsibility)
7. Resolution	Manager or Owner	See resolution options table below — Manager offers, Owner approves if monetary refund

Step	Who	Description
8. Client notified by system	System	Automatic WhatsApp at each status change: 'Your complaint TKT-C-0042 has been opened', 'Under in progress', 'Waiting for resolution', 'Complaint closed'
9. Closed	Manager	Complaint closed with documented outcome — stored permanently in client history and in manager report

2. Resolution options — what the pressing can offer:

Resolution	Who authorises	How CleanHub records it
Free re-treatment	Manager	New order created at 0 XOF — linked to original complaint — garment goes through the production process
Partial refund	Owner only	Refund amount entered — payment method selected (Wave / cash) — deducted from client's account balance
Full refund	Owner only	Full order amount refunded — order marked as refunded — client receives WhatsApp notification
Compensation voucher	Manager	A promo code or gift card created for a configurable amount — sent to client by WhatsApp
Store credit	Manager	Amount added to client's account balance — deducted automatically from their next order
Replacement service	Manager	Replacement garment or equivalent service offered — documented in the complaint file
No action — rejected	Manager (with proof)	Rejection reason documented — intake photo and signed reservations attached as evidence

3. Complaint file — what it contains:

Element	Content
Complaint reference	Unique number: TKT-C-2026-0042 — printed on all correspondence with the client
Original order	Link to the order: order number, date, articles, services, cashier who created it
Intake evidence	Intake photo of the garment + all reservations signed at intake + processing notes
Damage photo	Photo taken at the moment the complaint was opened — shows current state of the garment
Client statement	Verbatim note of what the client said — entered by the cashier
Processing history	Every status change the garment went through — who handled it at each stage
Resolution history	All actions taken: offers made, client responses, final resolution — with dates and staff names
Notification log	Every WhatsApp/SMS sent to the client about this complaint — with delivery status

4. Complaints reports — management tool:

Report	Description
Complaint rate	% of orders that generated a complaint — by period, by branch, by service type — is it increasing or decreasing?
Complaints by type	How many damage / lost / colour / shrinkage — identifies which type of incident is most frequent
Complaints by article family	Which garment types generate the most complaints: suits, delicate fabrics, coloured items — guide for production
Complaints by staff member	Which production staff member's work generates the most incidents — training and supervision tool
Average resolution time	How many days on average to resolve — service quality KPI
Financial impact	Total refunds, re-treatments and vouchers granted — monthly cost of complaints
Rejection rate	% of complaints rejected with proof — measures how well intake documentation protects the pressing

- Every complaint is visible on the Owner mobile dashboard — owner sees all open complaints from their phone
- If the same type of incident recurs frequently (e.g. colour bleeding on dark shirts), CleanHub flags a pattern alert: 'Unusual number of colour complaints this month — 7 cases vs average 1'

- Clients with an open complaint receive a priority flag on their profile — cashier sees it immediately at their next visit

■ **ONLY THE OWNER can grant a monetary refund on a complaint. The Manager can offer free re-treatment, compensation voucher or store credit. All complaint decisions are logged with server timestamp and mandatory reason. A complaint can never be deleted — only closed.**

3.6.6 Referral Programme — Client Recommends a Friend

A satisfied client recommends the CleanHub pressing to a friend or neighbour. Both are rewarded automatically. This is the best growth channel in Africa where word of mouth is more powerful than any advertising.

Step	What happens	Reward
1. Client A shares referral code	Client A receives a unique referral code (e.g. DIALLO2026) yet they need not share by WhatsApp, SMS or word of mouth	Client A automatically gets 500 bonus points credited on their first order (confirmed)
2. Client B comes to the pressing	Client B mentions Client A's referral code to the cashier	Client B gets a welcome discount on their first order (confirmed)
3. Bonus points to referrer	After Client B completes and pays their first order, Client A automatically gets 500 bonus points credited on their first order (confirmed)	
4. Notification automatique	Client A reçoit un WhatsApp : 'Votre ami NDIAYE Fatimata vient de rejoindre CleanHub. Renforcez votre réseau et gagnez des points bonus !'	
5. Multiple referrals	A client can refer as many friends as they want — each referral generates an additional bonus points	

Referral programme settings:

Setting	Description	Default
Referee welcome discount	% or fixed amount granted to the referred new client on their first order	10%
Referrer bonus points	Number of points credited to the referrer when their referee places their first order	500 points
Validation delay	Number of days after which points are definitively granted to the referrer (after referee's confirmed payment)	Immediate
Referral code	Referral code format: ClientName-Year or random code	Configurable
Referral limit	Maximum number of active referrals per client (optional — can be unlimited)	Unlimited
Referral report	Monthly tracking: number of new clients from referrals, revenue generated, top referrers	Monthly

■ *The referral programme is the most powerful growth tool for a pressing in Africa. A loyal client who recommends the pressing is more credible than any advertising. CleanHub fully automates tracking and rewards — the manager does nothing manually.*

3.6.7 Monthly Client Subscription — Service Package

The pressing can offer monthly packages to regular clients: e.g. '10 ironed shirts per month for 15,000 XOF'. The client pays in advance and uses their package progressively. Excellent for loyalty and predictable revenue.

Feature	Description
Create a package	Manager creates packages from the Settings panel: name, content (e.g. 10 shirts), monthly price, duration
Package examples	Shirts Package (10 ironed shirts/month), Family Package (20 items/month), Corporate Package (50 items/month)
Client subscription	Client subscribes to the package — pays in full or monthly — Wave, Orange Money or cash — agreed duration
Package usage	At each order, cashier selects the client's package — items automatically deducted from remaining monthly quota
Package balance visible	Cashier sees in real time: 'Client DIALLO — Shirts Package — 3 shirts remaining this month'
Quota alerts	When client reaches 80% of their quota, they receive a WhatsApp: 'You have 2 shirts remaining on your package this month'

Feature	Description
Package overage	If client exceeds their quota, additional items are charged at normal rate — cashier is informed
Automatic renewal	Package renews automatically each month — payment triggered automatically via Wave or Orange
Packages report	Package usage rate (% of quota used by subscribers), revenue generated, best-selling packages

Package example:

Client	Package	Price	Quota	Used this month	Remaining	Renewal
DIALLO M.	Shirts Package	15,000 XOF/month	10 shirts	7 shirts	3 shirts	In 12 days
NDIAYE F.	Family Package	35,000 XOF/month	20 items	20 items	0 — EXHAUSTED	In 8 days
FALL A.	Kilo Package	20,000 XOF/month	10 kg	4.5 kg	5.5 kg remaining	In 20 days

3.6.8 Gift Card — Offer a Service as a Gift

Un client peut offrir un montant en carte cadeau a un proche pour qu'il l'utilise au pressing. Simple, rapide, et excellent pour les occasions : fetes, mariages, naissance, Tabaski, Korité.

Step	Description
Gift card purchase	Client requests a gift card at the counter — chooses the amount (e.g. 10,000 XOF) — pays immediately
Card issuance	CleanHub generates a unique code (e.g. GIFT-2026-8472) — printed on a gift voucher with the pressing's logo
Usage	Recipient comes to the pressing and gives their gift code to the cashier — amount automatically deducted from bill
Partial balance	If the order is less than the card amount, the balance remains on the card for future use
Validity	Manager configures the validity period (e.g. 6 months) — code expires automatically after the deadline
Non-refundable	A gift card cannot be exchanged for cash — only usable for services at the pressing
Gift card report	Total sold, total used, expired unused cards (pure revenue), active cards in circulation

■ *Gift cards are an interesting additional revenue source: cards not used before expiration are net revenue for the pressing. Very popular for wedding gifts and celebrations in Africa.*

3.6.9 Instalment Payment — Internal Credit

A client can spread the payment of their order in 2 or 3 instalments directly at the pressing. This is not a bank loan — it is the pressing itself that grants an instalment schedule. CleanHub automatically tracks due dates and follows up with the client.

Feature	Description
Create instalment plan	Cashier (with manager authorisation) defines the schedule: deposit to pay now, dates and amounts of future payments
Example	Order of 30,000 XOF — Deposit: 10,000 XOF now — Payment 2: 10,000 XOF in 15 days — Payment 3: 10,000 XOF in 15 days
Instalment receipt	A detailed receipt is printed and sent by WhatsApp: dates and amounts of each payment — client knows what to expect
Automatic reminders	3 days before each due date, CleanHub sends a WhatsApp to the client: 'Reminder: a payment of 10,000 XOF is due on 15/08/2026'
Conditional delivery	Manager can configure whether garments are handed over after the deposit or only after full payment
Receivables tracking	All active instalment plans visible in the receivables dashboard — sorted by due date
Missed payment	If client does not pay on the scheduled date, system alerts the manager and offers to send a manual reminder
Authorisation required	Only Manager or Owner can grant instalment payment — Cashier cannot

■ Instalment payment is a service the pressing grants voluntarily — CleanHub provides automatic tracking but does not guarantee collection. The owner must assess the client's creditworthiness before granting an instalment plan. Only the Owner can cancel an instalment plan.

3.6.10 Simplified Accounting — VAT, Monthly Balance Sheet

CleanHub includes a simplified accounting module adapted to the needs of small and medium pressings — not a professional accounting software, but sufficient for the manager to track finances, calculate VAT and prepare declarations.

Module	Description
Revenue journal	List of all receipts by day, week, month — broken down by payment method (cash, Wave, Orange M
Expense journal	All expenses recorded by staff — rent, electricity, water, salaries, supplies, repairs — categorised
Net profit	Revenue minus expenses = net profit — calculated automatically by day, week, month — evolution o
VAT calculation	Manager configures the applicable VAT rate (e.g. 18% in Senegal) — CleanHub automatically calcul
Monthly VAT report	VAT report ready to give to the accountant: VAT collected, deductible VAT, net VAT to pay — PDF e
Monthly balance sheet	Monthly financial summary: total revenue, total expenses, gross profit, VAT, net profit — compared t
Accounting export	Export of all entries in Excel or CSV format — importable into professional accounting software
Annual report	Full-year balance: total revenue, total expenses, profit, month-by-month evolution — exportable PDF

■ *This simplified accounting module is not a certified accounting software. It is designed to help the manager track finances daily and prepare data for their accountant. For official accounting and tax declarations, a professional accountant is still required.*

3.6.11 Supplier & Purchasing Management

CleanHub allows management of consumable suppliers (detergent, hangers, cleaning products, packaging, tokens) and tracking of orders and purchases made from them.

Feature	Description
Supplier file	Create a file for each supplier: name, phone, address, products supplied, payment terms (credit, cas
Supplier catalogue	Link each stock product to the supplier who provides it — purchase price recorded per supplier
Supplier order	When a product's stock falls below minimum threshold, CleanHub suggests placing an order with the
Purchase order	Generate a PDF purchase order to send to the supplier by WhatsApp or print — formalises the busin
Order reception	When delivery arrives, staff record the reception — stock updated automatically — discrepancy with
Supplier payment tracking	If pressing buys on credit, CleanHub tracks what is owed to each supplier with due dates — alert if a
Price comparison	If several suppliers offer the same product, CleanHub compares them — useful for negotiating or cha
Purchase history	Total purchased per supplier per month — useful for volume negotiations and preferential conditions

3.6.12 Machine Management — Breakdowns, Maintenance, Lifespan

For laundries and pressings with industrial machines (washing machines, dryers, dry-cleaning machines, ironing presses), CleanHub manages maintenance tracking and breakdown alerts.

Feature	Description
Machine file	Create a file for each machine: name, brand, model, serial number, purchase date, warranty, supplie

Feature	Description
Usage counter	Number of cycles performed by each machine — calculated automatically from processed orders
Maintenance schedule	Manager schedules regular maintenance: oil changes, filter cleaning, seal checks — automatic alerts
Breakdown recording	When a machine breaks down: staff record it in CleanHub — date, breakdown description, technician
Machine status	Each machine has a status: In service / Under maintenance / Broken down / Out of service — visible
Automatic hiding	A broken down or under maintenance machine is automatically hidden from client assignment — pre
Maintenance cost	Record the cost of each maintenance or repair intervention — calculation of total maintenance cost p
Lifespan alert	If the cumulative maintenance cost of a machine exceeds a configurable threshold, CleanHub alerts
Machine report	Availability rate of each machine (% of time in service), number of breakdowns per machine, average

■ **These 12 new features make CleanHub the most complete pressing and laundry software in Africa. No existing competitor covers all these modules in a single application.**

3.6.13 Prepaid Account — Client Recharges in Advance

The client deposits money in advance into a prepaid account at the pressing. This balance is then deducted automatically from each order. The pressing receives cash upfront — the client gets a loyalty discount in return. Very popular in Chinese pressing chains — a powerful tool for both cash flow and client retention.

Feature	Description
Account top-up at counter	Client pays a top-up amount at the counter: 20,000 XOF, 50,000 XOF, 100,000 XOF — any amount. Pa
Top-up bonus	Manager configures a bonus for each top-up bracket: e.g. +5% bonus on any top-up of 50,000 XOF or m
Automatic deduction at payment	When the client pays for an order, CleanHub deducts from the prepaid balance automatically — cashier
Balance always visible	The client's prepaid balance is shown prominently on their profile — the cashier sees it at every visit. Cli
Low balance alert	When the prepaid balance drops below a configurable threshold (e.g. less than 5,000 XOF remaining), C
Top-up history	Full history of all top-ups and deductions on the prepaid account — date, amount, order reference — vis
Non-expiring balance	Prepaid balances do not expire (configurable) — the client's money is safe. If the pressing closes, unuse
Prepaid revenue report	Total prepaid balances held across all clients — represents future guaranteed revenue. Report shows: to

■ *The prepaid account is one of the most powerful loyalty and cash flow tools available. A client with 30,000 XOF prepaid will ALWAYS return to the pressing — their money is already there. For the pressing, this is guaranteed future revenue received today.*

■ **Prepaid balances are the client's money — the pressing must not lose or erase them accidentally. The developer must implement prepaid accounts as a separate ledger with full transactional integrity — every deduction must be atomic with the order payment. ONLY THE OWNER can manually adjust a prepaid balance.**

3.6.14 Automatic Conveyor — Numbered Rack System

For pressings with a motorised conveyor (carousel rack), CleanHub manages the numbering, assignment and retrieval of garments. When the client arrives, the cashier presses one button and the conveyor rotates to deliver the correct garment automatically.

Feature	Description
Conveyor slot assignment	When an order is stored on the conveyor, staff assign a slot number — CleanHub records it: order N202

Feature	Description
Retrieval in one press	Cashier finds the order by client name/phone — clicks 'Retrieve' — system displays the slot number and
Slot availability map	Real-time map of all conveyor slots: occupied (green) / empty (white) / reserved (orange) — manager se
Multi-row conveyor	Support for conveyors with multiple rows or sections — e.g. Row A: shirts, Row B: suits, Row C: heavy i
Garment arrival on conveyor	When production marks an order as 'Ready and stored', staff enter the slot number — CleanHub records
Slot freed on collection	When client collects their garment, the slot is automatically freed — available for the next order
Slot report	Daily report: conveyor occupancy rate, average time a garment spends on the conveyor before collection

■ The conveyor integration requires a serial (RS232/RS485) or HTTP API connection to the conveyor controller.

The specific protocol will be defined when the conveyor brand is selected. CleanHub will support the most common Chinese conveyor brands available on the African market.

3.6.15 Shoe Washing Module

Many pressing businesses also wash shoes — sneakers, leather shoes, suede, luxury. This is a separate service with its own pricing, processing time, storage and workflow. CleanHub has a dedicated shoe module with all the features of the garment pressing module adapted for shoes.

Feature	Description
Shoe types & pricing	Sneakers / Leather shoes / Suede / Boots / Sandals / Luxury (Louboutin, Gucci...) / Sports shoes — each
Condition at intake	At intake: agent records shoe condition (scratches, sole damage, colour issues), takes photo. Same def
Processing time per type	Sneakers (24h), Leather (48h), Suede (72h), Luxury (5 days) — pickup date calculated automatically pe
Label per pair	One matrix label per pair — attached to both shoes with a tie tag — waterproof — contains order numbe
Storage location	Shoes stored in numbered shelves — same storage system as garments — location recorded in CleanH
Mixed order	Client drops off 2 pairs of shoes + 3 shirts in the same visit — one combined order, one receipt, one pick
Packaging at collection	Staff can note packaging type at collection: bag, box, wrapping — visible on the order for production sta
Shoe report	Revenue per shoe type, number of pairs processed per day/week/month, most common shoe type, aver

- The shoe module uses the same order workflow as garments — no separate training required for staff
- Luxury shoe pricing can be set per individual model if needed — e.g. 'Louboutin' at a specific price separate from the general leather shoe rate

3.6.16 Fabric Care Code — Wash Instructions Per Garment

At intake, the cashier can record the CARE LABEL CODE of the garment — the international symbols indicating wash temperature, drying method, ironing level, dry-clean only. This information is stored in CleanHub and displayed to production staff when they process the garment — preventing costly mistakes on delicate or expensive items.

Feature	Description
Care code entry at intake	Cashier reads the garment's care label and selects the appropriate symbols from a visual icon-based
Predefined care icons	Standard international laundry care symbols available in CleanHub: Wash 30°C / 40°C / 60°C / Hand
Displayed on Kanban	The care code icons are displayed on the garment's Kanban card on the production board — produc
Printed on garment label	Care code icons printed on the matrix label — visible throughout the production process
Alert on incompatible service	If a garment with 'Dry-clean only' is assigned a 'Machine wash' service, CleanHub displays a warning

Feature	Description
Client note	Manager can add a text note alongside the care code: 'Client's instructions: cold water only, no tumble dry'

■ *The fabric care code module is a key quality and trust feature. When a client drops off an expensive suit or a silk dress, recording the care label at intake shows professionalism and protects the pressing against claims of damage caused by incorrect treatment.*

3.6.17 Daily Z Report — End-of-Day Cash Closing

At the end of each business day (or each shift), the cashier or manager runs the Z Report — the formal daily cash closing. This is the definitive record of all financial activity for the day. It cannot be modified after generation. Standard in every serious pressing in China and internationally.

Z Report section	Content
Opening balance	Cash in the drawer at start of day (or shift)
Receipts by payment method	Total received: Cash / Wave / Orange Money / Card / Prepaid balance / Gift card — each method shown
Total discounts granted	Sum of all discounts applied during the day — by cashier
Total cancellations	Orders cancelled and amounts voided during the day — with reasons
Total refunds	Amounts refunded to clients during the day — with order references
Cash drawer openings	Number and times of cash drawer openings — each with staff name
Closing balance	Cash expected in the drawer = Opening + Cash received – Cash refunds. Cashier counts physical cash
Net revenue	Total revenue after discounts and refunds — by service type and by payment method
Pending orders	Number of orders created today not yet paid — total outstanding amount
Signature	Cashier signs the Z Report (digital signature or PIN confirmation) — Manager can countersign

- Z Report generated automatically by CleanHub at closing time (configurable — e.g. every day at 9pm) OR manually triggered by the manager at any time
- Once the Z Report is generated, it is LOCKED — no one can modify the day's transactions after the Z Report is run — not even the Owner
- Z Report stored permanently in CleanHub — accessible at any time — the manager can view the Z Report from any previous day
- Z Report sent automatically by WhatsApp/email to the Owner at closing time — Owner sees the day's results without being on site
- Multi-shift support: if the pressing operates two shifts, a Z Report can be run at each shift change — the second report covers only the second shift

■ **The Z Report is a financial audit tool — its integrity is NON-NEGOTIABLE. The developer must ensure that once the Z Report is generated, no retroactive modification of transactions from that day is possible without a special Owner override logged in the audit trail.**

3.6.18 Dormant Client Reactivation — Clients Who Have Not Returned

CleanHub automatically identifies clients who have not placed an order in X days and sends them a personalised reactivation WhatsApp. This is one of the highest-return features in Chinese pressing software — a client who received one message and returned is pure revenue from the existing base with zero advertising cost.

Feature	Description
Dormancy threshold	Manager configures the number of inactive days that triggers a reactivation message — e.g. 30 days

Feature	Description
Automatic daily scan	Every day at 8am, CleanHub identifies all clients whose last order date exceeds the dormancy threshold
Personalised WhatsApp	Message sent automatically: 'Hello [First name], we miss you at [Pressing name]! It has been [X] days since your last order. We have a special offer for you!' — CleanHub new
Reactivation offer	Manager configures a special offer to attach to the reactivation message: fixed discount / bonus points
One message only	Each client receives a maximum of one reactivation message per dormancy period — CleanHub new
Opt-out	Client can opt out of reactivation messages — agent marks this preference in their profile
Reactivation report	Monthly report: number of messages sent, number of clients who returned after a message, revenue generated
Segment by value	Manager can choose to send reactivation only to high-value dormant clients: e.g. only clients who have

■ *Real-world impact: if a pressing has 500 dormant clients and 20% return after a reactivation message, that is 100 additional orders — at an average of 8,000 XOF per order = 800,000 XOF in revenue that would have been lost without this feature.*

3.6.19 Flash Promotions — Time-Limited Offers with Countdown

The manager can launch a FLASH PROMOTION — a special price valid for a limited time only (e.g. 30% off suits this weekend only). A countdown timer is displayed on the customer-facing screen. The promotion activates and expires automatically — no manual action required.

Feature	Description
Create a flash promotion	Manager selects: articles concerned (all / specific family / specific articles), discount type (% or fixed amount)
Countdown on customer screen	While the flash promotion is active, the customer-facing screen shows a prominent countdown: 'WEEKEND SPECIAL: 30% OFF - 2 hours left'
Automatic activation at POS	When the promotion period is active, CleanHub automatically applies the flash price on all qualifying items
WhatsApp campaign blast	When a flash promotion is created, the manager can send a WhatsApp announcement to all clients (opt-in only)
Flash offer on receipt	The receipt shows: original price struck through + flash promotion price + savings amount — full transparency
Automatic expiry	At the configured end time, the promotion stops automatically — prices return to normal — no manual action required
Flash promotion report	For each completed flash promotion: number of orders generated during the period vs a normal period

- Flash promotions are the most effective tool for filling slow periods — 'Dead Tuesday afternoon? Launch a 2-hour flash offer and fill the workshop'
- The countdown timer creates urgency — it is the most powerful visual element on the customer-facing screen during a promotion

3.6.20 Consolidated Invoice — Multi-Branch Client in the Same Group

For corporate clients or hotels that use multiple branches of the same pressing group, CleanHub generates a SINGLE consolidated monthly invoice covering all orders across all branches — one PDF, one payment, one account.

Feature	Description
Group detection	When a corporate client (e.g. Hotel Teranga) visits any branch of the pressing group, their profile is marked as 'Corporate'
Consolidated monthly invoice	At the end of the month, CleanHub generates one PDF invoice for the corporate client covering ALL branches
Invoice breakdown	The consolidated invoice shows: subtotal per branch, subtotal per service type, subtotal per week — full transparency
Single payment	The corporate client makes one payment for the entire consolidated invoice — recorded in CleanHub
Branch revenue attribution	Internally, the consolidated payment is automatically split and attributed to each branch proportionally

Feature	Description
Cross-branch reporting	Owner sees consolidated corporate client revenue across all branches — total revenue from Hotel T

3.6.21 Automatic Reminder for Uncollected Orders

If a client does not collect their ready order after a configurable number of days, CleanHub sends automatic reminders — escalating in urgency. This reduces the number of garments in reserve and improves cash collection.

Day	Trigger	Message	Channel
Day of notification	Order marked Ready	Your order N[XXX] is ready. Please come and collect it.'	WhatsApp + SMS
D+3	Order still not collected	Reminder: your order N[XXX] is ready and waiting for you. Amount N[XXX] XOF.'	WhatsApp
D+7	Still not collected	Your order N[XXX] has been ready for 7 days. Please collect it soon or it will be placed in reserve. Amount N[XXX] XOF limited.'	WhatsApp + SMS
D+15	Approaching reserve	Important: your order N[XXX] will be placed in reserve in [X] days if not collected. Storage fees may apply after it is placed in reserve. Amount N[XXX] XOF.'	WhatsApp + SMS
D+30	Placed in reserve	Your order N[XXX] has been placed in reserve. Contact us to arrange collection or refer to reserve ID.'	WhatsApp + SMS

- All reminder intervals are fully configurable by the manager — D+3, D+7, D+15 are examples that can be changed to match the pressing's policy
- Each reminder is logged in the order file — the manager can see when each reminder was sent and whether it was delivered
- The manager can disable individual reminders or the entire sequence for specific clients (e.g. a VIP client who should be called personally, not messaged automatically)
- Reminder report: orders collected within 24h of notification / 3 days / 7 days — helps optimise the timing of the reminders

■ **Reminders respect the pressing's configured business hours — no message is sent before opening time or after closing time, even if the automation triggers at night.**

3.6.22 Unified Client Profile — Visible Across All Branches, Orders Belong to the Depositing Branch Only

When a pressing group has multiple branches, a client registered at one branch is visible at all other branches of the same group — BUT their orders and their balance BELONG EXCLUSIVELY to the branch where they were deposited. The cashier at Branch B can identify the client and create a new order — but cannot see, collect or process orders from Branch A.

The fundamental rule — what is shared and what is NOT:

Data	Shared across all branches	Stays at the depositing branch ONLY
Client identity	■ Name, phone, category, loyalty points, photo	—
New order creation	■ Any branch can create a new order for this client	—
Loyalty points	■ Points earned anywhere — redeemed anywhere	—
Category pricing	■ VIP at Branch A = VIP at Branch B	—
Orders in progress	■ NOT visible at other branches	■ Visible only at the branch that created them
Outstanding balance per order	■ NOT visible at other branches	■ Only the depositing branch can collect payment for its own orders
Payment collection	■ Branch B CANNOT collect payment for a Branch A order	■ Only Branch A cashiers can collect for Branch A orders
Order history	■ Other branches cannot see past orders	■ Only at the branch where orders were placed

Data	Shared across all branches	Stays at the depositing branch ONLY
Financial reports	■ Branch managers never see other branches' revenue	■ Each branch sees only its own revenue

What the cashier sees when a client walks in from another branch:

Scenario	What Branch B cashier sees	What Branch B cashier CANNOT see or do
Client walks into Branch B — deposits clothes, category (VIP), loyalty points balance, Branch A order history	Client name, category (VIP), loyalty points balance, Branch A order history	Branch A revenue, Branch A balance, Branch A orders
Client wants to pay for a Branch A order displayed	This client has orders at Branch A	Cannot collect payment for a branch order details
Client creates a new order at Branch B	New order created at Branch B — linked to the client's shared profile	Nothing shared. Profile is processed by Branch B staff

- This strict separation PROTECTS EVERYONE: the client cannot pay at the wrong branch by mistake, Branch B cashiers cannot accidentally access Branch A revenue, and the Owner's branch accounts remain perfectly clean and separated
- Loyalty points are the ONLY financial element shared — because they belong to the client personally, not to any specific branch. A client earns points at Branch A and spends them at Branch B — normal behaviour
- If a corporate client has a monthly invoice, it covers orders from ALL branches — but it is issued and managed by the Owner, not by individual branch managers

■ Example: DIALLO Mamadou deposited 3 suits at Branch Almadies yesterday. Today he walks into Branch Plateau. The cashier types his phone number — his name appears: 'VIP client — 1,250 points'. The cashier can create a new order for today's visit. But the 3 suits at Almadies are invisible — Branch Plateau cannot touch them, collect for them or deliver them. DIALLO must go to Branch Almadies to collect his suits.

■ DEVELOPER REQUIREMENT: the order visibility rule must be enforced at the DATABASE LEVEL — not just at the UI. A branch cashier's API session must be scoped to their branch only. Even if a cashier attempts to access another branch's orders via a direct API call, the server must block the response. Row-level security on the orders table must enforce: WHERE branch_id = current_user_branch_id for all order reads, writes and payment operations.

A dedicated statistics module providing deep insights into customer behaviour, order patterns and business performance. All statistics filterable by date range, branch, staff member and customer category.

1. Customer History & Profile Statistics:

Statistic	Description
Full order history	Every order ever placed by a customer — date, articles, services, amount paid, staff who served the order
First & last visit	Date of first order and most recent order — shows customer loyalty duration
Total spent (lifetime)	Cumulative total amount spent by each customer since account creation
Visit frequency	Average number of days between visits — identifies regular vs occasional customers
Preferred services	Most ordered services per customer — useful for personalised promotions
Payment method preference	How each customer usually pays (Wave, cash, Orange Money, card)
Outstanding balance history	Full credit/debt history per customer with all payments recorded

2. Average Basket:

Metric	Description
Average basket — global	Average order value across all customers for the selected period
Average basket — by category	Average basket per customer category: Standard vs VIP vs Corporate vs Hotel

Metric	Description
Average basket — by staff	Average basket per cashier — identifies best-selling staff
Average basket trend	Month-by-month evolution of average basket — N vs N-1 comparison

3. Customer Rotation Rate:

Metric	Description
Active customers	Customers who visited at least once in the selected period (e.g. last 30 days)
Dormant customers	Customers with no visit in 30, 60 or 90+ days — configurable threshold — automatic WhatsApp win-back
New customer rate	% of orders from new (first-time) customers vs returning customers per period
Retention rate	% of customers who return within 30 / 60 / 90 days after their first visit
Churn rate	% of customers lost compared to previous period — month vs month
Top 10 customers	Ranking of top 10 customers by revenue, by number of orders and by loyalty points
Lost clients report	Monthly automatic report: list of all clients who have not visited since X days, sorted by value (best to worst)

3.6.23 Alterations & Tailoring Module

Many pressing businesses also offer clothing alterations and tailoring: hem shortening, zip replacement, button replacement, seam repair, waist adjustment, sleeve shortening, patching. CleanHub has a dedicated alterations module with its own workflow, separate from the pressing service.

Feature	Description
Alteration types	Manager configures the full list of alteration services offered: Hem shortening / Zip replacement / Button replacement / etc.
Alteration on a specific garment	Each alteration is linked to a specific garment in the order — e.g. 'Trousers: hem shortening 5cm + waist adjustment 2cm'
Measurements recorded	Agent records the exact measurements needed for the alteration: e.g. 'New hem length: 96cm', 'Waist measurement: 85cm'
Client measurement profile	Returning clients have their measurements saved in their profile — pre-filled automatically on the next alteration
Production label for alterations	A specific alteration label is printed for the tailor/seamstress: garment type, alteration type, measurements, etc.
Alteration workflow	Separate Kanban columns for alterations: Received → Cutting → Sewing → Finishing → Ready — tailor/seamstress manages
Alteration + pressing combined	Client can drop off a suit for dry-cleaning AND ask for hem shortening in the same visit — one combined order
Alteration pricing	Per alteration type: fixed price (e.g. Zip replacement: 2,500 XOF) or variable price set by the tailor based on measurements
Express alteration	Configurable express surcharge for same-day alterations — e.g. +50% for same-day hem shortening
Alteration report	Revenue per alteration type, number of alterations per week/month, most requested alteration, average turnaround time

- Alterations are managed identically to pressing garments — same labels, same storage, same pickup appointments, same partial delivery rules — no additional training required for staff

- If a garment needs both pressing AND alteration, the pressing step and the alteration step are tracked separately on the Kanban — the order is only marked Ready when BOTH steps are complete

■ *The alterations module is particularly relevant for markets like Senegal and Côte d'Ivoire where many pressing businesses also employ a tailor or seamstress. Having alterations fully integrated into CleanHub eliminates the need for a separate tracking system for tailoring work.*

3.6.24 Marketing Campaigns — Automated WhatsApp, SMS & Email

CleanHub has a dedicated marketing module for sending targeted campaigns to clients — promotional offers, seasonal messages, win-back campaigns, VIP rewards. The manager creates the campaign once, defines the target audience, and CleanHub sends automatically at the configured time.

Campaign type	Description	Example
Promotional campaign	Discount or offer sent to a selected client group — e.g. Special offer for all clients who have visited within 30 days	'Special offer! This week 20% off your visit! Come in today and enjoy your favourite dishes at only 80% of the usual price! Buy this
Seasonal campaign	Sent to all active clients on key dates: Tabaski, Korite, end-of-school-year	'Tabaski Mubarak! Prepare your delicious food for your family and friends. Enjoy the festive season with your favourite dishes at only 80% of the usual price. Buy this
VIP reward campaign	Exclusive offer sent only to VIP clients	'As a valued VIP client, you benefit from a private offer this week. Enjoy your favourite dishes at only 80% of the usual price. Buy this
Win-back campaign	Sent automatically to dormant clients — combined with a 30-day trial period	'We miss you! It has been 30 days since your last visit. Come back and enjoy your favourite dishes at only 80% of the usual price. Buy this
New service announcement	Inform all clients of a new service: shoe washing, car detailing, etc.	'CleanHub is now offering shoe washing! Bring your sneakers to the shop and enjoy a professional cleaning service. Buy this
Birthday campaign	Sent automatically on each client's birthday (see 3.6.25 Google Reviews & Client Satisfaction — Automatic Review Requests)	'Happy birthday! A gift is waiting for you at [Pressing Teranga]. Buy this
Loyalty milestone	Sent automatically when client reaches a loyalty point threshold	'Congratulations! You have reached 2,000 points — equivalent to a 20% discount on your next visit. Buy this

Campaign management tools:

Tool	Description
Target audience selector	Manager selects who receives the campaign: All clients / Specific category (VIP only) / Clients who have visited within 30 days
Channel selection	Send by: WhatsApp (primary) / SMS (fallback or additional) / Email (for clients who have an email address)
Scheduled sending	Schedule the campaign for a specific date and time — e.g. send Tabaski message on the Monday before Tabaski
Campaign preview	Manager sees a preview of the message exactly as it will appear on WhatsApp/SMS before sending
Personalised variables	Messages automatically include client name, loyalty points balance, last visit date, custom offer code
Send limit per day	Configurable maximum messages per day to avoid WhatsApp API rate limits — large campaigns automatically split into multiple days
Unsubscribe management	Clients who reply STOP are automatically removed from future campaigns — their preference saved
Campaign report	For each completed campaign: number sent, delivery rate (%), clients who placed an order within 7 days

- All campaign messages are personalised with the client's name — never generic bulk messages. A personalised WhatsApp from 'Pressing Teranga' with the client's first name has a 5x higher open rate than a generic SMS
 - The manager can save message templates for reuse — create once, reuse every Tabaski, every Korite, every end-of-school-year campaign
 - Campaign history: all past campaigns visible with their performance metrics — the manager can see which type of campaign generates the most return visits
- **WhatsApp Business API has daily and monthly sending limits depending on the plan level. The developer must implement proper rate limiting and queue management so that large campaigns (1,000+ clients) are sent progressively without triggering API blocks. The manager sees a progress indicator when a large campaign is being sent.**

3.6.25 Google Reviews & Client Satisfaction — Automatic Review Requests

After each completed order, CleanHub can automatically send the client a link to leave a Google review or a satisfaction rating. This builds the pressing's online reputation without any manual effort — a key competitive advantage in markets where clients search for pressing businesses on Google Maps.

Feature	Description
Automatic review request	When an order is marked as Delivered, CleanHub sends an automatic WhatsApp/SMS to the client: 'Thank you for your order! Please leave a Google review or a satisfaction rating. Click here: [link]'

Feature	Description
Direct Google Maps link	The message includes a direct link to the pressing's Google Maps page — one tap and the client is on the map
Configurable timing	Manager configures when the review request is sent: immediately at delivery / 2 hours after delivery
Send only to satisfied clients	Optional smart filter: the review request is only sent to clients who gave a high satisfaction score on the in-app rating
In-app satisfaction rating	Before the Google review link, CleanHub shows a quick 5-star rating screen on the customer-facing app
Review tracking	Manager sees how many review requests were sent this month and can monitor the evolution of the pressing's Google Maps rating
Custom message	Owner can customise the review request message from Settings — add the Google Maps link, adjust the tone of the message
Multi-platform support	Google Maps is the primary platform. The developer can also support Facebook page reviews as a secondary platform
Frequency limit	A client who already left a review in the last 90 days does not receive another review request — avoids spamming

■ *Real-world impact: a pressing with 50 orders per week that converts 20% of clients into Google reviews = 10 new reviews per week = 520 new reviews per year. A pressing with 500+ positive Google reviews ranks at the top of Google Maps searches for 'pressing' in their city — free organic marketing worth tens of thousands of XOF in advertising.*

- The in-app satisfaction rating (before the Google review link) acts as a filter — unhappy clients are directed to the complaints module, happy clients are directed to Google — protects the pressing's online reputation
- This feature is particularly valuable in Dakar and Abidjan where clients increasingly search for businesses on Google Maps before visiting — a pressing with 4.8 stars and 200 reviews wins over a pressing with no online presence

■ **The developer must integrate the Google My Business API or use a direct review link (shareable review URL from Google Maps). The direct review link approach requires no API key — simpler to implement. The manager enters their Google Maps business profile URL in Settings and CleanHub appends the review action parameter automatically.**

- Revenue reports: daily, weekly, monthly, by branch
- Order volume reports: total orders, by service type, by staff
- Customer reports: top customers, retention, new vs returning
- Inventory and expense reports
- Export reports to PDF and Excel

Automated Statistics & Owner Reports:

Report	Content	Delivery
Daily Cash Register (Z Report)	Total revenue by payment method, cash expected vs actual, discrepancy by branch, by time of day, by cashier	Auto-dispatched every morning to the owner's email
Weekly Performance Report	Revenue vs previous week, top services, top customers, staff performance, order volume trends	Auto-dispatched every Monday morning
Monthly Summary	Full month revenue, expenses, inventory costs, net margin, top customers, staff performance	Auto-dispatched at the end of each month
Low Stock Alert	List of consumables below minimum threshold with quantities and reorder points	Instant email + dashboard alert
Overdue Orders Alert	All orders past promised pickup date with customer contact details	Daily email if any overdue orders exist
Outstanding Receivables	All customer credit accounts with balance and last payment date	Available on demand from the app — manager can filter by status
Custom Report	Owner can configure any report with custom date range and filters	Available on demand — exported to PDF or Excel

3.7.0 Evolution Reports — Deposits & Collections (N vs N-1)

Two dedicated reporting screens to allow the owner to track business evolution month by month compared to the previous year and against targets (Prog).

Report	Description
Deposits Evolution	Monthly garment intake (deposits) — N vs N-1 vs Prog comparison
Collections Evolution	Monthly payment collections — N vs N-1 vs Prog — broken down by payment method (Wave, Orange, etc.)

Element	Specification
Period column	Month by month (January to December) with date range per row
N column	Current year figures — updated in real time
N-1 column	Previous year figures — same period comparison
Prog column	Budget / target figures — configurable by owner per month
Difference column	N minus N-1 — shown in red if negative, green if positive
% column	Percentage evolution — colour coded (red/green)
Bar chart	N vs N-1 vs Prog side-by-side bars per month
PDF export	One-click export of full report with chart embedded
Real-time update	Data refreshes automatically as new orders and payments are recorded

3.7.1 Customer Satisfaction Survey — Customer-Facing Screen

After each order payment or pickup, the system automatically triggers a satisfaction survey on a customer-facing screen. The survey is short, visual and usable by anyone regardless of literacy level.

#	Question	Response Format	Icon
1	Satisfaction generale	1 a 5 etoiles — appuyer	Souriant
2	Quality of cleaning	Smiley scale (sad to happy)	Garment
3	Speed / delay respected	Smiley scale (sad to happy)	Clock
4	Staff welcome & service	Smiley scale (sad to happy)	Handshake
5	Comment (optional)	Free text keyboard input	Chat bubble
6	Would you recommend us?	Thumbs up / Thumbs down	Megaphone

- Survey triggered automatically after payment confirmation — no staff action required
- Large colourful buttons and icons — fully usable without reading or typing
- Survey completes in under 30 seconds
- Client can skip any question or skip the entire survey with one tap
- Low satisfaction alert: if a customer gives 1 or 2 stars, owner receives an instant email notification

■ **This feature gives the owner real-time visibility on customer satisfaction without any manual process. Low ratings trigger instant alerts. All data feeds into the analytics dashboard.**

3.7.2 Optional Add-On Modules (Paid)

Module	Description	Starter	Pro / Business / Enterprise
TPE Integration	Connect physical payment terminals (PAX/SUNMI) — Wave, Orange, Money etc. card auto-synced	Yes	Yes

Module	Description	Starter	Pro / Business / Enterprise
Customer Satisfaction Survey	Customer-facing screen with post-payment satisfaction survey	Not available	Available (one-time)
Delivery & GPS Tracking	Live GPS tracking of delivery staff, route optimisation, client delivery notifications	Not available	Available (one-time)
Advanced Multi-Branch	Consolidated cross-branch reports, inter-branch stock transfers	Not available	Available (one-time)
Advanced BI & Analytics	Custom KPI dashboards, industry benchmarks, predictive demand forecasting	Not available	Available (one-time)
WhatsApp Business Advanced	Automated promotional campaigns, customer segmentation, WhatsApp (App) messaging	Not available	Available (one-time)
Portable Bluetooth Printer	Mobile printing support for delivery staff	No	Available (one-time)
Machine Maintenance Manager	Register machines, schedule maintenance, log downtime, dashboard	Not available	Available (one-time)
B2B Corporate Invoicing	Monthly auto-invoicing for hotel/corporate clients, VAT breakdown	Not available	Available (one-time)
Loyalty Card System	Physical barcode loyalty cards, points accumulation, redemption	Not available	Available (one-time)
Employee HR Module	Complete staff management: contracts, attendance, salaries, performance	Not available	Available (one-time)
AI Module	Stain and defect detection by AI, demand forecasting, dormancy detection	Not available	Available (one-time)

- All modules are activated instantly from the admin panel upon purchase — no developer intervention required
- Module purchase is a one-time fixed fee — no recurring monthly charge
- The Starter plan is strictly locked — no module can be added regardless of payment

3.7.2b CleanHub — Car Wash Mode

CleanHub is CleanHub version of CleanHub — same software platform, same subscription plans (Starter, Pro, Business, Enterprise), different branding. car wash businesses subscribe to CleanHub as a standalone product. It is NOT an optional add-on to CleanHub — it is a full product in its own right.

- A business that operates BOTH a pressing AND a car wash activates both modes under a single CleanHub subscription — configured at installation. Pricing for combined mode to be defined by the owner.

CleanHub Car Wash — Three Sub-Modules:

Sub-module	Workflow	Like pressing?
Vehicle Wash (on-site)	Client waits on site — waiting queue — estimated ready time — no appointment	Not quite the same workflow
Vehicle Pickup & Delivery	Staff collect the vehicle at the client's location, wash it and return it	Similar to the pressing workflow
Carpet & Mat Wash	Drop-off and collect later — matrix label — pickup appointment — storage location	Very different to pressing

Sous-module 1 — Lavage Vehicule (Service Immédiat — File d'Attente)

The client arrives with their vehicle, registers at the counter and waits on site while the vehicle is washed. There is NO appointment — the vehicle joins a waiting queue.

Feature	Description
Vehicle registration	Agent registers the vehicle at intake: licence plate, make, model, colour, type of wash (quick / full / intensive)
Vehicle file	Each vehicle has its own file in the system — full history of all previous washes accessible by licence plate
Waiting queue	Vehicle automatically takes a number in the queue: Q-01, Q-02, Q-03... — real-time dashboard shows queue status
Parking spot assignment	Each vehicle is assigned a numbered parking spot (P-01, P-02...) — agent knows exactly where each vehicle is
Vehicle intake photo	Agent takes a photo of the vehicle at intake — documents all pre-existing scratches, dents and damage

Feature	Description
Photo rules — configurable by manager	Manager can set photo rules from the Settings panel: (1) ALWAYS mandatory — photo required
How the agent takes the photo	The agent uses the webcam (on PC) or the tablet/smartphone camera (on Android terminal). The photo is taken at registration.
Photo sent to client	Optional but recommended: after intake, the photo(s) are automatically sent to the client by WhatsApp.
Photos stored permanently	All intake photos are stored permanently in the vehicle file — accessible even after the order is closed.
Pre-existing damage record	Staff record all visible damage before washing: scratch location, dent, broken mirror — signed by client.
Fuel level at intake	Agent records the FUEL LEVEL when the vehicle arrives — selected from a visual gauge on screen.
Releve des objets du client	L'agent enregistre tous les objets de valeur ou articles laissés dans le véhicule par le client à la réception.
Belongings signed by client	The full list of declared belongings is printed on the client receipt — client signature confirms what was declared.
No belongings declared — disclaimer	If client declares no items, the receipt shows: 'No valuables declared by client at intake — the car was washed as is.'
Estimated ready time	System calculates and displays the EXACT ESTIMATED READY TIME immediately at registration — based on current queue.
Receipt with ready time	A receipt is printed at intake showing: order number, vehicle details, wash type, ESTIMATED READY TIME.
WhatsApp at registration	Client receives WhatsApp/SMS immediately at registration: 'Your vehicle [plate] is registered. Estimated ready time: 14h30.'
Delay update	If a delay occurs, system recalculates and sends a new WhatsApp/SMS with updated ready time.
Real-time status board	Large dashboard showing all vehicles in the centre: waiting (red) / in progress (orange) / ready (green).
Ready notification	When vehicle is ready, client automatically receives WhatsApp/SMS: 'Your vehicle [plate] is ready. A staff member will assist you.'
Payment at collection	Client returns, pays at the counter — Wave, Orange Money, cash or card — receipt printed and sent via WhatsApp.
Queue history	Full history of all vehicles washed: date, wash type, amount, staff who performed the service.
Daily queue report	End of day: total vehicles washed, revenue by wash type, average time per wash, peak hours.

Real-time waiting queue dashboard:

Queue No.	Plate	Make/Model	Colour	Wash Type	Status	Est. Ready	Parking
Q-01	SN 1234 AB	Toyota Corolla	Blanche	Lavage complet	En cours	14h30	P-03
Q-02	SN 5678 CD	Peugeot 308	Noire	Lavage rapide	Pret — client appelle	—	P-07
Q-03	SN 9012 EF	Mercedes C200	Argent	Detailing	Pending	15h45	P-01
Q-04	SN 3456 GH	Renault Logan	Rouge	Nettoyage interieur	Pending	15h15	P-05
Q-05	SN 7890 IJ	Hyundai i20	Bleue	Lavage rapide	Pending	14h50	P-02

Vehicle Intake Record — Fuel Level & Client Belongings:

At every vehicle intake (whether on-site or pickup at client location), the agent must record FIVE things that protect the car wash legally: pre-existing damage, fuel level, client belongings, digital signature and any special service warnings. All are signed by the client on the customer-facing screen and printed on the receipt.

What is recorded	How	Why it matters
Pre-existing damage	Staff select damage locations on a vehicle diagram (dents, scratches, etc.)	Protects car wash from liability if damage occurs during wash. If damage is not recorded, the client could claim the damage was caused by the wash.
Fuel level	Agent selects from a visual gauge: Empty / 1/4 / 1/2 / 3/4 / Full	Full was observed visually by the wash operator at intake. If fuel is not recorded, the client could claim the wash was not completed.
Client belongings in vehicle	Agent lists all items left by the client: money (cash/cards), phone, keys, etc.	Protects car wash from liability if items are lost or damaged. If belongings are not recorded, the client could claim the items were lost or damaged by the wash.

What is recorded	How	Why it matters
Digital signature	Client signs directly on the touchscreen (tablet or customer-facing screen), rather than a physical signature	The digital signature is saved as an image file linked to the order — stored permanently in the vehicle file
Engine / electronics warning (if requested)	Engine cleaning or washing of electronic parts	Specific electronic parts protection disclaimer is displayed and must be signed separately

Signature Numerique — Comment ça Fonctionne :

Step	Description
1. Intake form completed	Agent fills in all intake information: damage, fuel level, belongings, special services requested
2. Summary displayed	The customer-facing screen shows the complete intake summary for the client to review: vehicle details, damage notes
3. Client signs on screen	Client signs directly with their finger on the touchscreen — a clear signature box is displayed on the customer-facing screen
4. Signature saved	The digital signature is saved as an image file linked to the order — stored permanently in the vehicle file
5. Receipt printed	The signature receipt includes a small image of the client's digital signature — legally valid proof
6. WhatsApp confirmation	Client receives a WhatsApp message with the intake summary — 'You have confirmed the following at intake: [details]

■ If the customer-facing screen is not a touchscreen, the agent can sign on behalf of the client using a signature pad (optional hardware) or the client can sign on the agent's tablet. In all cases, a signature is required before any service begins.

Nettoyage Moteur & Pieces Electroniques — Decharge Speciale & Signature :

Washing a vehicle engine or electronic components (sensors, control units, wiring, cameras, speakers) is a HIGH-RISK service. Water or cleaning products can damage these parts even with care. If a client requests this service, CleanHub must display a mandatory disclaimer that the client must read and sign separately BEFORE the service is performed.

Scenario	What the system does
Client requests engine cleaning	System displays a full-screen warning: 'ENGINE CLEANING IS A HIGH-RISK SERVICE. Water or cleaning products can damage engine components.'
Client requests washing of electronic parts	System displays disclaimer adapted: 'Washing electronic components (sensors, cameras, speakers, control units) is a high-risk service. Water or cleaning products can damage these parts.'
Client signs the disclaimer	Client signs on the customer-facing touchscreen — digital signature saved separately from the main intake signature
Disclaimer printed on receipt	The disclaimer text and client signature are printed on the receipt — clearly separated from standard intake information
Service can only begin after signature	The system BLOCKS the service from starting until the disclaimer has been digitally signed — agent cannot proceed
Disclaimer stored permanently	The signed disclaimer is stored permanently in the vehicle file — even after the order is closed — accessible for legal evidence

- The engine/electronics disclaimer is a separate signature from the main intake signature — the client signs TWICE: once for the general intake (damage, fuel, belongings) and once specifically for the high-risk service
- Manager configures which services trigger the disclaimer from the Settings panel — e.g. Engine cleaning, Underbody wash, Interior steam cleaning, Electronic component cleaning
- If the client refuses to sign the disclaimer, the service is automatically blocked — agent cannot proceed — the order is saved without that specific service
- The disclaimer can be customised by the owner from the Settings panel — add the business name, adjust the text, add specific exclusions

■ This double signature system (general intake + specific service disclaimer) is the car wash centre's complete legal protection. In case of any dispute about engine damage or missing items, the signed receipts stored in CleanHub are the definitive legal evidence.

What is printed on the client receipt at intake:

Section	Content on receipt
Vehicle	Plate: SN 1234 AB Make: Toyota Corolla Colour: White Wash type: Full wash + Engine cleaning
Estimated ready time	Ready at: 2:55pm
Fuel level	Fuel level at intake: 1/2 tank
Dommages preexistants	Pare-choc avant : petite rayure Portiere droite : bosse legere
Belongings declared	1x mobile phone (Samsung), 1x pair of sunglasses, 500 XOF coins
General intake signature	[Client digital signature image] — Date/time: 2:10pm — Server timestamp
Engine cleaning disclaimer	ENGINE CLEANING AUTHORISED AT CLIENT'S OWN RISK — Client has read and accepted the
Disclaimer signature	[Client digital signature image] — specific to engine cleaning service
Amount to pay	5,500 XOF (3,000 full wash + 2,500 engine cleaning) — payable at collection

■ Recording fuel level, client belongings and obtaining digital signatures is MANDATORY at every vehicle intake. The agent cannot proceed without completing the full intake form including all signatures. This is CleanHub's primary legal protection for car wash businesses.

The client calls or sends a WhatsApp to the car wash centre. A staff member goes to the client's location, collects the vehicle, washes it at the centre and then returns it. This is a premium service — the client never needs to come to the centre.

Step	Action	System Response
1. Client request	Client calls or WhatsApps to request vehicle collection	Agent creates a PickUp Order. Client type name, phone number, vehicle plate
2. Pickup scheduled	Agent assigns the job to a delivery staff member	WhatsApp/SMS sent to client: pickup time vehicle pickup is scheduled for [time]. Our
3. Staff departs	Delivery staff member sees the pickup job	App shows CleanHub mobile address, vehicle plate address type, special instruction
4. Staff arrives	Staff member arrives, inspects the vehicle	Takes intake photo and records any pre-existing damage on delivery staff's smartphone
5. Vehicle collected	Staff member drives the vehicle to the car wash	Vehicle status: 'Collected — in transit to centre'
6. Vehicle washed	Vehicle enters the normal washing queue	Vehicle joins the queue — staff at centre see it on the dashboard
7. Wash complete	Vehicle is washed and ready	Status updated to 'Ready for return delivery'
8. Staff departs	Delivery staff member assigned for return	WhatsApp/SMS sent to client: 'Your vehicle is clean and ready. Our staff will r
9. Vehicle returned	Staff delivers the vehicle to the client's location	Client inspects — confirms delivery on staff's smartphone — order closed
10. Payment	Client pays on delivery — Wave, Orange	Receipt sent by WhatsApp payment fully closed

Vehicle Pickup & Delivery — Key Features:

Feature	Description
Booking by phone or WhatsApp	Client does not need to come to the centre — books pickup by calling or sending a WhatsApp messa
Staff assignment	Manager assigns pickup and delivery jobs to specific staff members from the dashboard
Mobile app for delivery staff	Staff member uses CleanHub mobile app to see their assigned jobs, navigate to the client, take vehi
Real-time tracking	Manager sees on the dashboard: which staff member has which vehicle, current status (going to clie
Client notification at every step	Automatic WhatsApp/SMS sent to client at each stage: pickup confirmed, vehicle collected, wash co
Intake photo on mobile	Delivery staff takes vehicle intake photo directly from the mobile app at the client's location — same
Portable payment terminal	Delivery staff can accept Wave, Orange Money or cash payment at the client's location using a porta

Feature	Description
Estimated return time	System calculates and displays the estimated return time based on travel time + wash duration — shown in the app
Pickup & delivery zone	Manager configures the service zone (radius around the centre) — only clients within the zone can book
Pickup & delivery pricing	Manager sets a pickup/delivery fee (e.g. 1,000 XOF) added automatically to the order total — configurable by zone
History per vehicle	Every pickup and delivery recorded in the vehicle file — client can see full service history by their license plate

Pickup & Delivery Dashboard (Manager View):

Agent	Status	Vehicule	Client	Lieu	Retour Estimé
Moussa D.	Going to client	SN 1234 AB — Toyota	DIALLO M.	Almadies, Dakar	2:45pm
Ibou F.	Vehicle collected — in transit	SN 5678 CD — Peugeot	NDIAYE F.	Plateau, Dakar	3:00pm
Abdou K.	Returning to client	SN 9012 EF — Mercedes	FALL A.	Ouakam, Dakar	3:30pm
Cheikh B.	Disponible	—	—	Au centre	—

■ The vehicle pickup and delivery service uses the same portable Bluetooth printer already used for pressing delivery slips — no additional hardware required. The delivery staff prints a vehicle collection slip at the client's location confirming the vehicle details and pre-existing damage recorded.

■ The intake photo and damage record taken at the client's location are **MANDATORY** before the staff member drives away with the vehicle. This protects the car wash against any false claims of damage during transport. **ONLY THE OWNER** can modify or delete a vehicle pickup order once it has been registered.

Sub-module 3 — Carpet & Mat Wash (Drop-off and Collect Later — Matrix Label)

The client drops off their carpets, mats or car seats. They are given a PICKUP APPOINTMENT (exactly like pressing). The carpets are marked with a matrix label, treated, stored and collected later.

Step	Action	What the system does
1. Drop-off	Client drops off carpets/mats	Order created — agent measures dimensions and attaches price tag
2. Matrix label	Label printed on MATRIX PRINTER (water resistant barcode, QR code)	Label contains order number, service details, name, carpet type, dimensions, pickup date, piece count
3. Treatment	Carpets washed and dried	Staff scan label at each stage to update status
4. Storage	Carpets stored in numbered location (marked in the system)	Location recorded in the system — accessible by client name or order number
5. Pickup appointment	System assigns a pickup date	WhatsApp SMS sent to client with their pickup date and order number
6. Ready for pickup	When carpets are ready: WhatsApp SMS sent to client	Pickup appointment: [date]. Amount: [XOF]
7. Collection	Client arrives — cashier finds location	Delivery finalized — location freed — payment collected

Carpet & Rug Pricing — Per Square Metre (m²):

Carpets and rugs vary enormously in size. Charging a flat rate per item is unfair and unprofitable. CleanHub supports pricing per square metre (m²) for carpets — the agent measures the length and width at intake, CleanHub calculates the price automatically.

Feature	Description
Pricing unit: per m²	The manager configures a price per m² for each type of carpet service: Basic wash (XOF/m²), Deep wash (XOF/m²)
Dimension entry at intake	Agent enters length (m) and width (m) at intake — CleanHub calculates the area automatically: e.g. 3m x 2m = 6m²
Automatic price calculation	Price = area × rate: 6 m² × 2,000 XOF/m² = 12,000 XOF — displayed instantly — no manual calculation

Feature	Description
Dimensions on label	Carpet dimensions printed on the matrix label: '3.5m × 2.0m — 7 m ² — 14,000 XOF' — staff see the
Dimensions on receipt	Full calculation printed on customer receipt: Carpet 1: 3.5m × 2.0m = 7 m ² × 2,000 XOF/m ² = 14,000
Minimum charge	Manager can configure a minimum charge per carpet (e.g. minimum 3,000 XOF regardless of size) —
Mixed order	Client drops off 2 large carpets (billed per m ²) + 3 car mats (billed per item) — CleanHub handles bo
Category pricing per m ²	VIP clients or hotel clients can have their own per-m ² rate — individual client pricing applies to m ² -ba

■ *Example: client drops off 1 large Persian rug (4m × 3m = 12 m²) at 2,500 XOF/m² = 30,000 XOF + 2 car mats at 3,500 XOF each = 7,000 XOF. Total order: 37,000 XOF. CleanHub prints one combined receipt and one label per item.*

- The carpet wash workflow is IDENTICAL to the pressing workflow — same matrix labels, same storage system, same pickup appointments, same partial delivery and payment rules
- A client can drop off BOTH a vehicle for immediate washing AND carpets for later collection in the same visit — one combined invoice
- Carpet orders appear in the daily pickup list alongside pressing orders — staff manage everything from one screen

■ *The matrix printer used for pressing garment labels is the SAME printer used for carpet labels — no additional hardware required.*

■ **ONLY THE OWNER can reprint carpet marking labels, modify or delete a registered carpet order — same security rules as the pressing.**

CleanHub vs CleanHub — Full Comparison:

Fonctionnalite	Pressing	Lavage Moquettes/Tapis	Lavage Vehicule
Order creation	Yes	Yes	Yes
Matrix marking label	Yes — on garments	Yes — on carpets/mats	No — ticket in vehicle
Pickup appointment	Yes	Yes	No
Waiting queue	No	No	Yes
Storage location	Yes — rack/locker/conveyor	Yes — rack/zone	Yes — parking spot
Pickup date auto-assigned	Yes	Yes	No — immediate service
Ready notification (WhatsApp/SMS)	Yes	Yes	Yes
Photo dommages preexistants	Oui — etat vetement	Oui — etat moquette	Oui — rayures/bosses vehicule
Partial delivery	Yes	Yes	No — full vehicle only
Partial payment / deposit	Yes	Yes	Yes
Daily inventory list	Yes	Yes	Yes — queue board
Payment method	At collection	At collection	At collection after wash

■ **CleanHub is the ONLY software in Africa that manages pressing AND car wash (vehicle wash + carpet/mat wash) in a single application with a single subscription. One software, one subscription, one training — for pressing businesses, car wash centres, or both.**

3.7.2c Car Wash — Advanced Features

3.7.2c.1 Vehicle History by Licence Plate

Every vehicle that has ever been washed at the centre has its own permanent file in CleanHub, identified by its licence plate. The agent simply scans or types the plate at intake and the full history appears instantly — even if the client never gave their name.

Feature	Description
Full history by plate	Every wash ever performed on a vehicle: date, wash type, amount paid, agent who performed the wash
History without client name	The vehicle file exists independently of the client profile. A vehicle can have a history even if the owner never gave their name
Client linked to plate	If the client is registered, their profile is linked to all their vehicles. The agent sees: 'DIALLO Mamadou has 3 vehicles'.
Multiple vehicles per client	A client can own several vehicles — all linked to their profile. The agent selects which vehicle is being washed.
Vehicle notes	The agent can add permanent notes to a vehicle file: 'Sensitive paintwork — use pH-neutral shampoo'.
New client detection	If a plate is not in the system, CleanHub prompts the agent: 'New vehicle — would you like to register it?'.
History on receipt	The client's previous visit information is printed on the receipt: 'Last wash: 12 days ago — Full wash'.

3.7.2c.2 Automatic Recall — Vehicle Not Washed Since X Days

CleanHub automatically monitors how many days have passed since each registered vehicle was last washed. When the configured threshold is reached, CleanHub sends an automatic WhatsApp or SMS to the client to bring them back.

Feature	Description
Recall threshold	Manager configures the number of days after which a recall message is triggered — e.g. 10 days, 15 days, 30 days.
Automatic WhatsApp/SMS	When the threshold is reached, CleanHub automatically sends: 'Hello [First name], your [Toyota Corolla] is due for a wash'.
Personalised message	The message includes the client's name, vehicle plate and make — personalised, not generic. Configurable by manager.
Optional promo offer	The manager can add an automatic promotional offer to the recall message: 'Come back this week and get 10% off'.
Do not disturb rule	If the client has already booked or visited since the recall was triggered, the message is automatically suppressed.
Recall report	Monthly report: number of recall messages sent, number of clients who returned following a recall, revenue generated.
Opt-out	Clients can opt out of recall messages at any time — the agent marks this preference in the client profile.

■ *The automatic recall feature is one of the most powerful loyalty tools for a car wash. A client who visited once and forgot about the car wash will return simply because they received a personalised WhatsApp message. No advertising cost — pure revenue from the existing client base.*

3.7.2c.3 Wash Packages — Basic / Premium / VIP

The manager creates wash packages with different service levels, contents and prices. The client chooses their package at intake. Each package has a clear name, price and list of included services — no ambiguity.

Package	Example Content	Example Price
Basic Wash	Exterior hand wash + rinse + quick dry	2,000 XOF
Full Wash	Exterior wash + rinse + full dry + tyre shine	3,500 XOF
Premium Wash	Full wash + interior vacuum + dashboard wipe + windows	6,000 XOF
VIP / Detailing	Full wash + complete interior clean + wax + tyre shine + perfume	15,000 XOF
Express (15 min)	Quick exterior wash — fast turnaround	1,500 XOF
Engine Clean	Engine degreasing — requires signed disclaimer	2,500 XOF

Package management feature	Description
Fully configurable	Manager creates, edits and deletes packages at any time from the Settings panel — no developer intervention
Package content list	Each package has a detailed content list visible to the agent at intake — ensures consistent service
Estimated duration per package	Manager sets the estimated duration per package: Basic (15 min), Full (30 min), Premium (45 min)
Package printed on receipt	The selected package name and full content list are printed on the intake receipt — client knows exactly what to expect
Package on customer-facing screen	When the agent selects a package, the customer-facing screen displays the package name, content list and estimated duration
Package statistics	Report: which package is most popular, revenue per package, average time per package — helps manager make decisions
Promotional package	Manager can create a temporary promotional package (e.g. 'Ramadan Special — Full wash + interior detail for 30% off')

3.7.2c.4 Agent Performance Management

Each wash is assigned to one or more agents. CleanHub tracks who washed what, how long it took and what client satisfaction score was given. This is the manager's primary tool for monitoring team productivity and quality.

Feature	Description
Agent assignment	At intake, the agent who will perform the wash is selected — or assigned automatically by the manager
Per-agent tracking	Every wash is linked to the agent(s) who performed it — full traceability. If a client complains about quality, manager can find out who washed the vehicle
Agent dashboard	Real-time view per agent: number of vehicles washed today, vehicles currently in progress, average time per wash
Time tracking per agent	CleanHub records the start and end time of each wash per agent — calculates average time per wash
Agent satisfaction score	After each wash, the client's satisfaction score (if survey completed) is linked to the agent who performed the wash
Performance report	Weekly and monthly report per agent: total vehicles washed, revenue generated, average satisfaction score
Agent targets	Manager can set daily or weekly targets per agent: e.g. 'Wash minimum 15 vehicles per day'. CleanHub alerts manager if target is not met
Bonus calculation	If the manager pays bonuses based on performance, CleanHub exports the agent performance data to a CSV file

3.7.2c.5 Real-Time Wash Timer — Step-by-Step Tracking

Each vehicle wash is divided into steps. The agent marks the start and end of each step directly in CleanHub. The manager sees in real time exactly what stage each vehicle is at and how long it has been there.

Step	Agent action	CleanHub records
1. Pre-rinse	Agent taps 'Start pre-rinse' on tablet	Timer starts — vehicle status: Pre-rinse in progress
2. Shampooing	Agent taps 'Start shampoo'	Previous step time recorded — new timer starts
3. Rinsing	Agent taps 'Start rinse'	Shampoo duration recorded
4. Drying	Agent taps 'Start drying'	Rinse duration recorded
5. Interior (if included)	Agent taps 'Start interior'	Exterior duration recorded
6. Finishing	Agent taps 'Start finishing' (wax, tyre shine, perfume)	Interior duration recorded
7. Complete	Agent taps 'Wash complete'	Full wash duration recorded — client notified automatically

- If a step takes longer than the configured maximum time, CleanHub alerts the manager: 'Vehicle Q-03 — shampooing in progress for 18 minutes (maximum: 10 min)'

- The manager can see all vehicles on the dashboard with their current step, elapsed time and colour-coded urgency: green (on time), orange (running late), red (significantly overdue)
- At the end of each day, CleanHub generates a step-by-step timing report: average time per step per wash type — identifies where the team loses time
- The timer data feeds directly into the agent performance report — no double entry

3.7.2c.6 Monthly Vehicle Subscription

The car wash can offer monthly subscriptions to regular clients: e.g. '8 washes per month for 20,000 XOF'. The client pays in advance and uses their quota progressively. Excellent for predictable revenue and guaranteed loyalty.

Feature	Description
Create a subscription plan	Manager creates plans from the Settings panel: name, number of washes included, price, duration (1
Example plans	Basic Monthly (8 quick washes / 15,000 XOF), Full Monthly (8 full washes / 25,000 XOF), VIP Month
Client subscription	Client subscribes at the counter — pays by Wave, Orange Money or cash — subscription recorded in
Usage tracking	At each visit, cashier selects the client's subscription — one wash deducted from the monthly quota
Quota alert	When the client uses their last wash, CleanHub alerts the agent: 'Client DIALLO — last wash on this
Overage	If the client visits after exhausting their quota, additional washes are charged at normal rate — the ag
Automatic renewal	Subscription renews automatically each month — payment triggered via Wave/Orange Money — or
Multi-vehicle subscription	A client with several vehicles can apply their subscription to any of their registered vehicles — or hav
Subscription report	Active subscriptions, usage rate per plan (% of quota used), revenue from subscriptions, most popul

3.7.2c.7 Corporate Fleet Partnership — Monthly B2B Invoicing

Companies with a fleet of vehicles (banks, NGOs, taxi companies, delivery companies, government) can sign a partnership agreement with the car wash. CleanHub manages all their vehicles, tracks every wash per plate and generates a single monthly invoice for the company.

Feature	Description
Corporate client profile	Create a corporate client: company name, contact person, phone, email, billing address, negotiated
Fleet registration	Register all the company's vehicles: plate, make, model, colour, department (e.g. 'Director's vehicle',
Negotiated corporate rates	Corporate clients get their own price list: e.g. Basic wash 1,500 XOF (vs. 2,000 XOF standard) — co
No payment at wash	When a corporate vehicle arrives, the agent registers the wash and marks it as 'Invoice — [Company
Monthly invoice generation	At the end of each month, CleanHub automatically generates a detailed PDF invoice for the compan
Invoice breakdown by vehicle	The company can see exactly how much each vehicle cost that month — useful for their own interna
Payment tracking	When the company pays the invoice (bank transfer, cheque, Wave Business), the manager records
Overdue alert	If an invoice is not paid by the agreed date, CleanHub alerts the manager and optionally sends an au
Corporate report	Monthly: total corporate revenue, revenue per company, top corporate clients by volume, outstanding

■ *The corporate fleet partnership is the most lucrative revenue stream for a car wash in Africa. A single contract with a bank or NGO with 20 vehicles can represent more revenue than 200 individual clients. CleanHub automates the entire process — registration, tracking and invoicing — with zero manual work.*

■ **Corporate clients who pay by invoice must be approved by the OWNER before their first visit on credit. The manager can register corporate washes but cannot activate the 'Invoice' payment mode for a new company without owner authorisation. All credit limits are set by the Owner from the Settings panel.**

Clients who own multiple pressing locations must have access to a unified Back Office dashboard where they can monitor and manage ALL their branches from a single screen.

Feature	Description	
Unified dashboard	One screen showing all branches side by side — revenue, orders, staff activity, stock alerts	
Branch switcher	Switch instantly between branches with one click — or view all consolidated	
Cross-branch orders	Transfer a garment order from one branch to another — tracked with full history	
Consolidated reports	Revenue, orders, staff, stock — all branches combined in one report or compared side by side	
Inter-branch stock	Transfer stock (consumables) between branches — tracked with quantity and date	
Staff management	Manage all employees across all branches from one place	
Unified customer base	One customer profile visible across all branches — loyalty points shared or separate (configurable)	
Branch performance ranking	Automatic ranking of branches by revenue, order volume and customer satisfaction score	
Alert centre	All alerts (low stock, overdue orders, unpaid accounts) from all branches in one feed	
Role	Access	Capabilities
Owner / Admin	ALL branches	Full back office — all reports, all staff, all financials, all settings
Responsible Regional	Assigned group of branches	Reports and staff for assigned branches only
Branch Manager	Single branch only	Local operations — no cross-branch visibility
Cashier / Staff	Single branch only	POS and order entry only — no back office access

■ **The back office is accessible from any device (application, tablet, smartphone) — the owner can monitor all branches in real time from anywhere in the world.**

3.8 SaaS Administration Panel

- Tenant onboarding: create new business accounts, assign subscription plans
- Subscription management: Free Trial, Starter, Pro, Business, Enterprise — for both pressing and car wash businesses
- Billing and invoice generation for SaaS subscriptions
- Platform-level analytics: total tenants, active users, MRR
- Support ticketing and tenant communication tools

3.8.0a In-App Technical Support — Bug Reporting to CleanHub Support Team

The manager or owner can contact the CleanHub SUPPORT TEAM directly from inside the application — without leaving the software, without searching for a phone number, without sending an email manually. One button, accessible at any time. The CleanHub support team receives all tickets, analyses each problem and decides what action to take — the client never deals directly with the developer team.

Feature	Description
Support button — always visible	A 'Help / Report a bug' button is permanently visible in the main menu or navigation bar of the application
Bug report form	When the button is pressed, a simple form opens: (1) Type of problem — dropdown: Bug / Printer issue
Automatic technical info attached	CleanHub automatically attaches to every ticket: pressing code (e.g. SN-0042), app version, device type
Ticket number issued instantly	As soon as the form is submitted, a ticket number is generated (e.g. TKT-2026-0147) and displayed

Feature	Description
WhatsApp/SMS confirmation	The owner receives an automatic WhatsApp/SMS with the ticket number and a summary of what was submitted.
Real-time ticket tracking	The owner can see all their open tickets from a 'My Support Tickets' screen: ticket number, date submitted, status.
CleanHub support response in-app	When the CleanHub support team responds to a ticket, the owner receives a WhatsApp/SMS notification.
Ticket history	Full history of all tickets ever submitted — useful to track recurring issues, verify that a bug was fixed.
Emergency ticket	If the problem is critical (application completely blocked, cannot process payments, data loss), the owner can mark a ticket as 'Emergency'.
Auto-capture on crash	If the application crashes or freezes, CleanHub automatically captures the error log and creates a draft ticket.

Support ticket flow — from bug to resolution:

Step	What happens
1. Bug detected	Manager notices something is not working correctly — presses 'Report a bug' button
2. Form filled	Manager selects problem type, writes a short description, optionally adds screenshot — submits in under 60 seconds
3. Ticket created	Ticket number generated instantly — technical info automatically attached — WhatsApp confirmation sent to owner
4. Developer notified	Developer team receives the ticket on their support dashboard — sees all technical info — can reproduce the issue
5. Developer responds	Developer sends a response via the ticket system — owner receives WhatsApp notification — sees response inside app
6. Fix deployed	Developer fixes the bug and deploys the update — closes the ticket with a message: 'Fixed in version X.X.X — your issue is resolved'
7. Owner confirms	Owner confirms the issue is resolved — ticket closed — added to resolution history

- The 'Report a bug' button is available in ALL plan levels — Starter, Pro, Business, Enterprise. Every CleanHub client has access to technical support directly from the app regardless of their subscription level
- The manager does not need to know the pressing code, app version or device type — CleanHub attaches all this information automatically. The developer has everything needed to diagnose without a back-and-forth
- All ticket communications are archived permanently — the CleanHub support team can search ticket history to track recurring issues and identify patterns across all clients

■ **IMPORTANT — Support chain: the client reports to the CleanHub SUPPORT TEAM. The support team analyses the ticket. If the problem requires a code fix, the support team escalates internally to the developer — the client never contacts the developer directly. The developer has NO access to the client ticket portal. This separation is contractual and must be enforced at the technical level: the ticket system has two separate interfaces — one for clients (inside CleanHub) and one for the CleanHub support team (back office). The developer interface is separate and only accessible by the developer for tickets escalated to them by support.**

3.8.0b Remote Technical Assistance — TeamViewer Integration

When a bug cannot be resolved remotely by exchanging messages, the developer needs to take direct control of the client's computer or tablet to diagnose and fix the problem. CleanHub integrates TeamViewer to enable this — the client launches a remote session directly from the application without installing anything separately.

Feature	Description
TeamViewer embedded in CleanHub	TeamViewer (or equivalent: AnyDesk, Zoho Assist) is integrated directly into the CleanHub application
One-button remote session	From the 'Help / Support' screen, the Owner or Manager presses 'Allow remote assistance' — CleanHub launches the session
Session ID communicated to CleanHub support	The Owner reads or sends the session ID and password to the CleanHub support agent via WhatsApp

Feature	Description
Session only with owner's consent	The remote session can ONLY be started by the owner or manager pressing the button themselves
Session visible to client	During the remote session, the client can see everything the developer does on their screen in real time
Session automatically ends	The remote session closes automatically after the developer disconnects or after a configurable time
Session log	Every remote session is logged in CleanHub: date, time, duration, support agent name, ticket reference
Works on Windows and Android	TeamViewer integration works on both the Windows desktop application (Electron) and the Android app
No installation required by client	The TeamViewer module is bundled inside CleanHub — the client does not need to install TeamViewer
Unattended access — Owner only	For emergency interventions (server crash, critical bug), the Owner can optionally grant unattended access

Remote assistance workflow:

Step	Owner / Manager	Developer
1. Bug reported	Owner submits a ticket from the app	Developer receives ticket — diagnoses remotely from logs
2. Remote session initiated	CleanHub support sends WhatsApp: 'I need to access your screen to fix this'	
3. Session started	Owner presses 'Allow remote assistance' in CleanHub	Developer opens TeamViewer on their computer
4. ID shared	Owner sends session ID and password by WhatsApp	Support agent enters ID and password — connects instantly
5. Intervention	Owner watches the screen — can see everything the support agent does	Support agent navigates, diagnoses, fixes the issue — or escalates if needed
6. Session closed	Owner presses 'End session' when done — or session ends automatically	Support agent confirms fix — updates the ticket
7. Logged	Session recorded: date, duration, support agent — visible in ticket history	

- TeamViewer is the world's most used remote access tool — widely known and trusted. Many clients in Africa are already familiar with it from IT support contexts. No learning curve required.
- Alternative to TeamViewer: AnyDesk (lighter, faster on slow connections) or Zoho Assist (web-based, no install). The developer chooses the best option for the African network context — AnyDesk is recommended for Senegal/West Africa due to better performance on 3G/4G connections.
- The remote session button is available to Owner and Manager roles only — Cashier and Delivery staff cannot initiate a remote session

■ **The developer must embed the remote access module (TeamViewer / AnyDesk / Zoho Assist SDK) inside the CleanHub application at both the Windows (Electron) and Android levels. Remote sessions are initiated by the client and connected to by the CleanHub SUPPORT TEAM — not directly by the developer. The developer has no direct remote access to any client installation. This is a contractual boundary that must be respected.**

3.8.0b.1 Support Chain — Who Sees What

CleanHub operates a strict three-level support chain. The client only ever communicates with the CleanHub support team. The developer is an internal resource — invisible to the client.

Level	Who	Role	Access to client tickets
1	Client (Owner / Manager)	Reports the bug from inside CleanHub — describes what happened	Seen only if attached to a ticket shot
2	CleanHub Support Team	Receives ALL client tickets — analyses every bug — provides first response	Sees ALL tickets from all clients — can initiate session
3	Developer Team	Receives ONLY tickets escalated by CleanHub support — fixes code bugs the developer is responsible for	NO code bugs the developer is responsible for

Rule	Detail
Client → Support only	The client NEVER contacts the developer directly. The developer's contact details are not shared with the client

Rule	Detail
Support sees everything	The CleanHub support team has a back-office dashboard showing ALL tickets from ALL clients in real time.
Developer sees nothing by default	The developer has NO access to the client ticket portal. They only receive a ticket when the CleanHub support team escalates an issue to them.
Escalation decision by support	The support team decides what is a configuration issue (solved internally), what is a user error (explained to client), and what is a bug (escalated to developer).
Bug dashboard for support	The support back office shows: number of tickets per client, most common bug types, bugs by version, bugs by device, bugs by location.
Developer receives bug reports	When the support team escalates to the developer, they send a structured bug report: bug description, client information, device information, location, screenshots, and any relevant logs.

■ **TECHNICAL REQUIREMENT: the ticket system must have TWO completely separate interfaces with separate authentication: (1) Client portal — inside CleanHub, accessible to Owner/Manager roles. (2) Support back office — separate web interface, accessible only to CleanHub support team accounts. The developer interface is a third separate system used only for escalated tickets. These three interfaces share the same database but have strict role-based access control — no cross-access is permitted.**

3.8.0c Self-Repair Tools — Fix Common Problems Without Contacting Support

Most day-to-day problems at a pressing or car wash are minor: a search result that seems wrong, a counter that doesn't match, a printer that stopped responding, a sync that got stuck. CleanHub provides a dedicated 'Maintenance & Repair' screen where the Owner or Manager can fix these issues themselves in one click — without calling support, without waiting, without losing time.

Tool	What it fixes	What it does
Re-index search	Client or order search returns incomplete or incorrect results	Rebuilds the full search index from the database — all names, phone numbers, addresses, etc.
Recalculate counters	Daily totals, loyalty points balance, or stock quantities are incorrect	Recalculates all counters from raw transaction data — daily revenue, loyalty points, stock levels, etc.
Force sync	App shows 'Offline' or data entered on device is not updating	Forcibly updates the app's sync status and triggers a full sync of data between the device and the server.
Clear printer queue	Printer is blocked — jobs piling up, not printing	Clears all pending print jobs from the printer queue — resets the printer to a ready state.
Reset printer connection	Printer was working but suddenly stopped responding (Bluetooth or WiFi)	Disconnects the printer from the system and re-establishes the connection.
Rebuild appointment list	Daily pickup list is empty or showing wrong information	Regenerates the day's pickup list from scratch based on all active appointments.
Fix payment status	A Wave or Orange Money payment shows as pending or failed	Resends the payment confirmation to the Wave/Orange Money API and updates the status.
Re-index vehicle history	Vehicle history by plate is incomplete or incorrect	Rebuilds the vehicle history index — links all historical orders back to the correct vehicle.
Clear app cache	Application is slow, lagging or displaying errors	Clears the local application cache — forces a fresh data load from the server.
Restart sync engine	Background sync is stuck — new orders not coming in	Restarts the background sync engine — reconnects to the server and resumes syncing.
Fix loyalty points	A client's loyalty points balance appears wrong	Recounts the client's loyalty points from their complete order history.
Recount stock	Stock quantities appear incorrect after a sale	Recounts all stock levels from the full transaction history — corrects discrepancies.

How the Maintenance & Repair screen works:

Element	Description
Access	Settings menu → 'Maintenance & Repair' — accessible to Owner and Manager roles only — not visible to Client.
Tool list	Each tool is displayed as a large button with a clear name, a one-line description of what it fixes and an estimated time to complete.
Before running	Each tool shows a warning if needed: 'This operation may take 2 minutes. The application will remain usable but may be slightly slower.'
Progress indicator	When a tool is running, a progress bar is displayed — the manager can see the operation in progress — the tool cannot be interrupted.
Result confirmation	When the tool completes, CleanHub displays: 'Re-indexing complete. 4,847 records indexed. Search is now accurate.'

Element	Description
Automatic log	Every tool execution is logged automatically: tool name, date, time, duration, result, user who ran it — visible
One click, no parameters	No technical knowledge required — the manager does not need to know what an index is or how a database
Safe operations	All self-repair tools are READ operations or safe RECALCULATION operations — they never delete data, n

Recommended use — when to use each tool:

Symptom observed	Tool to use
'I searched for a client by phone and they didn't appear'	Re-index search
'My daily total seems lower than what I actually collected'	Recalculate counters
'I added an order on the tablet but it doesn't show on the PC'	Force sync
'The printer is blocked and nothing is printing'	Clear printer queue
'The printer was working but stopped suddenly'	Reset printer connection
'The pickup list for today is empty but I have orders due'	Rebuild appointment list
'A Wave payment shows Pending but the client got the confirmation'	Fix payment status
'I searched for a vehicle plate and old washes are missing'	Re-index vehicle history
'The app is very slow and some information looks outdated'	Clear app cache
'New orders on one device are not appearing on the other'	Restart sync engine
'A client says their points are wrong'	Fix loyalty points
'Stock quantity is wrong after a sale'	Recount stock

■ *The self-repair tools resolve approximately 80% of day-to-day minor issues without any contact with CleanHub support. Every operation a manager runs is logged and visible to the CleanHub SUPPORT TEAM — if the problem persists after a self-repair, the support team can see exactly what was attempted, which speeds up diagnosis considerably.*

■ **DEVELOPER REQUIREMENT: all self-repair tools must be idempotent — running the same tool twice must produce the same result as running it once. No tool must ever delete data, modify transaction records or alter order history. All tools must complete within 5 minutes maximum on a database of 100,000 orders. Each tool must display a clear success or failure message with a count of records processed. NOTE: the self-repair tool execution log is visible ONLY to the CleanHub support team in their back office — the developer does not have direct access to client-side tool execution logs.**

CleanHub offers two separate subscription plan grids — one for pressing businesses and one for car wash centres. Both grids follow the same Starter / Pro / Business / Enterprise structure but are adapted to each business type.

3.8.0 Free Trial Policy — 3-Month Trial for Own-Equipment Clients

CleanHub has a clear two-track commercial policy depending on whether the client provides their own hardware or uses hardware supplied by iWash Group.

Client profile	Trial period	How it works	Rationale
Client provides their OWN hardware (tablet, PC, printer, scanner)	3 months (free)	Client receives a time-limited download link for the software	The client has invested in their own hardware
Client uses hardware SUPPLIED by iWash Group	No trial - iWash Subscription	Client pays directly from month 1.	iWash Group supplies the hardware, iWash

How the 3-month trial link works — technical requirements:

Feature	Description
Link generation	The iWash Group owner generates a trial link from the CleanHub admin panel: selects the client, sets the link format, and activates the trial.
Download link format	A short, simple URL (e.g. cleanhub.app/trial/XK7F9) that the client can type manually or receive by WhatsApp.
90-day countdown	From the moment the client activates the app with the link, a 90-day countdown begins. The expiry date is displayed on the app.
Automatic deactivation	On day 91, the app displays a clear message: 'Your 3-month trial has ended. Contact iWash Group to activate your subscription.'
Data preservation	All orders, clients, and data created during the trial are fully preserved after expiry. When the client subscribes, the data is seamlessly transferred to the subscription.
Conversion to subscription	The owner activates the client's subscription from the admin panel — the app unlocks immediately, and the client can use the app.
Link expiry (not activated)	If the client has not activated the app within 30 days of the link being generated, the link expires automatically.
One trial per client	A client can only benefit from one 3-month trial. The system prevents generating a second trial link for the same client.
Trial report	The admin panel shows all active trials: client name, activation date, days remaining, status (active / expired), and the trial link.

- The trial link is sent by WhatsApp in one tap from the admin panel — the owner selects the client, clicks 'Send trial link', and the system sends the link automatically with a pre-written welcome message.
- The 90-day countdown runs on the server clock — not on the device clock — so the client cannot extend the trial by changing the date on their device.
- The download link works for both Android (APK) and Windows (Electron installer) — the same link detects the device type and delivers the correct installer.

■ **DEVELOPER REQUIREMENT:** The trial system must be fully automated — no manual intervention required for expiry or activation. The trial countdown must use the NTP server clock (non-negotiable). The link generation, download, activation, countdown display, automatic deactivation and data preservation must all work without any action from the developer or the CleanHub support team. The admin panel must show the full trial dashboard at all times.

Pressing Plans — CleanHub Pressing:

Feature	Starter Press	Pro Press	Business Press	Enterprise Press
Boutiques	1	1	Jusqu'a 5	Unlimited
Utilisateurs / Personnel	2	5	15	Unlimited
Notifications SMS / WhatsApp	No	Yes	Yes	Yes
Suivi vêtements (QR / étiquette matricielle)	Yes	Yes	Yes	Yes
Pickup appointments & daily list	Yes	Yes	Yes	Yes
Storage locations (racks/lockers)	Yes	Yes	Yes	Yes
Washing by kg (weighing + label + appointment)	Yes	Yes	Yes	Yes
Formule laverie (lavage + repassage + pliage)	Yes	Yes	Yes	Yes
Laverie à jetons (self-service assisté)	Yes	Yes	Yes	Yes
Retail product sales (detergent, hangers...)	Yes	Yes	Yes	Yes
Fidelite client & points	Yes	Yes	Yes	Yes
Rapports multi-boutiques	No	No	Yes	Yes
Modules complémentaires optionnels	No	Yes	Yes	Yes
Conveyor integration (future)	No	No	Yes	Yes
Accès API	No	No	No	Yes

Feature	Starter Press	Pro Press	Business Press	Enterprise Press
White-label	No	No	No	Yes

■ All laundry service packages (washing by kg, ironing, folding, assisted self-service with tokens) are available from the Starter plan — a small pressing can offer all these services without a higher subscription.

Car Wash Plans — CleanHub Car Wash:

Feature	Starter Car	Pro Car	Business Car	Enterprise Car
Places parking / Capacite file	Jusqu'a 5	Jusqu'a 15	Jusqu'a 30	Unlimited
Utilisateurs / Personnel	2	5	15	Unlimited
File d'attente vehicules	Yes	Yes	Yes	Yes
Estimated ready time (displayed & WhatsApp)	Yes	Yes	Yes	Yes
Vehicle intake photo & damage record	Yes	Yes	Yes	Yes
Lavage moquettes / tapis (etiquette matricielle + rendez-vous)	Yes	Yes	Yes	Yes
Notifications SMS / WhatsApp	No	Yes	Yes	Yes
Retail product sales (oil, wax, perfume...)	Yes	Yes	Yes	Yes
Rapports multi-boutiques	No	No	Yes	Yes
Modules complementaires optionnels	No	Yes	Yes	Yes
Acces API	No	No	No	Yes
White-label	No	No	No	Yes

Combined Plan — Pressing + Car Wash:

A business that operates both a pressing AND a car wash centre subscribes to a single combined CleanHub plan that activates both modes. Pricing for the combined plan to be defined by the owner — generally a discounted bundle vs two separate subscriptions.

Plan	Pressing Mode	Car Wash Mode	Price
Starter Combo	Yes (Starter Press features)	Yes (Starter Car features)	To be defined by owner
Pro Combo	Yes (Pro Press features)	Yes (Pro Car features)	To be defined by owner
Business Combo	Yes (Business Press features)	Yes (Business Car features)	To be defined by owner
Enterprise Combo	Yes (full Press features)	Yes (full Car features)	To be defined by owner

■ Implement plan limits as middleware checks, not in business logic. This makes plan upgrades instant without code changes. The plan type (pressing / car wash / combined) is set at tenant creation and can be upgraded at any time from the owner admin panel.

3.8.1 Manual Subscription Management — Multi-Region & Multi-Country Clients

For clients who cannot pay online (Wave/Orange Money/card) because they are in a different region or country, or who prefer to pay in cash or bank transfer, the owner must be able to manage their subscription manually.

Feature	Description
Record manual payment	Owner records a subscription payment manually: tenant name, amount, period covered, payment method

Feature	Description
Period activation	After recording payment, owner clicks Activate — tenant subscription extended instantly
Partial payment	Record partial payment — system tracks remaining balance — subscription activates only when fully paid
Currency flexibility	Payment can be recorded in any currency (XOF, USD, EUR, NGN, KES) — system converts to reference currency
Receipt generation	System generates a PDF subscription receipt automatically — sendable by WhatsApp or email to the client
Subscription status dashboard	One screen showing ALL subscribed clients: country, plan, expiry date, payment status — colour coded by status
One-click suspend/reactivate	Owner can manually suspend or reactivate any tenant instantly from this screen

Automatic Reminders:

Timing	Channel	Message
D-10	WhatsApp + Email	Subscription renewal reminder — amount due, payment instructions
D-3	WhatsApp + SMS	Urgent reminder — subscription expires in 3 days
Day 0 (expiry)	WhatsApp + Email	Subscription expired — please renew to keep access
D+7	WhatsApp + Email	Account suspended — contact owner to reactivate

3.8.2 Two-Model Onboarding — Own Hardware vs Owner-Provided Hardware

CleanHub operates TWO distinct client onboarding models depending on whether the client purchased their own hardware or received hardware provided by iWash Group. Each model has its own subscription activation logic — the owner manages both models from the same admin panel.

	Model A — Client buys own hardware	Model B — iWash Group provides hardware
Who provides hardware	The client purchases their own PC, tablet, printer or smartphone	iWash Group sources and provides the hardware to the client
Trial period	Optional — owner decides case by case whether to grant a trial	Not applicable — subscription starts immediately on delivery
Trial duration	3 months maximum — non-renewable	Not applicable
Trial activation	Owner sends a unique time-limited trial link by WhatsApp or email — the client clicks to activate their 3-month trial	Not applicable
Trial expiry	The trial link expires automatically on the configured date — access is blocked — owner must convert to paid subscription	Not applicable
Subscription start	After trial: owner converts the trial account to a paid subscription	Automatic on delivery — no trial period
Payment collection	Wave, Orange Money, card or manual — client pays their own subscription	Payment is built into the hardware agreement

Model A — Trial Link System (Own Hardware Clients):

The trial is NOT a default — it is a deliberate decision by the owner for specific clients who request it. The owner controls who gets a trial and who does not.

Feature	Description
Trial is optional	By default, a new client account goes directly to paid subscription. The owner explicitly chooses to grant a trial
Trial link generation	Owner creates a new tenant in the admin panel, selects 'Grant Trial — 3 months', and the system generates a unique link
Link expiry date	The link displays a clear countdown to the client: 'Your trial expires on [date] — X days remaining'. W
Link is non-transferable	The trial link is tied to one specific pressing (tenant ID) — it cannot be used to activate another pressing
One trial per client	Each client can only receive one trial. After the 3-month trial expires, the owner can convert the account to paid

Feature	Description
Trial features	The trial gives access to the FULL CleanHub feature set — not a limited version. The client experiences the full system.
Trial-to-paid conversion	When the trial expires or before, the owner converts the trial account to a paid subscription from the admin panel.
Trial not renewed	If the trial expires and the client does not convert to paid, the account is suspended. All data is kept for 30 days.
Trial report	Owner dashboard shows all active trials: client name, trial start date, expiry date, days remaining, when converted to paid.

Model B — Automatic Subscription (iWash Group-Provided Hardware):

When iWash Group provides the hardware to a client, the subscription is activated automatically at delivery. There is no trial period — the client has agreed to subscribe as part of accepting the hardware.

Feature	Description
Subscription starts on delivery	When the technician installs the hardware and activates CleanHub at the client's premises, the subscription is activated.
No trial possible	A client who received hardware from iWash Group cannot request a trial period. They are on a paid subscription from Day 1.
Automatic billing	The subscription is billed automatically via Wave or Orange Money on the monthly due date. No manual intervention required.
Hardware agreement link	The subscription plan is linked to the hardware agreement signed by the client — the plan level (Starter, Professional, Enterprise) is determined by the hardware package.
First payment	First payment is collected on delivery day or within the first billing cycle, as agreed with the client. The system tracks payment status.

■ *Example: Client A calls and says they want to try CleanHub before committing — they have their own tablet. Owner creates a trial account in the admin panel, sends a trial link valid for 3 months. After 2 months, the client is satisfied and the owner converts to a Starter paid subscription. Client B is a pressing that ordered a hardware package from iWash Group — when the technician installs on Day 1, subscription starts automatically. Client B cannot request a trial.*

■ **DEVELOPER REQUIREMENT: The trial link system must be implemented as a time-limited JWT token tied to the tenant ID. The token encodes the expiry date and cannot be extended without an admin override. On expiry, all client sessions are invalidated automatically. The owner admin panel must clearly separate 'Trial accounts' from 'Active paid accounts' and 'Suspended accounts' in the tenant dashboard.**

3.9 Subscription Billing & Automated Suspension

The billing system is fully automated. No manual intervention is required. The system handles payment collection, reminders, suspension and reactivation entirely on its own.

Payment Method	How it works	Automation
Wave Senegal	Payment request sent automatically via Wave API	100% automatic — webhook confirmation
Orange Money	Payment request sent automatically via Orange Money API	100% automatic — webhook confirmation
Credit / Debit Card	Card stored securely, charged automatically on due date (if not auto-paid)	100% automatic — recurring charge
Manual Override	Owner can manually mark any invoice as paid from the admin panel	Manual — for exceptional cases only

Timing	Trigger	Automated Action
D-5	5 days before due date	Auto-send WhatsApp + SMS payment reminder
D-2	2 days before due date	Auto-send urgent WhatsApp + SMS reminder
Day 0	Due date	Auto-trigger payment request on all active methods (Wave, Orange Money, Card)
D+3	3 days overdue	Auto-send final warning — suspension in 4 days
D+7	7 days overdue	Account AUTO-SUSPENDED — all access blocked instantly

Timing	Trigger	Automated Action
Upon payment	Webhook / card charge confirmed	Account AUTO-REACTIVATED instantly — no human action needed

4. Database Design

The database is PostgreSQL. All tables must include a `tenant_id` foreign key for multi-tenancy. Timestamps (`created_at`, `updated_at`) are required on every table.

Table	Key Fields
tenants	id, name, pressing_code, plan, subscription_status, country, created_at
users	id, tenant_id, name, email, password_hash, role, branch_id, is_active
customers	id, tenant_id, name, phone, email, address, category, loyalty_points, date_of_birth
orders	id, tenant_id, order_number, customer_id, branch_id, status, total_price, created_by, due_date, appointment
garments	id, order_id, qr_code, type, service_type, colour, notes, status, storage_location, conveyor_slot, photo_image
payments	id, order_id, tenant_id, amount, method, status, mobile_ref, paid_at
products	id, tenant_id, name, category, price, unit
inventory	id, tenant_id, product_id, quantity, min_threshold, branch_id
suppliers	id, tenant_id, name, phone, address, email
expenses	id, tenant_id, branch_id, category, amount, description, date, created_by
appointments	id, tenant_id, customer_id, type (pickup/delivery), scheduled_at, assigned_to, status
branches	id, tenant_id, name, address, phone, manager_id
birthday_campaigns	id, tenant_id, customer_id, sent_at, channel (whatsapp/sms), promo_code, discount_value, discount_type
storage_locations	id, tenant_id, zone, rack, position, full_code, size, status (free/occupied), current_garment_id

5. Security Requirements

4.9 User Rights & Access Management — Admin Control Panel

The administrator must have complete control over what each staff member can see and do in CleanHub. Rights are managed from a dedicated access management panel — no technical knowledge required. Changes apply instantly.

Action	Description
Create user account	Admin creates a new staff account: name, phone, PIN, role, branch assignment — takes under 1 min
Assign role	Assign one of the predefined roles (Owner, Manager, Cashier, Delivery) OR create a custom role with
Assign to branch	Each staff member assigned to one or multiple branches — they only see data from their assigned b
Activate / deactivate	Admin can deactivate a staff account instantly — fired employee loses access immediately, no data
Reset PIN	Admin resets any staff member's PIN in one click
View activity log	Admin sees every action performed by each staff member: orders created, payments processed, dis
Set working hours	Configure each staff member's working schedule

Granular Permission Control:

Permission	Owner	Manager	Cashier	Delivery
Create / edit orders	Yes	Yes	Yes	No
Process payments	Yes	Yes	Yes	No
Partial order delivery	Yes	Yes	Yes	Yes
Partial payment / deposit	Yes	Yes	Yes	No
Apply discounts	Yes	Yes	Limited (max 5%)	No
Cancel / delete an order	Yes	No	No	No
View revenue reports	Yes	Branch only	No	No
Edit prices	Yes	With auth	No	No
Edit client list	Yes	Yes	No	No
Manage stock	Yes	Yes	No	No
Scan garments	Yes	Yes	Yes	Yes
Manage deliveries	Yes	Yes	No	Yes
Create pre-orders	Yes	Yes	Yes	No
Access back office	Yes	Branch only	No	No
Manage users	Yes	No	No	No
View satisfaction surveys	Yes	Yes	No	No
Export reports	Yes	Yes	No	No
Delete / modify registered orders	Yes — mandatory reason	No	No	No
Reprint garment marking labels	Yes — logged	No	No	No
Discounts (max %)	Unlimited	Up to 20%	Up to 5%	None

■ **SECURITY RULE: ONLY THE OWNER can delete, modify or reopen a registered order. ONLY THE OWNER can reprint garment marking labels. ONLY CLEANHUB SUPPORT can modify the date and time used by the software. These rules are NON-NEGOTIABLE.**

5.0 Session Security — Auto-Lock & Staff Protection

The system must automatically lock after a configurable period of inactivity to prevent one staff member from working under another's name. This is critical in a multi-staff pressing environment where the same POS terminal is shared.

Feature	Description
Auto-lock timer	Screen automatically locks after X minutes of inactivity — configurable per role
Lock screen	Shows a PIN entry screen with the last logged-in user's name — another staff member must enter the PIN to unlock
Quick PIN login	Each staff member has a 4-digit PIN for fast re-authentication at the POS
Manual lock button	Staff can manually lock the screen instantly with one tap before stepping away
Session log	Every login, logout and auto-lock event recorded with timestamp and user name — visible to admin
Failed PIN alert	After 3 failed PIN attempts, screen locks for 5 minutes and manager is notified

Role	Default auto-lock	Configurable range
Caissier / Agent	2 minutes	1 to 10 minutes
Branch Manager	5 minutes	1 to 30 minutes
Owner / Admin	10 minutes	1 to 60 minutes
Delivery Staff (mobile)	5 minutes	1 to 15 minutes

■ **This feature is mandatory. It is strictly forbidden for one staff member to process orders or payments under another staff member's account. All transactions must be traceable to the exact staff member who processed them.**

5.1 Authentication

- JWT-based authentication with short-lived access tokens and refresh tokens
- Secure password hashing using bcrypt (min. 12 rounds)
- Optional: Two-factor authentication (2FA) for Admin accounts
- Session invalidation on logout and on suspicious activity detection

5.2 Authorisation — Role-Based Access Control (RBAC)

Feature	Admin	Manager	Cashier	Delivery
All reports & financials	Yes	Branch only	No	No
User management	Yes	No	No	No
Order creation	Yes	Yes	Yes	No
Payment processing	Yes	Yes	Yes	No
Partial order delivery	Yes	Yes	Yes	Yes
Partial payment / deposit	Yes	Yes	Yes	No
Garment scanning	Yes	Yes	Yes	Yes
Delivery management	Yes	Yes	No	Yes

■ *Important rule: Partial delivery is accessible to ALL roles including delivery staff. Partial payment (deposit) is accessible to all counter staff except delivery staff.*

5.3 Data Protection

- HTTPS/TLS 1.3 enforced on all endpoints
- Database encryption at rest for sensitive fields (phone, email, payment references)
- Tenant data must never be accessible by other tenants (strict row-level isolation)
- Regular automated database backups (daily, 30-day retention minimum)

6. UI / UX Screens Required

The interface must be fully in French. It must be designed for users with little or no formal education — many pressing managers in Senegal have never used a computer or smartphone application before. The UI must be intuitive enough to use without training or reading.

Principle	Implementation Requirement
Large icons & buttons	Every action must have a large icon — users navigate by images, not text
Minimal text	Use icons, colours and images as primary communication — text is secondary
Colour-coded status	Order statuses shown as colours: red = urgent, green = done, orange = in progress
Voice guidance (optional)	Audio feedback or spoken confirmation of actions in French/Wolof
1-tap actions	Most common actions (new order, scan, mark ready) must require 1 or 2 taps maximum
No typing required	Use dropdown selectors, toggles and preset options — minimise keyboard use
Error-proof design	Confirmation dialogs before any irreversible action — undo available where possible
Onboarding tutorial	First-launch visual walkthrough with illustrations — no text-heavy instructions

Screen	Key Features
Dashboard (Home)	Live KPIs: daily revenue, active orders, pending deliveries, stock alerts. Fully visible at a glance without scrolling.
POS / Order Entry	Split-screen: client info left, article catalogue right (grid with family tabs). One-click article selection. Running total visible.
Order Search	Universal search bar: searches simultaneously across order number, barcode, client name, phone. Results in list view.
Customer File	Tabs: Profile Active Orders History Loyalty Balance. All current orders visible with status badges. One-click to add new order.
Production Board	Kanban-style: columns per production status. Each card shows garment type, client name and due date. Click to view details.
Planning Calendar	Day view default. Pickup and delivery events in different colours. Click event to see full order details.
Reports	Date range picker at top. Charts (bar, line, pie) with drill-down. Export buttons always visible.
SaaS Admin Panel	Tenant management, subscription plans, platform metrics.
Settings	Tab structure: Business Info Pricing & Services Users & Roles Notifications Printers Branches Subscriptions

7. Third-Party Integrations

7.1 Mobile Payments

- Wave Senegal API: initiate payment requests, receive webhook confirmation, refund handling
- Orange Money Senegal API: same as Wave; support USSD and API-based flows
- Display payment status in real time on the order screen

7.2 Communication

- WhatsApp Business API (Twilio or 360dialog): send order status messages and invoices
- SMS Gateway (e.g. Sendigit or local Senegalese provider): fallback for customers without WhatsApp
- Message templates: order received, ready for pickup, delivery confirmed

8. Local & Technical Requirements

- Offline Mode: POS and order creation screens must function without internet; sync when reconnected
- Language: the platform must be fully bilingual — French (fr-FR) AND English (en). Staff and customers can switch language at any time.
- Language switcher: visible on every screen — one click to switch between French and English instantly in the application

- All content translated: menus, buttons, notifications, SMS/WhatsApp messages, PDF invoices, receipts and reports — all available in both languages
- Default language configurable per tenant: Francophone pressing sets French as default, Anglophone pressing sets English as default
- Performance: all screens must load in under 3 seconds on a 3G connection
- Flexible pricing: the system must support dynamic pricing by garment type, branch and service — no hardcoded prices
- QR/Barcode printing: direct integration with thermal and matrix printers (sourced from China) via USB print driver integrated in Windows and Android applications — no browser, one-click printing
- Currency: configurable per tenant — West African CFA Franc (XOF) for Francophone markets, Nigerian Naira (NGN), Ghanaian Cedi (GHS), Kenyan Shilling (KES) or any local currency

8.1 Portable Bluetooth Printer — Field Collection Slip

Requirement	Specification
Printer type	Portable Bluetooth thermal printer (58mm roll) — sourced from China
Connection	Bluetooth — paired directly with delivery staff smartphone
Triggered from	Native Android app — delivery staff screen
Printed document	Collection Slip — generated at client location upon garment pickup
Offline support	Must print even without internet — order data stored locally on phone
Auto-sync	Order synced with main system when internet is restored

The collection slip must include:

- Order number + QR code (scannable at the pressing on arrival)
- Client name, phone number and address
- List of garments collected with service type (wash, iron, dry-clean)
- Date and time of pickup
- Estimated delivery date
- Staff name and signature line for client to sign
- Business name and logo of the pressing

■ **The Android application must support Bluetooth printer pairing natively. The developer must confirm Bluetooth printing capability with React Native or Flutter before submitting the quotation.**

9. Delivery Plan

CleanHub will be delivered as a complete product. The developer will deliver all features defined in the functional specifications document (90 pages). The owner may agree with the developer on a delivery sequence for the features — but all features must be fully delivered and validated before the final payment is released.

Milestone	Description	Condition
Software delivery	Developer delivers the complete CleanHub software for testing	All features in the specifications document are implemented
Trial period	Owner tests the software — minimum 1 month — on real data at two pressing locations	Software is fully operational
Corrections	Developer fixes all issues identified during the trial period	All issues resolved to owner's satisfaction
Final sign-off	Owner signs the formal written acceptance document	All features of the specifications document are delivered

Milestone	Description	Condition
Final payment	Last payment instalment released to developer	Conditional on written final sign-off by the owner

■ **Key principle: the developer receives the final payment ONLY after the owner has tested and formally accepted the complete software. The CleanHub functional specifications document (90 pages) is the sole reference for what must be delivered.**

9.1 Continental Expansion Strategy — Africa

Phase	Region	Countries	Language
inclusion	West Africa Francophone	Senegal, Ivory Coast, Mali, Burkina Faso, Guinea, Cameroon, Togo, Benin	French
Phase 2	West Africa Anglophone	Nigeria, Ghana, Sierra Leone, Liberia, Gambia	English
integration	East Africa	Kenya, Tanzania, Uganda, Ethiopia, Rwanda	English + Swahili
Phase 4	North Africa	Morocco, Tunisia, Algeria, Egypt	French + Arabic
Phase 5	Southern Africa	South Africa, Zimbabwe, Zambia, Mozambique	English + Portuguese
Phase 6	Afrique Centrale	RDC, Congo, Gabon, Tchad, RCA	Francais

Requirement	Specification
Multi-language	French, English, Arabic, Swahili, Portuguese — switchable per tenant with one click
Multi-currency	XOF, NGN, GHS, KES, TZS, MAD, EGP, ZAR, USD — configurable per tenant
Multi-timezone	Each tenant operates in their local timezone — reports and notifications use local time
Mobile money per country	Wave + Orange Money (Senegal/West Africa), M-Pesa (Kenya/Tanzania), MTN Mobile Money (Ghana)
Tax compliance	Configurable VAT/tax rate per country — invoice format adapts to local tax requirements
Pressing code per country	Country prefix: SN-XXXX (Senegal) / CI-XXXX (Ivory Coast) / NG-XXXX (Nigeria) / KE-XXXX (Kenya)

■ **Vision — CleanHub: The #1 Pressing SaaS Platform in Africa. Starting in Senegal, expanding systematically to West Africa, then East, North and Southern Africa. With 1,000+ pressing businesses as potential clients per country, the total addressable market across Africa represents tens of thousands of potential subscribers.**

10. Pre-Development Research — ABSOLUTE PRIORITY

PRIORITY INSTRUCTION TO THE DEVELOPER

Before writing A SINGLE line of code, the developer must purchase and use the most complete pressing software available on the Chinese market. The objective is simple: copy all its features, then complete them with this CleanHub specification. CleanHub must be AT LEAST as good as the best Chinese pressing software — and better on African-specific features.

■ **Winning strategy: the best Chinese pressing software has been developed and improved over years by pressing professionals. Rather than inventing from scratch, the developer must be fully inspired by it, then add the African specificities of this specification (Wave, Orange Money, offline mode, multilingual, Africa expansion).**

10.1 Step 1 — Purchase the Best Chinese Pressing Software on the Market

The developer must research, test and purchase the COMPLETE VERSION (not a demo) of the most advanced pressing software available in China. Where to look:

Platform	How to search	Keywords to use
Taobao (taobao.com)	Search in Chinese — largest selection of professional software	专业服装熨斗 / 专业熨斗
1688.com	B2B version of Alibaba — professional software	专业服装熨斗 / 专业熨斗
Baidu (baidu.com)	Chinese search engine — find market-leading publishers	专业服装熨斗 / 专业熨斗
WeChat / QQ	Contact Chinese publishers directly for demo and purchase	Professional pressing groups in China
China Trade Shows	Texcare Shanghai, China Clean — software presented by exhibitors	Contact by exhibitors

Priority Chinese pressing software to evaluate:

Software	Why to study it
eXiYi (eXiYi)	One of the Chinese market leaders — complete pressing management with conveyor
XiBei (XiBei)	Popular in pressing chains — good loyalty module
专业服装熨斗	Specialised in multi-branch chains — good back office
YiGuanjia (YiGuanjia)	Advanced touchscreen interface — good for small pressings
专业服装熨斗	Complete: POS + production + delivery + accounting
Any other software rated 4.5+ on Taobao	Developer chooses the most complete and highest-rated on the market

Developer MUST purchase the full commercial version — not a limited demo. The cost of the Chinese software purchase (generally 500 to 3,000 RMB) must be included in the quotation. This is an indispensable investment to build CleanHub correctly.

10.2 Step 2 — Use the Software for at Least 2 Weeks

The developer must use the Chinese software as if they were themselves a pressing manager. They must simulate real orders, deliveries, reports, payments — to understand every detail of the trade workflow.

What to test	Detail
Garment intake	Create orders, print labels, scan barcodes
Production tracking	Move garments from Received to In Progress to Ready via scan
Storage & lockers	Assign physical locations, use conveyor if available
Payment	Test all payment methods, deposits, discounts
Reports & statistics	Explore all available reports — daily, monthly, by cashier
Gestion clients	Fidelite, historique, recherche sans ticket, categories
Notifications	SMS, WeChat, rappels automatiques
Parametres avances	Tarification, utilisateurs, droits d'accès, imprimantes
Special cases	Reserve orders, refunds, corrections, reprints

10.3 Step 3 — Mandatory Benchmark Report

After use, the developer writes a detailed benchmark report. This report must be submitted and approved by the owner BEFORE starting CleanHub development.

Report Section	Expected Content
1. Presentation of purchased software	Name, publisher, price, version, screenshots of main interface
2. Complete feature list	All features found in the Chinese software — exhaustive list
3. Features to reproduce in CleanHub	Everything in the Chinese software that must be in CleanHub — with screenshots
4. Features to improve	What the Chinese software does poorly or incompletely — CleanHub must do better
5. Missing features	What the Chinese software lacks that the CleanHub spec adds (Wave, Orange Money, Africa...)
6. UX recommendations	Which interfaces and workflows are best designed — to reproduce in CleanHub
7. Planning impact estimate	Which Chinese features impact the development schedule — those that take longer

- Report must include annotated screenshots of each important feature
- Report must be in French or English — not in Chinese
- Report must be validated and signed by the owner before any development begins
- Cost of the software + benchmark time (minimum 2 weeks) are included in the global quotation

■ **BLOCKING CONDITION: CleanHub development CANNOT begin before this report is submitted and approved. Any developer who starts coding without completing this step is in breach of contract. This step guarantees that CleanHub will be built on a solid, proven foundation — not assumptions.**

10.4 Step 4 — Pressing Visits in China

In parallel with the software study, the developer must visit real pressing businesses in China to observe field operations.

Pressing type	Objective
Small independent pressing (x1 min)	Observe daily counter operations — closest to our African clients
Medium pressing with multiple staff (x1 min)	Observe multi-staff coordination, garment tracking, production management
Large chain or industrial (x1 min)	Observe conveyor systems, advanced software, B2B workflows

- Photograph the counter setup, label printers, scanners, customer displays, conveyors
- Document which software they use, how they use it, what they like and don't like
- The field visit report is attached to the benchmark report — submitted together for approval

■ **Cost of field visits (transport, entry) included in the quotation. Minimum 3 pressings visited of different types.**

10.5 International Software to Study as Supplement

Software	Market	What it brings specifically
CleanCloud	Global (UK)	Most used worldwide — comprehensive general benchmark
Geelus	Australia	Excellent mobile UX and touchscreen interface
Cents	USA	Modern SaaS architecture — good subscription management
SPOT Business System	USA	Advanced reporting and multi-branch management
Fabricare Manager	USA	Very detailed POS workflow and garment tracking

■ *These international platforms are a supplement to the Chinese software — not a replacement. The Chinese software remains the primary reference because it is closest to the African pressing context (small business, simple interface, basic but complete management).*

11. Evolution Policy & Versioning — CleanHub

Version Type	Frequency	Content
Major version (v1.0, v2.0...)	Once per year	Major new features, new integrations, new technologies, UI overhaul if needed
Minor version (v1.1, v1.2...)	Once per quarter	Functional improvements, performance optimisations, new languages or currencies
Patch releases (v1.1.1...)	As needed	Bug fixes, security patches — deployed without delay

Source	Mechanism
Client Feedback	Built-in suggestion system in the app — clients vote for desired features — most voted are prioritised
Technician Feedback	Field technicians report uncovered needs during installations and training sessions
Satisfaction Surveys	Customer satisfaction data feeds product decisions
Technology Watch	Developer submits an annual technology watch report: new technologies applicable to CleanHub (e.g. AI, AR, VR, etc.)
New Markets	Adaptation to specific requirements of each new target country (currency, language, tax, mobile money, etc.)

- Deliver a major version of CleanHub every year — schedule defined at the start of the year with the owner
- Present in January each year a roadmap of planned evolutions for validation
- Maintain backward compatibility — existing client data and configurations are never lost during an update
- Deploy new versions without service interruption (zero-downtime deployment)
- Notify clients of each new version via in-app notification, email and WhatsApp

11.2 Managing Changes During Development

The CleanHub owner reserves the right to request additions, modifications or new features at any time during development. These requests are welcome and are an integral part of the product philosophy.

Step	Description
1. Owner Request	Owner submits a new idea or feature request in writing (email or project management tool)
2. Developer Analysis	Developer analyses the request and responds within 48h with: time estimate, planning impact and cost if applicable
3. Validation	Owner approves or declines — no modification is developed without written approval
4. Planning Integration	Feature is integrated into the next sprint or scheduled according to jointly defined priority
5. Delivery & Testing	Feature is delivered in test environment — owner validates before production deployment

- Small modifications (< 4h of development): included at no extra cost in the maintenance contract
- Medium features (4h to 2 days): express quote within 48h — billed separately per agreement
- Large features (> 2 days): detailed quote with planning impact — may delay phase delivery
- No modification is developed without written owner approval — protects both parties

■ **The owner can always add new ideas. This process ensures all new ideas are considered, quickly evaluated and developed transparently — with no surprises on costs or timelines.**

11.3 Technician Training Programme — Installation & Client Support

CleanHub is sold and deployed by field technicians who travel to client locations to install the software, configure it, train the pressing manager, and provide on-site troubleshooting. The developer MUST deliver a complete technician training programme as a mandatory contractual deliverable — without it, field

deployment is impossible.

■ **MANDATORY DELIVERABLE: The complete technician training programme (online platform + field manual + certification) must be delivered no later than 2 weeks after the final sign-off. It is a condition of the final payment release. Without this programme, CleanHub cannot be deployed commercially.**

What a CleanHub field technician must be able to do:

Competency	Description
Software installation	Install CleanHub on a Windows PC (Electron), Android tablet and Android smartphone from the APK or i
Initial tenant configuration	Create a new pressing account (tenant) in the CleanHub admin panel: pressing name, logo, address, ph
Printer setup	Install and configure all printer types: thermal receipt printer (USB/Bluetooth/WiFi), matrix garment label
Barcode scanner pairing	Pair Bluetooth HID scanner to tablet/smartphone/PC. Test scanning speed and accuracy. Reconnect if p
Cash drawer connection	Connect cash drawer via RJ11/RJ12 to printer. Test automatic opening on cash payment. Troubleshoot
Payment configuration	Configure Wave and Orange Money: enter API keys, test a payment, verify webhook notifications. Confir
WhatsApp Business setup	Configure WhatsApp Business API credentials. Test sending order ready notification to a test client. Cor
Article catalogue setup	Enter pressing's articles, prices and service types into CleanHub. Create client categories (Standard, VI
User account creation	Create Owner, Manager and Cashier accounts. Assign roles and permissions. Show owner how to creat
Client-facing screen setup	Configure customer-facing screen on second monitor or tablet. Test display of order summary, payment
Offline mode verification	Disconnect the internet and verify POS continues to work. Create an order offline. Reconnect and verify
Owner mobile app setup	Install CleanHub on the owner's personal smartphone. Log in with Owner role. Verify dashboard shows
Client handover training	Train the pressing manager in 60-90 minutes: how to receive a client, create an order, print labels, mana
On-site troubleshooting	Diagnose and fix common problems during installation: printer not detected, scanner not pairing, app no

Technician Training Programme — 6 Online Modules:

The developer must build an online training platform accessible 24/7 from any smartphone or PC. Each module is a short video (5-10 minutes) followed by a practical exercise and a validation quiz. A technician must pass all 6 modules to receive their CleanHub Technician Certificate.

Module	Title	Content
Module 1	Hardware Installation	Step-by-step installation of CleanHub on Windows PC (Electron) and Android tablet/smartphone
Module 2	Printer & Scanner Configuration	Configuring each printer type: thermal, matrix, A4 laser, portable Bluetooth, adhesive label. Blue
Module 3	Tenant Setup & Configuration	Creating a new pressing account in the admin panel. Configuring articles, prices and service typ
Module 4	Client Handover Training	How to train a pressing manager in 60-90 minutes. Step-by-step: first client reception, creating a
Module 5	Troubleshooting & Field Support	Diagnosing the 20 most common problems: printer disconnected, scanner lost pairing, app not l
Module 6	Updates & Maintenance	How to update CleanHub to a new version without losing data. Pre-update backup procedure. P

- Each module includes: intro video (5-10 min), step-by-step illustrated PDF guide, practical exercise on a test environment, validation quiz (80% minimum to pass)
- Downloadable CleanHub Technician Certificate after all 6 modules are validated — PDF with technician name, date and CleanHub logo
- Platform accessible 24/7 from mobile and web — a technician in Dakar can train at 11pm before an installation the next morning

- Videos in FRENCH with French narration — English subtitles available
- Platform updated automatically when a new CleanHub version introduces changes to the installation or configuration process
- The developer must host the training platform — it must remain accessible for at least 3 years after delivery

Technician Field Manual — Printed Deliverable:

In addition to the online platform, the developer must deliver a printed Field Manual that a technician can bring to a client site without needing internet access.

Section	Content
1. Pre-Installation Checklist	Equipment to bring: laptop or tablet, USB stick with CleanHub installer, thermal and matrix printer, Bluetooth scanner
2. Step-by-Step Installation	Numbered steps with screenshots: install Windows app → configure printers → pair scanner → connect to server
3. Client Training Script	60-minute training script with talking points for each step. Key things to say and demonstrate to the person installing
4. Troubleshooting Quick Reference	1-page reference card: symptom → diagnosis → fix → escalate if not resolved. Covers the 15 most common issues
5. Support Contacts	Level 1: CleanHub support WhatsApp number. Level 2: support ticket portal URL. Level 3: emergency contact details
6. Client Sign-Off Sheet	Document the client signs confirming: installation complete, training received, hardware working, support understood

- Field Manual delivered in French AND English — two separate bilingual versions
- Printable PDF format + editable Word (.docx) — both delivered
- Printed A5 format recommended — fits in a technician's bag
- Updated with each major CleanHub version — old versions archived

■ *A well-trained technician can complete a full CleanHub installation, configuration and client handover training in 3 to 4 hours. The field manual and online training are what make this possible at scale — for every new technician, in any city, without the developer needing to be present.*

■ **DEVELOPER REQUIREMENT: The online training platform and field manual are MANDATORY deliverables — they must be included in the development quotation and budget. The developer must allocate specific development time to building the training platform (estimated 3-4 weeks). This is not optional and will not be accepted as out-of-scope.**

11.4 Support Chain — 3 Levels

CleanHub operates a strict 3-level support chain. Clients never contact the developer directly. Every issue passes through this chain.

Level	Who	Response Time	Handles
Level 1	CleanHub Field Technician (on-site or remote)	Immediate (same day)	Installation problems, printer configuration, scanner pairing, basic training
Level 2	CleanHub Support Team (via ticket system)	4 hours (business days)	Software bugs, API/AnyDesk, WhatsApp/SMS configuration issues, payment problems
Level 3	Developer team — remote	1 day (emergency 24/7)	Total server outage, security breach, data corruption, critical bug affecting all clients

- A client with a problem calls or WhatsApps their technician — NOT the developer
- The technician uses the 12 self-repair tools built into CleanHub to resolve most issues without escalating
- If the technician cannot resolve, they submit a Level 2 ticket via the support portal — the developer never sees client data unless a ticket is submitted
- The developer must NEVER be given direct access to client WhatsApp groups or contacts — all communication goes through the ticket portal

■ **This 3-level support chain is NON-NEGOTIABLE. The developer must not give their personal WhatsApp or phone number to pressing clients. All client support passes through the CleanHub support team. The**

developer's role is purely technical — they respond to tickets submitted by the CleanHub support team.

5. Automated Notifications (SMS & WhatsApp)

The platform sends automatic SMS/WhatsApp notifications for key order events. This dramatically reduces inbound phone calls asking 'Is my order ready?' All messages must be in French, personalised with the customer's name.

Event Trigger	Message Template (French)
Order created	"Hello [First name], your drop-off N[XXXX] has been registered. Pickup planned for [Date]. Thank you"
Order ready for pickup	"Hello [First name], your order N[XXXX] is ready! Come and collect it. [Pressing name]"
Order overdue / reminder	"Reminder: your order N[XXXX] has been waiting since [Date]. Please come and collect it."
Delivery confirmed	"Your order N[XXXX] has been handed over. Thank you for your trust! [Pressing name]"
Promotional campaign	Custom message — configurable by the manager per campaign. Promotions are entirely free: the pr
Birthday greeting (AUTOMATIC)	"Happy birthday [First name]! On this special day, [Pressing name] offers you -15% onur votre proch

- SMS Gateway: integrate a Senegalese SMS provider (e.g. Sendigit, ExpressWifi SMS or Orange SMS API)
- WhatsApp Business API: primary channel (Twilio or 360dialog); SMS as automatic fallback if WhatsApp fails
- Notification log: every sent message stored with timestamp, delivery status and recipient
- Manager can disable/enable notifications per event type from Settings

5.1 Birthday Message System — Automatic Customer Messages

The platform must automatically send a personalised birthday message to every customer on their date of birth. This feature is inspired by leading Chinese pressing software where birthday promotions are standard. No existing pressing SaaS in West Africa currently offers this — it is a key differentiator for CleanHub.

- Customer date of birth collected at profile creation — optional field, but staff must always ask at intake
- Every day at 8:00 AM, the system automatically checks which customers have a birthday that day and sends them a WhatsApp or SMS message
- The birthday message includes a personalised promotional offer: configurable discount (%) or fixed XOF amount) valid for a configurable number of days (default: 7 days)
- The promo code generated is unique per customer — cashier sees it when the customer presents it at the counter
- If the customer has no WhatsApp, the system falls back to SMS automatically
- Manager can customise the message template and the discount offer from the Settings panel at any time

Birthday Campaign Settings:

Setting	Description	Default
Discount type	Percentage (%) or Fixed amount (XOF)	%
Discount value	Amount of the discount offered	15%
Promo code validity	Number of days the code remains valid	7 days

Setting	Description	Default
Message template	Custom text for the WhatsApp/SMS message	Configurable
Notification channel	WhatsApp first, SMS fallback, or both simultaneously	WhatsApp + SMS
Active / Inactive	Enable or disable birthday messages globally	Active
Per-branch override	Each branch can have its own birthday offer	Global by default

- Monthly birthday campaign report: number of messages sent, customers who used their promo code, revenue generated from birthday orders
- Report filterable by branch and by period — accessible from the Reports Dashboard

Database Requirements:

Table / Field	Specification
customers.date_of_birth	DATE field — optional — collected at profile creation
birthday_campaigns table	id, tenant_id, customer_id, sent_at, channel (whatsapp/sms), promo_code, discount_value, discount_code
Scheduled job	Cron job runs daily at 08:00 local time — queries customers whose date_of_birth matches today —

■ The date_of_birth field must be added to the customers table from inclus. The birthday message system itself can be delivered later — but the data must be collected from day one so no customer birthday is missed.

9. MVP Scope (inclus — Minimum Viable Product)

Feature	MVP (inclus)	Later Phase
Gestion clients (CRUD + recherche)	Yes	
Order creation & garment tracking (QR)	Yes	
Paiements especes & mobile (Wave, Orange Money)	Yes	
Basic inventory tracking	Yes	
Basic revenue & order reports	Yes	
Auth JWT + RBAC (4 roles)	Yes	
UI Francais/Anglais + mode hors ligne (POS)	Yes	
Native Android app (installed on terminal via APK)	REQUIRED	
Multi-branch management		Included
WhatsApp / SMS notifications		Included
Advanced scheduling & delivery calendar		Included
SaaS subscription billing panel		Included
Advanced BI analytics dashboards		integration future

10. Delivery Timeline & Milestones

Phase	Duration	Deliverables
inclus — MVP	2-3 months	Core modules: customers, orders, garment tracking, payments, basic reports, auth, Fr

Phase	Duration	Deliverables
Advanced Features	TBD	Multi-branch, WhatsApp/SMS, scheduling calendar, SaaS billing, advanced reports, lo
integration future — Scale	Ongoing	Advanced BI, API marketplace, white-label for resellers, conveyor integration, perform

11. Expected Deliverables from Developer

- Full source code hosted in a private Git repository (GitHub / GitLab)
- Docker deployment configuration (docker-compose for local + production)
- API documentation (Swagger / OpenAPI 3.0)
- Database migration scripts (using a migration tool such as Flyway or Alembic)
- User manual in French for end users
- Administrator guide in English
- Automated test suite: minimum 70% code coverage on core business logic
- CI/CD pipeline configuration for automated deployment

This document must be used as the primary technical reference during development. The developer must raise questions or flag blockers within 48 hours of receiving this document. Any deviation from the requirements described here must be approved in writing before implementation.

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Open to Innovation & Additional Features: We are open to any suggestions the developer may have based on their experience and knowledge of the latest technologies. If you identify features, tools or innovations that could add value to this platform — whether related to AI, automation, analytics, user experience or any other area — we strongly encourage you to propose them.